

Global Arthur Enhancements of Cuspidal Excursion Parameters

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Abstract

Under the finite-level Ramanujan–Arthur decomposition stated in Theorem 1.1, we construct a global Arthur enhancement for every occurring cuspidal excursion parameter at a specified full finite level for a split connected semisimple group over a global function field. A single algebraic SL_2 and a commuting Weil centralizer map recover the given parameter by diagonal specialization on the entire Weil group, including inertia. The result does not assert ellipticity, Arthur-packet classification, or a multiplicity formula.

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1 Introduction

An Arthur parameter separates two kinds of information in an automorphic parameter: a part of weight zero and an algebraic SL_2 that accounts for its nonzero weights. Over a global function field, the excursion construction attaches a semisimple Galois parameter to cuspidal automorphic functions. The question considered here is whether its weights come from one SL_2 commuting with its remaining global monodromy. A decomposition of the Frobenius conjugacy class at each place does not supply such a commuting pair. The pair must recover a fixed excursion parameter on the entire Weil group.

1.1 Excursion parameters and the statement

Let X be a smooth projective geometrically connected curve over $\mathbb{F}_q$, let $F = \mathbb{F}_q(X)$, and let $G/\mathbb{F}_q$ be split, connected and semisimple. Fix an effective $\mathbb{F}_q$ -defined divisor D , with arbitrary multiplicities, and set $U = X \setminus \mathrm{supp}(D)$. Write $\mathbb{A}_F$ for the adeles and $\mathcal{O}_{\mathbb{A}}$ for their integral subring. The full level subgroup is

$$K_D = \ker(G(\mathcal{O}_{\mathbb{A}}) \longrightarrow G(\mathcal{O}_D)).$$

For $D = 0$ it is $G(\mathcal{O}_{\mathbb{A}})$.

Fix a prime $\ell \neq \mathrm{char}(\mathbb{F}_q)$, put $k = \overline{\mathbb{Q}}_{\ell}$, and choose an embedding $j : \overline{\mathbb{Q}} \hookrightarrow k$. Let $L = \widehat{G}$ be the split Langlands dual group, equipped with its split rational form. Let C_D be the k -space of compactly supported functions on $G(F) \backslash G(\mathbb{A}_F) / K_D$ whose constant terms along all proper

F -parabolic subgroups vanish. Thus, if N is the unipotent radical of such a subgroup, the condition is

$$\int_{N(F) \backslash N(\mathbb{A}_F)} f(n g) dn = 0 \quad (g \in G(\mathbb{A}_F)).$$

It is independent of the nonzero normalization of the Haar measure. The space C_D is finite dimensional.

Let $\mathcal{B}_D$ be the commutative image of the excursion operators on C_D . An *occurring excursion character* is a character $\nu : \mathcal{B}_D \rightarrow k$ with nonzero simultaneous generalized eigenspace. Its parameter is a conjugacy class of continuous homomorphisms

$$\sigma_\nu : \pi_1(U) \longrightarrow L(k)$$

defined over a finite extension of $\mathbb{Q}_\ell$, with reductive Zariski closure. The parameter theorem characterizes it by all excursion evaluations: for every finite set I , regular function h on $L \backslash L^I / L$, and tuple $(\gamma_i) \in \pi_1(U)^I$,

$$\nu(S_{I,h,(\gamma_i)}) = h((\sigma_\nu(\gamma_i))_{i \in I}). \quad (1.1)$$

Here the quotient denotes simultaneous left and right invariance. This is the excursion parameterization, with its full tuple data [Laf18, Theorems 0.1 and 11.11].

Let $\Gamma = W_U$ be the inverse image of $\mathbb{Z}$ under $\pi_1(U) \rightarrow \text{Gal}(\overline{\mathbb{F}}_q / \mathbb{F}_q) = \widehat{\mathbb{Z}}$, with its Weil topology. We use the Frobenius and normalized Satake conventions of [Ope26]; in particular

$$\deg(\text{Frob}_x) = d_x = [k(x) : \mathbb{F}_q], \quad q_x = q^{d_x}.$$

Fix $a = j(\sqrt{q})$ once and for all. A Frobenius lift and a path to the base point are chosen when an element, rather than a conjugacy class, is needed.

The finite-level Ramanujan–Arthur decomposition is our spectral input. We state precisely the consequence used below.

Theorem 1.1 (Finite-level Ramanujan–Arthur decomposition [Ope26, Theorem 1.1]). *For every occurring excursion character ν , there is a unique nilpotent L -orbit $\mathcal{O}_\nu \subset \text{Lie } L$ such that, at every closed point $x \in U$, the semisimple class of $\sigma_\nu(\text{Frob}_x)$ has the form*

$$\left[\phi_{\mathcal{O}_\nu} \begin{pmatrix} a^{d_x} & 0 \\ 0 & a^{-d_x} \end{pmatrix} c_x \right]. \quad (1.2)$$

Here $\phi_{\mathcal{O}_\nu} : \text{SL}_2 \rightarrow L$ is a Jacobson–Morozov homomorphism for $\mathcal{O}_\nu$ and c_x is semisimple, centralizes its image, and is algebraic and conjugate into a compact subgroup at every complex embedding. The same orbit occurs at every x .

The theorem is a statement about good-place semisimple classes. We use it at that scope. The choices of c_x in (1.2) are not required to arise from a homomorphism. Our result supplies the global homomorphism and retains the specified character ν .

Theorem 1.2 (Global Arthur enhancement). *Assume Theorem 1.1. For every X, G, D, ℓ, j and every occurring excursion character ν as above, there are an algebraic homomorphism*

$$\phi_\nu : \text{SL}_{2,k} \longrightarrow L$$

and a continuous homomorphism

$$\tau_\nu : \Gamma \longrightarrow Z_L(\phi_\nu(\text{SL}_2))(k),$$

both defined over a finite extension of $\mathbb{Q}_\ell$, with the following properties.

- (i) *The differential of ϕ_ν sends $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ to $\mathcal{O}_\nu$.*
- (ii) *The Zariski closure of $\tau_\nu(\Gamma)$ is reductive. For every finite-dimensional algebraic representation r of the split rational form of L and every closed point $x \in U$, each eigenvalue of $r(\tau_\nu(\text{Frob}_x))$ lies in $j(\overline{\mathbb{Q}})$ and its inverse image under j has absolute value one under every embedding $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$.*
- (iii) *After one fixed $L(k)$ -conjugation of σ_ν ,*

$$\sigma_\nu(w) = \tau_\nu(w) \phi_\nu \begin{pmatrix} a^{\deg(w)} & 0 \\ 0 & a^{-\deg(w)} \end{pmatrix} \quad (w \in \Gamma). \quad (1.3)$$

The centralizer is the full algebraic centralizer; it may be disconnected.

Equivalently, $\Psi_\nu(w, h) = \tau_\nu(w) \phi_\nu(h)$ is a homomorphism $\Gamma \times \text{SL}_2 \rightarrow L$ with the prescribed diagonal specialization. Equation (1.3) holds on geometric monodromy and on the inertia groups at the points of D , as well as on Frobenius elements. The conclusion concerns the original curve and level.

1.2 Historical context and related work

Arthur’s conjectures explain the non-tempered part of the discrete spectrum by supplementing a tempered parameter with an algebraic SL_2 whose image commutes with it. This structure appeared in his Maryland lectures, published in 1984, and was developed in his subsequent formulation of unipotent automorphic representations [Art84, Sections 1.3 and 2.1] [Art89, Section 8]. The diagonal specialization of this additional SL_2 accounts for the prescribed powers of the residue-field cardinality in the associated Langlands parameter. The corresponding global conjectures also concern packets and their multiplicities; the existence of a commuting pair is one part of that larger structure.

Over function fields, Drinfeld’s work for GL_2 and Laurent Lafforgue’s work for GL_n realized the Langlands correspondence in the cohomology of shtukas [Dri80, Laf02]. Laurent Lafforgue’s purity theorem for irreducible lisse sheaves with finite-order determinant supplies the weight input used here [Laf02, Theorem VII.6]. Vincent Lafforgue subsequently constructed excursion operators and attached semisimple global parameters to cuspidal automorphic functions for reductive groups at arbitrary finite level [Laf18, Theorems 0.1 and 11.11]. These operators retain invariants of tuples of Galois elements, so the resulting parameter contains more information than its individual unramified Frobenius classes. His Arthur conjecture [Laf18, Section 12.2.2 and Conjecture 12.7] asks that each occurring parameter arise from an elliptic global Arthur parameter.

In the everywhere unramified setting, Gaitsgory–Lafforgue–Raskin formulate the Ramanujan–Arthur decomposition by requiring the same nilpotent-orbit spectral type at every unramified place [GLR26, Section 3.4.3, Conjectures 3.4.5–3.4.6, and Remark 3.4.7]. They distinguish this condition from support on global Arthur parameters, which is formulated separately in [GLR26, Section 3.6]. The finite-level theorem of [Ope26, Theorem 1.1], recalled in Theorem 1.1, supplies the former condition for arbitrary full level. Passing from that condition to one commuting pair requires compatibility with the complete excursion parameter, rather than a separate choice of a triple at each closed point.

A global enhancement in the everywhere unramified setting has already been announced by Raskin as part of joint work with Gaitsgory and Lafforgue [Ras26, Theorem C]. That announcement recovers the specified cuspidal excursion parameter from a commuting $W_X \times \text{SL}_2$ parameter and asserts irreducibility of the latter, under the standing characteristic hypotheses of its Section 1.1.1.

Theorem 1.2 treats arbitrary full finite level under the finite-level decomposition hypothesis. Its conclusion is the existence and purity of the commuting pair stated there; the conclusion does not impose ellipticity or give a packet multiplicity formula.

The local geometry in the proof combines several established constructions. Geometric Satake identifies spherical perverse sheaves with representations of the dual group [MV07]; the derived calculation of Bezrukavnikov–Finkelberg adds the polynomial directions on its Lie algebra [BF08]. Gaitsgory’s nearby-cycles construction produces central sheaves in the affine flag category [Gai01]. Bezrukavnikov’s comparison of the constructible and coherent realizations of the affine Hecke category [Bez16] and the coherent trace theorem of Ben-Zvi–Chen–Helm–Nadler [BZCHN24] then provide the local coherent models. The comparisons developed below retain Frobenius, central labels, and change of level simultaneously. These compatibilities are needed to apply the local models at several places to one global support module.

The global cohomological input comes from Xue’s finiteness and smoothness theorems for stacks of shtukas [Xue20a, Xue20b]. They allow the excursion and partial-Frobenius formalism to be used on the cohomology carrying the local tests. The parahoric nearby-cycles comparison of Salmon [Sal23, Theorem 5.2(1)] relates those tests to fibers at distinct points. The more general restricted-coefficient comparison is developed by Eteve–Xue [EX24]. The parahoric comparison, together with the constituent purity theorem, brings the cohomological weight bound into the simultaneous local calculation.

1.3 The proof mechanism

Put $A = L_{\text{ad}}$ and $\mathfrak{g} = \text{Lie } L = \text{Lie } A$. We first attach to a nonzero ν -eigenvector $f \in C_D$ a finite-type affine scheme $\text{Spec } R$. A point of this scheme records a homomorphism $b : \Gamma \rightarrow A(k)$ and an element $e \in \mathfrak{g}$ satisfying

$$\text{Ad}(b(w))e = q^{\deg(w)}e \quad (w \in \Gamma). \quad (1.4)$$

The invariant evaluations of every tuple of b are those of $\sigma_{\nu, \text{ad}}$. The construction uses matrix entries of partial Weil actions on regular Satake labels, together with a degree-two operation. It is this tuple information that eventually recovers the chosen global parameter.

At suitably chosen good places x , choose a representative t_x of the semisimple Frobenius class and consider the resonance space

$$E_{t_x} = \{e \in \mathfrak{g} : \text{Ad}(t_x)e = q_x e\}.$$

The connected centralizer of t_x has a unique open orbit, whose elements belong to $\mathcal{O}_\nu$. If a point of $\text{Spec } R$ reaches this open orbit at even one test place, the scaling relation yields a global commuting pair. Levi projection and the full tuple identities give the required simultaneous conjugation; a norm argument then proves weight zero. Consequently, failure of Theorem 1.2 would force the image of $\text{Spec } R$ into the boundary at every test place.

The rest of the proof rules out this boundary condition. At the i th test place, a coherent comparison identifies the hyperspecial object with the structure sheaf $B_i = \mathcal{O}_{Y_i}$ of an ordinary complete intersection Y_i . Let P_i^{bad} be the sum of the derived direct images of structure sheaves from those local flag spaces whose vector images miss the open orbit. Their structure-sheaf units define

$$Q_i = \text{fib}(B_i \longrightarrow P_i^{\text{bad}}), \quad q_i : Q_i \longrightarrow B_i.$$

Every boundary point supplied by the global support lies under one of these flag spaces. On its derived residue-field fiber, evaluation at a flag splits the unit map, so the fiber map q_i is zero.

Shtuka smoothness, nearby cycles and constituent purity give one lower cohomological bound, independent of the number of test places. This permits the local tests to act on the global support in ordinary quasi-coherent complexes. If every local image of $\mathrm{Spec} R$ lies in the boundary, each q_i therefore vanishes on every derived residue-field fiber of this support. A noetherian tensor argument gives a finite product $q_{\Pi} = \boxtimes_{i=1}^n q_i$ for which $q_{\Pi}^* f = 0$.

On the other hand, a simple quotient of the cuspidal Iwahori Hecke module detects f and is annihilated by the idempotents of all the summands in P_i^{bad} . Applying shtuka cohomology to the defining fiber diagrams for Q_i shows that f survives their product: $q_{\Pi}^* f \neq 0$. This contradiction forces an open-orbit point and completes the global enhancement. Figure 1 records the dependencies of these two conclusions.

1.4 Technical contributions and organization

Three comparisons carry most of the additional work. The first retains matrix entries and the degree-two operation simultaneously in a finite-type global support. The second constructs the enhanced Frobenius-compatible Iwahori realization and compares the particular holonomy and higher morphisms used by the spherical trace. These are stronger requirements than a comparison of Hecke algebras or characters. The third obtains a simultaneous bound through a one-leg nearby-cycles theorem, using Weil restriction to place the chosen degenerations in one fiber. The argument also isolates a tensor-vanishing lemma for a sequence of maps from pseudo-coherent bounded-above complexes; its proof is independent of the automorphic setting.

We also use the local constructions and constituent purity proved in [Ope26, Sections 2–3 and Lemma 6.4], with their stated compatibilities. In particular, spherical Satake and path rotation are enhanced [Ope26, Theorem 2.2 and Proposition 2.5]. The central-sheaf functor takes values in the Drinfeld center of the homotopy category. For each fixed central label its interchange is a natural equivalence of enhanced exact functors in the other kernel variable; the central tensor identities hold in the homotopy category [Ope26, Theorem 2.3]. The resulting Hecke algebra action is on cohomology [Ope26, Proposition 2.6]. The additional enhanced Iwahori comparison and its compatibility with the trace transfers are proved in Sections 4–6.

Section 2 constructs the cyclic support, and Section 3 proves the open-orbit criterion for the global enhancement. Section 4 constructs the enhanced local realization. Sections 5 and 6 establish the transfer maps and coherent tests. Section 7 proves their simultaneous cohomological bound. Section 8 combines the tests and proves Theorem 1.2.

1.5 Conventions

All categories used in traces or tensor products have their stable dg enhancements; all coends are derived. The notation Coh means bounded derived coherent complexes, and shifts are cohomological. A *neutral* representation of L is one factoring through A . On these representations the component parity correction in geometric Satake is trivial. Satake kernels are normalized relative to the positions of their legs, with the shifts and Tate twists of [Ope26]; thus $k(-1)$ has Frobenius q . The comparison of actual trace operations always retains these normalizations. When a single complex absolute value is used in a weight estimate, we fix a complex realization extending the chosen embedding of algebraic coefficients.

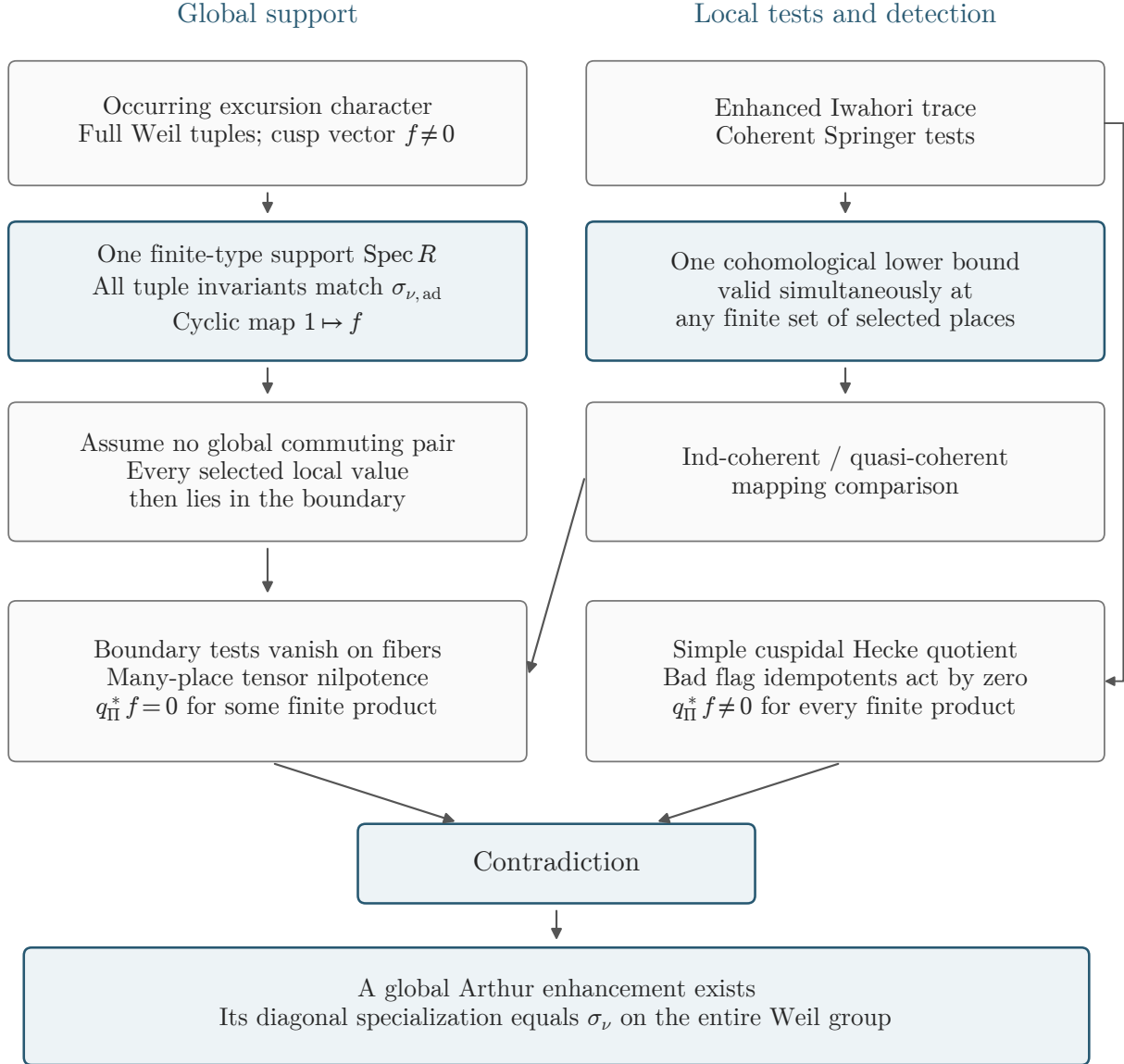


Figure 1: The two uses of the global cyclic support. Full excursion tuple data identify the parameter when an open-orbit point exists. If every test sees boundary support, simultaneous coherent tests give incompatible vanishing and nonvanishing statements for the same cusp vector.

2 A global support from shtuka cohomology

Fix an occurring excursion character ν and its parameter σ_ν . The first step is to retain the matrix entries of this parameter in a commutative algebra acting on shtuka cohomology. Derived Satake supplies an additional Lie-algebra coordinate e . The resulting algebra will parametrize homomorphisms b with the full excursion invariants and the equation $\mathrm{Ad}(b(w))e = q^{\deg(w)}e$. Its finite type is important: later we will test one and the same support at arbitrarily many closed points.

Write $L = \widehat{G}$, $A = L_{\mathrm{ad}}$, and $\mathfrak{g} = \mathrm{Lie} L = \mathrm{Lie} A$. A representation of L is called *neutral* if it factors through A . On these representations the component-parity correction in geometric Satake is trivial. We use the shifts relative to the positions of the legs and the Frobenius convention of [Ope26, Section 2]; in particular, Frobenius on $k(-1)$ is q . All representation categories in this section have coefficients in $k = \mathbb{Q}_\ell$.

2.1 The cohomology and its Weil actions

We first allow either the prescribed full level D , or that level together with Iwahori level at a finite set $M \subset |U|$. Let U° be the corresponding good open, namely U or $U \setminus M$, and put $\Gamma = W_{U^\circ}$. For a finite set J and $V \in \mathrm{Rep}(L^J)$, denote by $H_{J,V}^b$ the degree- b compact cohomology of the stack of shtukas with Satake coefficient V , in the filtered limit over Harder–Narasimhan bounds. We use the same notation for its fibers, specifying paths when they matter. The normalization is relative to $(U^\circ)^J$, so fusion introduces no change in b .

Lemma 2.1 (Cohomological inputs). *The groups $H_{J,V}^b$ have the following properties.*

- (i) *Their cohomology sheaves are ind-smooth on $(U^\circ)^J$. Geometric transport and partial Frobenius give an action of Γ^J on the transported fibers, compatible with fusion. When several legs are fused, the Weil group acts diagonally on those legs.*
- (ii) *For every auxiliary closed point $y \in |U^\circ|$, the generic fiber of $H_{J,V}^b$ is finitely generated over the spherical Hecke algebra $\mathcal{H}_y$. This action commutes with the partial Weil actions. Quotients by finite-codimension ideals of $\mathcal{H}_y$ are finite-dimensional continuous Weil representations, defined over finite extensions of $\mathbb{Q}_\ell$ whenever the data are so defined.*
- (iii) *The identity expressing a spherical Hecke operator as creation, partial Frobenius, and annihilation holds with arbitrary spectator legs, on the full cohomology. On cusp vectors its excursion operations agree with those defining $\mathcal{B}_D$.*

Proof. We use the fusion, partial Frobenius, and Hecke constructions of V. Lafforgue [Laf18, Propositions 4.12, 4.14, and 6.2], the finiteness and excursion theorems of Xue [Xue20a, Theorem 2 and Sections 3.2–3.7], and Xue’s ind-smoothness theorem [Xue20b, Theorem 4.2.3 and Proposition 5.0.4]. These results permit an arbitrary finite level subscheme. The central lattice quotient in their statements may be taken trivial because G is semisimple. Additional Iwahori level is obtained from full level at the added points by invariants under a finite level group. In characteristic zero this is a direct summand, so the same assertions hold there, with the added points removed from U° .

For clarity, the product Weil action does not require a product formula for the geometric fundamental group of an open curve. The path group with partial Frobenius has quotient $\mathbb{Z}^J$ and geometric kernel the ordinary geometric fundamental group of $(U^\circ)^J$. It maps onto Γ^J ; on degree-zero kernels, surjectivity follows by fixing all but one coordinate. Drinfeld’s lemma for finite-dimensional continuous representations factors its action through Γ^J [Xue20a, Lemmas 3.2.10

and 3.3.2]. One may see the finite-dimensional assertion from its finite-cover version as follows. The compact geometric image is a compact ℓ -adic analytic group and has open characteristic subgroups forming a neighborhood basis. Attach the degrees to the image and quotient by one of these subgroups. The result is finite-by- $\mathbb{Z}^J$ and is residually finite: after a finite-index passage the finite kernel is central, and sufficiently divisible powers of the remaining generators give an abelian subgroup of finite index. The finite-cover lemma therefore kills the kernel of the map to Γ^J in every such quotient, hence in the original representation.

Apply this to the finite-dimensional quotients in (ii). Their geometric actions are continuous by ind-smoothness, or by taking images of finite stages at a geometric generic point. Finite generation over the noetherian algebra $\mathcal{H}_y$ implies

$$\bigcap_{\mathfrak{m}} \bigcap_{n \geq 1} \mathfrak{m}^n H_{J,V}^b = 0,$$

where $\mathfrak{m}$ ranges over maximal ideals. Thus these quotients detect the action on the whole module, as in [Xue20a, Lemma 3.2.13]. Hecke operators commute with the partial actions in disjoint position; smoothness then gives the transported assertions. Finally, fusion identifies the action on a fused leg with the product of its partial Frobenius maps. The creation and annihilation morphisms are the same ones on full and Hecke-finite cohomology, which proves (iii). $\square$

We will also need the usual compact-cohomology weight bound, in the normalization just fixed. Here and below a weight is measured using one fixed complex realization of the coefficient field.

Lemma 2.2 (Complete Frobenius weights). *Let the legs be at pairwise distinct points of U° rational over $\mathbb{F}_{q^a}$, with external Satake labels. Every eigenvalue of complete q^d -Frobenius on a finite-dimensional subquotient of $H_{J,V}^b$ has weight at most b .*

Proof. With the path normalization of [Ope26, Section 2], the coefficient pulled back from Satake is a mixed complex of weights at most zero: its degree- a stalk cohomology has weights at most a . On each finite-type Harder–Narasimhan open, compact direct image consequently has weights at most b in degree b . The bounded-leg shtuka stacks have Deligne–Mumford opens with finite stabilizers; the same bound follows either directly for these stacks or by stratifying and using coarse spaces with exact finite-group invariants. The complete Frobenius obtained from the partial actions is the cohomological Frobenius, by the cyclicity of the path functor [Ope26, Section 2, Proposition “Coherent path rotation”]. The bound passes to the filtered limit and its finite-dimensional subquotients. $\square$

2.2 Matrix entries and a Lie-algebra coordinate

Choose a geometric base point in U° and use it in every factor. For representatives V_λ of the simple objects of $\text{Rep}(A)$, form the rational A -module

$$M_b^{\text{reg}} = \bigoplus_{\lambda} H_{\{1\}, V_\lambda}^b \otimes V_\lambda^*.$$

The action of A is on the coefficient factors V_λ^* . Semisimplicity and fusion give, naturally in the neutral labels V_i ,

$$\left(M_b^{\text{reg}} \otimes \bigotimes_{i \in J} V_i \right)^A \simeq H_{J, \boxtimes_{i \in J} V_i}^b. \quad (2.1)$$

We now keep the individual matrix entries of the Weil actions on the right-hand side.

For $V \in \text{Rep}(A)$ and $w \in \Gamma$, define the A -equivariant map

$$T_V(w) : M_b^{\text{reg}} \otimes V \longrightarrow M_b^{\text{reg}} \otimes V$$

by the following tests. After tensoring with any $W \in \text{Rep}(A)$ and taking invariants, use (2.1) for $V \boxtimes W$ and act by w on the V -leg. These maps are natural in W . Since rational A -modules are semisimple, such tests determine a unique equivariant map, also when their multiplicity spaces are infinite.

Let $\mathcal{B}_\Gamma$ be the coordinate algebra of the scheme $\text{Hom}(\Gamma, A)$, with Γ regarded as an abstract group. Equivalently, for a commutative k -algebra B , its B -points are the homomorphisms $\Gamma \rightarrow A(B)$. This algebra need not be of finite type. Denote its universal homomorphism by $\mathbf{b}$.

Lemma 2.3 (The matrix-entry action). *The matrices $T_V(w)$ define an A -equivariant action of $\mathcal{B}_\Gamma$ on each M_b^{reg} , with A acting on $\text{Hom}(\Gamma, A)$ by conjugation. For a good closed point y , the function $\text{Tr}_V(\mathbf{b}(\text{Frob}_y))$ acts by the spherical Hecke operator with label V .*

Proof. The entries of $T_V(w)$ and $T_W(w')$ commute: test both maps using separate V - and W -legs and use the product Weil action of Lemma 2.1. Fusion and the group law give

$$T_{V \otimes W}(w) = T_V(w)T_W(w), \quad T_V(w)T_V(w') = T_V(ww'), \quad T_1(w) = 1.$$

In the first identity the two matrices act on their respective tensor factors. Together with naturality in representations and duality, these are the tensor, invertibility, and group-law relations presenting $\mathcal{B}_\Gamma$. The construction is equivariant by definition of the tests. Contracting the V - and V^* -labels identifies its invariant trace with creation, partial Frobenius, and annihilation. The Hecke identity with spectators proves the last assertion. $\square$

The graded module $M_\bullet^{\text{reg}}$ carries a further commuting operation. Choose a nonempty open $U_0 \subset U^\circ$ and an étale coordinate $t : U_0 \rightarrow \mathbb{A}^1$ over $\mathbb{F}_q$. Derived Satake identifies its local spherical category with $\text{Perf}^L(\text{Sym}(\mathfrak{g}^*[-2]))$ and all morphisms monoidally [Ope26, Section 2, Theorem “Derived Satake with Frobenius”]. In particular, the identity tensor in $\mathfrak{g}^* \otimes \mathfrak{g}$ defines

$$\epsilon : \mathbf{1} \longrightarrow \text{Sat}(\mathfrak{g})[2]. \quad (2.2)$$

Frobenius multiplies this morphism by q .

Lemma 2.4 (The degree-two operation). *The morphism (2.2) induces an equivariant map*

$$E_b : M_b^{\text{reg}} \longrightarrow M_{b+2}^{\text{reg}} \otimes \mathfrak{g}.$$

Its coordinates commute with each other and with the $\mathcal{B}_\Gamma$ -action. Writing $\mathbf{e}$ for the resulting Lie-algebra coordinate, so that its linear functions have degree two, one has the identity of operators

$$\text{Ad}(\mathbf{b}(w))\mathbf{e} = q^{\deg(w)}\mathbf{e} \quad (w \in \Gamma). \quad (2.3)$$

These operations commute with all good-place Hecke actions.

Proof. First globalize the local morphism over the chosen coordinate open. For a moving leg v , the parameter $z = t(x) - t(v)$ identifies its formal disc with the standard formal disc. The local equivariant kernel and ϵ therefore define a relative kernel morphism over U_0 . With other legs present, apply this morphism in one step of the iterated Hecke stack. On bounded modifications the twisted products and their morphisms descend along the trivialization torsors, using a sufficiently large

finite jet quotient. Pullback to the Frobenius path stack and compact direct image in the bounds limit give maps of cohomology sheaves over powers of the geometric open. The direct image merging successive modifications is proper, so proper base change identifies the restriction to coincident legs with convolution by ϵ .

More explicitly, write $\mathcal{H}_{J,V}^b$ for the ind-smooth cohomology sheaf whose fibers are $H_{J,V}^b$. Fix a distinguished leg $i \in J$ and an external neutral label W on the remaining legs. The construction just given, applied to the Frobenius-invariant Tate twist $\mathbf{1} \rightarrow \text{Sat}(\mathfrak{g})[2](1)$ of ϵ , is a map of the relative coefficient complexes on the iterated shtuka stack. Compact direct image gives the underlying geometric map

$$\mathcal{H}_{J,1_i \boxtimes W}|_{U_0^J} \longrightarrow \mathcal{H}_{J,\mathfrak{g}_i \boxtimes W}^{b+2}|_{U_0^J}(1). \quad (2.4)$$

Thus this map exists on the entire coordinate product, including the diagonals. It is not defined by extension from disjoint legs.

Using this map with a spectator label and then (2.1) gives E_b . Its naturality in spectator representations is the convolution naturality on the diagonal. The tensor and composition rules in derived Satake show that the coordinates of E commute. More explicitly, the two ways of applying ϵ are both the tensor square of the identity tensor in $\mathfrak{g}^* \otimes \mathfrak{g}$; interchanging the two neutral adjoint labels is the ordinary flip. The enhanced monoidal identification includes these morphisms, and degree two introduces no Koszul sign.

We now check the partial Weil structures on (2.4). On the target, denote by (1_i) the Tate twist equipped with partial Frobenius multiplied by q^{-1} in the i th variable and unchanged in the other variables. Its total Frobenius is the ordinary Tate-twisted Frobenius. The map is compatible with these structures. After forgetting the twist, this says that it commutes with partial Frobenius on every spectator leg, whereas on its own leg

$$F_{\text{target}} \epsilon = q \epsilon F_{\text{source}}.$$

To check these equalities of cohomology-sheaf maps, work first where all modifications, including their Frobenius moves, are disjoint. A spectator partial Frobenius changes only its own factor of the twisted product and commutes with the morphism on the distinguished factor. The distinguished partial Frobenius instead pulls that morphism through Frobenius and its Weil structure. Since the coordinate is defined over $\mathbb{F}_q$, the latter comparison is exactly the local scaling by q for ϵ . Evaluation on a geometric generic fiber is faithful for maps of ind-smooth sheaves, as is seen on their finite-dimensional smooth stages. Both equalities therefore hold on all of U_0^J . Restriction to a fusion diagonal uses the proper-base-change identification already established, so the same equalities hold for the fused tests. Because (2.4) is a geometric sheaf map, it also commutes with geometric transport. Lemma 2.1 then gives the asserted compatibility with each partial Weil action.

Spectator compatibility, tested against the matrix entries of $T_V(w)$, proves that those entries commute with the coordinates of E . On the distinguished adjoint leg the source label is the unit; hence the second compatibility gives $T_{\mathfrak{g}}(w)E_b = q^{\deg(w)}E_b$. Since $T_{\mathfrak{g}}(w) = \text{Ad}(\mathbf{b}(w))$, this is (2.3). It suffices to prove these assertions on U_0 : the map $W_{U_0} \rightarrow \Gamma$ is surjective. Finally, a good-place Hecke correspondence can be taken disjoint from the moving legs at a geometric generic point. It then commutes with the construction, and transport gives the stated Hecke compatibility. $\square$

We have thus constructed a graded action of the commutative algebra

$$\mathcal{R}_{\Gamma} = \frac{\mathcal{B}_{\Gamma} \otimes \text{Sym}(\mathfrak{g}^*)}{(\text{coefficients of } \text{Ad}(\mathbf{b}(w))\mathbf{e} - q^{\deg(w)}\mathbf{e}, w \in \Gamma)}.$$

The matrix coordinates have degree zero and the linear $\mathbf{e}$ coordinates have degree two. We retain this grading on cyclic quotients, but use the ordinary underlying affine scheme when speaking of their points or support.

2.3 Comparison with fixed-place path traces

The preceding action must agree with the local coordinates used in the later tests. We establish the comparison here while the cohomological construction is explicit.

Let $x_1, \dots, x_n$ be distinct closed points of U_0 , put $d_i = \deg(x_i)$ and $r_i = q^{d_i}$, and choose paths defining $\gamma_i = \text{Frob}_{x_i} \in \Gamma$. In each geometric Frobenius cycle, place $\text{Sat}(V_i)$ at one selected point and the unit at the other points. The spectral description of the hyperspecial path trace [Ope26, Section 2, Proposition “Spectral form of a Frobenius trace”] uses the dg algebra

$$B_r = \mathcal{O}(L) \otimes \text{Sym}(\mathfrak{g}^*[-2] \oplus \mathfrak{g}^*[-1]), \quad d\eta_\alpha = \langle \alpha, \text{Ad}(s)e - re \rangle.$$

Here $s \in L$ is the holonomy coordinate; e has linear functions in degree two, and η_α has degree one. Write $\mathcal{M}_n$ for the cohomology of the corresponding equivariant spectral module, restricted to the sector on which $Z(L)^n$ acts trivially.

Lemma 2.5 (Frames and local coordinates). *As graded rational A^n -modules,*

$$\mathcal{M}_n \simeq \text{Ind}_A^{A^n} M_\bullet^{\text{reg}} = (\mathcal{O}(A^n) \otimes M_\bullet^{\text{reg}})^A. \quad (2.5)$$

If $g_i \in A$ are the frame coordinates, with diagonal change of frame $g_i \mapsto g_i h^{-1}$, then the adjoint holonomy and degree-two coordinates act as

$$s_{i,\text{ad}} = g_i \mathbf{b}(\gamma_i) g_i^{-1}, \quad e_i = g_i \mathbf{e} g_i^{-1}. \quad (2.6)$$

Proof. Testing the left side by $\boxtimes_i V_i$ gives the compact path cohomology of the selected labels. Transport and (2.1) identify this with $(M_\bullet^{\text{reg}} \otimes \boxtimes_i V_i)^A$. Frobenius reciprocity identifies the same test on the right side of (2.5); semisimplicity proves that equation. The description of e_i follows from the construction of ϵ on the selected leg.

For holonomy one must check the entire matrix, not only its trace. The local full-cycle slide acts on a representation label by $V_i(s_i)$, by the cited spectral trace proposition. Keep a spectator label at the same place, and before fusion retain their ordered successive modifications, with the sliding leg at the rotating end. Moving that leg through Frobenius to the other end is precisely the partial Frobenius morphism in the iterated-modification definition of shtukas. The Weil structure identifies the induced slide on cohomology with the positive-degree partial Frobenius action. The passage between orders uses the same proper convolution direct images and Satake fusion maps as coalescence. If $d_i > 1$, the intermediate moves are disjoint from the spectator, and the returning move uses this Satake interchange. The full cycle consequently acts by $T_{V_i}(\gamma_i)$ with every spectator, proving the matrix identity in (2.6). All comparisons use the product Weil actions of Lemma 2.1, so are compatible as the finite set of legs varies. $\square$

2.4 The finite-type cyclic support

Return to level D and hence $\Gamma = W_U$. The nonzero generalized ν -eigenspace contains a simultaneous eigenvector $0 \neq f \in C_D$: a commuting finite-dimensional algebra over an algebraically closed field has a nonzero socle in every nonzero local module. Empty legs identify C_D with the cuspidal subspace of $(M_0^{\text{reg}})^A$. Define

$$R = \mathcal{R}_\Gamma / \text{Ann}(f).$$

Proposition 2.6 (Global support). *The algebra R is a nonzero finitely generated graded k -algebra with rational A -action. It has a universal homomorphism $\mathbf{b} : \Gamma \rightarrow A(R)$ and an equivariant element $\mathbf{e} \in \mathfrak{g} \otimes R$ satisfying (2.3). There is an equivariant graded injection*

$$R \hookrightarrow M_{\bullet}^{\text{reg}}, \quad 1 \mapsto f. \quad (2.7)$$

For every finite tuple $(w_i)_{i \in I}$ and every $F \in \mathcal{O}(A^I)^A$, its value on the universal tuple is the scalar

$$F((\mathbf{b}(w_i))_{i \in I}) = F((\sigma_{\nu, \text{ad}}(w_i))_{i \in I}) \quad \text{in } R.$$

In particular the full simultaneous-conjugacy data of the excursion parameter are retained by $\text{Spec } R$.

Proof. All assertions except finite type and the invariant evaluations follow from the construction. The annihilator is graded because f is homogeneous and is A -stable because f is invariant. For the invariant evaluation, append an identity variable to convert F into a function invariant under simultaneous left and right multiplication. The action of this function is the corresponding excursion operation, by creation and annihilation in (2.1). Lemma 2.1(iii) identifies its action on f with ν . Commutativity then gives the asserted identity throughout the cyclic module R .

Choose one auxiliary good point y_0 . All operations producing (2.7) commute with $\mathcal{H}_{y_0}$, so their values on f belong to its simultaneous Hecke eigenspace. Choose finitely many representations whose matrix coefficients generate $\mathcal{O}(A)$. As w varies in Γ , the corresponding matrix entries applied to f have degree zero and lie in a fixed finite list of A -isotypic components. In each such component the multiplicity space is finitely generated over the noetherian algebra $\mathcal{H}_{y_0}$. Its submodule annihilated by the Hecke maximal ideal is therefore finite-dimensional over k . Hence the entries under consideration span a finite-dimensional subspace of R . Finitely many of them generate all degree-zero matrix coordinates as an algebra. Adding the finitely many linear coordinates of $\mathbf{e}$ proves that R is of finite type. $\square$

We will use y_0 as the fixed auxiliary Hecke place. The construction up to this final cyclic quotient remains available at every auxiliary Iwahori level, a fact needed for the weight argument in Section 7.

3 An open-orbit criterion for enhancement

The algebra R of Proposition 2.6 records abstract homomorphisms with the full excursion invariants of $\sigma_{\nu, \text{ad}}$. Its points need not initially be continuous. We prove that a point whose Lie coordinate lies in one specified open orbit already yields the continuous enhancement of Theorem 1.2. The remaining sections will force such a point to exist.

3.1 Places with connected centralizers

For $x \in |U_0|$, let $t_x \in L$ be a torus representative of the spherical class acting on f , in the holonomy convention of [Ope26, Section 2]. Its image $t_{x, \text{ad}} \in A$ represents the semisimple class of $\sigma_{\nu, \text{ad}}(\text{Frob}_x)$. The full-group convention may differ from another normalized Satake presentation by a finite central sign. This does not change the adjoint class, any absolute value exponent, or any centralizer. Write $L//L$ and $A//A$ for the affine conjugation quotients. Let Ω be the finite group of length-zero elements of the extended affine Weyl group of G .

Lemma 3.1 (Choice of test places). *There are infinitely many distinct points $x_i \in |U_0| \setminus \{y_0\}$, with $d_i = \deg(x_i)$ divisible by $|\Omega|$, such that, for $t = t_{x_i}$,*

- (i) $H = Z_L(t)$ and $Z_A(t_{\text{ad}})$ are connected;
- (ii) near the class of t_{ad} , the map $L//L \rightarrow A//A$ is a $Z(L)$ -torsor, and the central torsor $L \rightarrow A$ is its pullback. In particular, $Z_A(t_{\text{ad}}) = H/Z(L)$.

Proof. Choose a faithful representation of A . The parameter has compact image in its matrices over a finite extension of $\mathbb{Q}_\ell$. Chebotarev applied to a sufficiently small finite quotient of this image gives infinitely many Frobenius classes in any prescribed identity congruence neighborhood. Include the constant-field quotient modulo $|\Omega|$ to impose the degree condition. We may discard y_0 and the finitely many points outside U_0 .

All eigenvalues of these matrices are as close to one as required. The weights of a faithful representation generate the character lattice of a maximal torus of A , so a torus representative t_{ad} lies in a sufficiently small neighborhood of one. Choose the neighborhood so that the torus logarithm is injective there and on its Weyl transforms. The Weyl stabilizer of t_{ad} then equals the stabilizer of its logarithm. The stabilizer of that vector is generated by the reflections in the roots vanishing on it. To apply the usual real reflection-group statement, one may replace the logarithm by a real vector having the same stabilizer and root vanishings: these conditions are membership and nonmembership in finitely many rational linear subspaces. Injectivity of logarithm identifies the vanishing roots with those α for which $\alpha(t_{\text{ad}}) = 1$.

The identity component of a semisimple centralizer is generated by the torus and these root groups, and its component group is the corresponding quotient of Weyl stabilizers [DM18, Propositions 1.11, 1.12, and 1.14]. Thus $Z_A(t_{\text{ad}})$ is connected. Every generating reflection also fixes any lift $t \in L$, because its root evaluates to one on t . This proves connectedness of $Z_L(t)$ and shows that no nontrivial central translate of t is conjugate to t .

The torus quotient by the Weyl group computes the conjugation quotient. The last assertion says exactly that $Z(L)$ acts freely on $L//L$ over the class in question. After shrinking around that class, its quotient map is a torsor. The natural map

$$L \longrightarrow A \times_{A//A} (L//L)$$

is a map of $Z(L)$ -torsors and hence an isomorphism on this open. The centralizer identity follows, either from this description or from the equality of Weyl stabilizers just proved. $\square$

Fix this sequence of places. For one of them, suppress the index and put

$$r = q^{\deg(x)}, \quad H = Z_L(t), \quad E_t = \{e \in \mathfrak{g} : \text{Ad}(t)e = re\}.$$

Theorem 1.1 supplies a homomorphism $\phi : \text{SL}_2 \rightarrow L$, with raising element in $\mathcal{O}_\nu$, and a commuting semisimple element c such that

$$t = m(r^{1/2})c, \quad m(z) = \phi(\text{diag}(z, z^{-1})).$$

Here $r^{1/2} = a^{\deg(x)}$, and the conjugacy class of c is compact at every complex embedding.

Lemma 3.2 (The open resonance orbit). *The cocharacter m is central in H , and E_t has m -weight two. Every element of E_t is nilpotent and admits a triple compatible with t . There are finitely many H -orbits in E_t . The raising element supplied by Theorem 1.1 lies in the unique open H -orbit, and its stabilizer in H is reductive. These assertions include the zero orbit, when $E_t = 0$.*

Proof. In a maximal torus containing t and m , the logarithmic absolute-value direction of t , using base r , is $m/2$. Indeed the compact factor has direction zero. If a root evaluates to one on t , it

therefore has m -weight zero; if it evaluates to r , it has m -weight two. Since H is connected, this proves the assertions about m and E_t .

The compatible-triple and finite-orbit statements are the resonance orbit lemma of [Ope26, Section 3, Lemma “Resonance orbits”]. Nilpotence can also be seen directly: a homogeneous invariant polynomial P of positive degree j satisfies $P(e) = P(re) = r^j P(e)$, and hence vanishes. For the specified raising element, decompose $\mathfrak{g}$ into irreducible $\mathfrak{sl}_2$ -modules and simultaneous c -eigenspaces. The Lie algebra of H is the part with m -weight zero and c -eigenvalue one, while E_t is the part with m -weight two and c -eigenvalue one. Raising maps weight zero onto weight two in every such irreducible module. Thus the orbit tangent map

$$\mathrm{Lie} H \longrightarrow E_t, \quad X \longmapsto [X, e]$$

is surjective. The orbit is open, and an irreducible vector space has at most one open orbit. An element of H fixing e also fixes the neutral element dm ; uniqueness of the lowering element then makes it centralize the triple. Its stabilizer is consequently the centralizer of the remaining semisimple factor c in the reductive triple centralizer, and is reductive. If the triple is zero, the same absolute-value calculation gives $E_t = 0$. $\square$

Write E_t^{op} for this open orbit and $\partial E_t = E_t \setminus E_t^{\mathrm{op}}$ for its boundary. The next argument explains why reaching E_t^{op} at a single place is enough. It uses simultaneous invariant functions, not only conjugacy classes of individual group elements.

3.2 Recovering a global commuting pair

We recall the precise invariant-theoretic fact needed to compare homomorphisms of an arbitrary abstract group.

Lemma 3.3 (Simultaneous tuple criterion). *Let B be a connected reductive group over an algebraically closed field of characteristic zero, and let Δ be an abstract group. Suppose $\rho_1, \rho_2 : \Delta \rightarrow B(k)$ have reductive Zariski closures. If every finite tuple has the same values under all simultaneous conjugation-invariant regular functions for ρ_1 and ρ_2 , then the two homomorphisms are conjugate by one element of $B(k)$.*

Proof. The closed-orbit criterion for tuples identifies complete reducibility of the generated subgroup with closedness of its simultaneous conjugation orbit; in characteristic zero, reductive subgroups are completely reducible [BMR05, Section 2.2 and Theorem 3.1]. There is a finite tuple of each image that tests the same parabolic and Levi containments as the whole image. For example, embed B in $\mathrm{GL}(V)$ and choose a finite tuple spanning the associative algebra generated by the image; this is a generic tuple in the sense of [BMRT13, Definition 5.4, Lemma 5.5, and Theorem 5.8(iii)]. Choose finitely many elements of Δ that give such tuples for both homomorphisms. Enlarging any finite test tuple by these elements makes both orbits closed. Equality of invariant values then makes the two enlarged tuples conjugate, because an affine reductive quotient separates closed orbits.

For each $\delta \in \Delta$, the equation $g\rho_1(\delta)g^{-1} = \rho_2(\delta)$ defines a closed subset of B . What we have proved gives the finite intersection property for this family. Noetherianity implies that its total intersection is nonempty, yielding one conjugator for all of Δ . $\square$

Proposition 3.4 (Open-orbit enhancement criterion). *Let x be a place selected in Lemma 3.1 and $z \in \mathrm{Spec} R(k)$. Write $b : \Gamma \rightarrow A(k)$ and $e \in \mathfrak{g}$ for the specialization of $\mathbf{b}$ and $\mathbf{e}$ at z . Conjugate so that the semisimple part of $b(\mathrm{Frob}_x)$ is $t_{x,\mathrm{ad}}$. If $e \in E_{t_x}^{\mathrm{op}}$, then there exist ϕ_ν and τ_ν with all the properties of Theorem 1.2: the prescribed orbit, reductive closure, finite field of definition, continuity*

on W_U , purity at every complex embedding, and equality with the given parameter on the entire Weil group.

Proof. We first construct a reductive commuting pair in A with the same full tuple invariants as b , then recover σ_ν and lift the pair through the finite center.

Removing the unipotent part of the scaling normalizer. The assumed orbit gives an $\mathfrak{sl}_2$ -triple with raising element e in the prescribed orbit. Let $\phi : \mathrm{SL}_2 \rightarrow A$ be its homomorphism and $h(z) = \phi(\mathrm{diag}(z, z^{-1}))$. The relation (2.3) at z says

$$\mathrm{Ad}(b(w))e = q^{\deg(w)}e \quad (w \in \Gamma).$$

The centralizer $Z_A(e)$ is contained in the Jacobson–Morozov parabolic $P(h)$, whose weights are nonnegative. Indeed, the unipotent radical of $Z_A(e)$ has positive h -weights, while a reductive factor of $Z_A(e)$ is the triple centralizer [BIY24, Proposition 3.3]. Since $h(a^{\deg(w)})$ has precisely the indicated scaling on e , every $b(w)$ lies in $P(h)$. Project to its Levi subgroup $Z_A(h)$. Removing $h(a^{\deg(w)})$ from that projection leaves an element centralizing e and h , hence the whole triple. We obtain

$$b_0(w) = c(w)h(a^{\deg(w)}), \quad c : \Gamma \rightarrow Z_A(\phi(\mathrm{SL}_2)).$$

Because the cocharacter is central in the Levi, c is a homomorphism. A Levi projection is a limit of conjugations by a cocharacter. It therefore preserves the values of all invariant functions on every simultaneous tuple.

Making the closure reductive. Consider the image of the product homomorphism $c \times \phi : \Gamma \times \mathrm{SL}_2(k) \rightarrow A(k)$. Choose a parabolic minimal among those containing this product image, allowing A itself, and project into a Levi. The projected image lies in no proper parabolic of that Levi and consequently has reductive closure in characteristic zero. This is the usual complete-reducibility construction [BMR20, Corollary 3.5 and Example 4.8]. Denote the projected maps by c_1, ϕ_1 and the diagonal of ϕ_1 by h_1 . The image of ϕ was already reductive, so its Levi projection is conjugate to it. In particular $d\phi_1$ has the same raising orbit as $d\phi$.

The closure of $c_1(\Gamma)$ is a normal subgroup of the reductive closure of the product image, and is reductive. Moreover

$$b_1(w) = c_1(w)h_1(a^{\deg(w)})$$

has reductive closure. To verify the last assertion, take the closure of $w \mapsto (c_1(w), a^{\deg(w)})$ in $\overline{c_1(\Gamma)} \times \mathbb{G}_m$. Both projections have reductive closure, so its unipotent radical projects trivially to both factors and is trivial. The multiplication map $(c, z) \mapsto ch_1(z)$ is a homomorphism because the factors commute, and its image is reductive. Both projections made in the proof have preserved all invariant tuple evaluations.

Recovering the given parameter. Proposition 2.6 identifies those tuple evaluations with the evaluations of $\sigma_{\nu, \mathrm{ad}}|_\Gamma$. The latter has reductive closure: W_U is dense in $\pi_1(U)$, and the original parameter is semisimple. Lemma 3.3 therefore gives one conjugation making

$$\sigma_{\nu, \mathrm{ad}}(w) = c_1(w)h_1(a^{\deg(w)}) \quad (w \in W_U).$$

Lift that conjugation to L . The homomorphism ϕ_1 lifts through $L \rightarrow A$ because SL_2 is simply connected. Call its lift ϕ_ν and its diagonal cocharacter $\tilde{h}_1$. Define

$$\tau_\nu(w) = \sigma_\nu(w)\tilde{h}_1(a^{\deg(w)})^{-1}. \quad (3.1)$$

Its adjoint image is $c_1(w)$ and centralizes the adjoint triple. Commutation modulo a finite center implies commutation with the lifted triple: the corresponding commutator map from connected SL_2

to the finite center is constant. Thus $\tau_\nu(w)$ lies in the full group $Z_L(\phi_\nu(\mathrm{SL}_2))$. It commutes with the diagonal cocharacter, and (3.1) is a homomorphism. Its closure is reductive, since its adjoint image has reductive closure and the kernel of $L \rightarrow A$ is finite.

All algebraic maps, conjugators, and a involved are defined over some finite extension of $\mathbb{Q}_\ell$; enlarging the field of definition of σ_ν accommodates them all. Equation (3.1), together with continuity of the degree map in the Weil topology, proves continuity of τ_ν . It also proves the required equality on all of W_U , including geometric monodromy and the inertia remaining in $\pi_1(U)$. No connectedness assumption on the triple centralizer has been made.

Purity at all embeddings. Fix an arbitrary closed point $y \in |U|$ and an arbitrary complex embedding of $\overline{\mathbb{Q}}$. Eigenvalues of every rational representation of L on $\sigma_\nu(\mathrm{Frob}_y)$ are algebraic, by Theorem 1.1. The diagonal factor has eigenvalues powers of $a^{\deg(y)}$ and commutes with $\sigma_\nu(\mathrm{Frob}_y)$. Simultaneous triangularization therefore proves algebraicity of all eigenvalues of $\tau_\nu(\mathrm{Frob}_y)$ as well.

Take a complex realization of k extending the chosen embedding on $j(\overline{\mathbb{Q}})$, and use a maximal torus containing the semisimple part of $\tau_\nu(\mathrm{Frob}_y)$ and $\tilde{h}_1(\mathbb{G}_m)$. Let λ_τ be the real cocharacter direction obtained by taking logarithms of absolute values, with base q_y . The direction for $\sigma_\nu(\mathrm{Frob}_y)$ is

$$\frac{1}{2}\tilde{h}_1 + \lambda_\tau.$$

These summands are orthogonal for a positive Weyl-invariant form on the real cocharacter space, for example the form induced by the adjoint representation. Indeed, the semisimple part of $\tau_\nu(\mathrm{Frob}_y)$ centralizes the triple. Every root occurring in its raising or lowering element evaluates to one on this semisimple part, hence vanishes on λ_τ . Consequently λ_τ centralizes the triple as a Lie element. Invariance of the form and the expression of the neutral element as a bracket give $\langle \lambda_\tau, \tilde{h}_1 \rangle = 0$. This argument also applies when the centralizer is disconnected.

By Theorem 1.1, the norm of the logarithmic direction of $\sigma_\nu(\mathrm{Frob}_y)$ is the norm of one half of a cocharacter for $\mathcal{O}_\nu$. The cocharacter $\tilde{h}_1$ belongs to that same orbit, so

$$\left\| \frac{1}{2}\tilde{h}_1 + \lambda_\tau \right\|^2 = \left\| \frac{1}{2}\tilde{h}_1 \right\|^2.$$

Orthogonality and positive definiteness force $\lambda_\tau = 0$. Every weight in every rational representation therefore has absolute value one on $\tau_\nu(\mathrm{Frob}_y)$. Since the place and the embedding were arbitrary, this is the required purity. $\square$

Corollary 3.5 (Boundary under nonexistence). *If the enhancement in Theorem 1.2 does not exist for ν , then at every selected place x_i and every geometric point of $\mathrm{Spec} R$, the coordinate $\mathbf{e}$ belongs to $\partial E_{t_{x_i}}$ after alignment of the semisimple holonomy with $t_{x_i, \mathrm{ad}}$.*

Proof. Invariant evaluations in Proposition 2.6 place the semisimple holonomy in the specified conjugacy class. Write the aligned holonomy as $t_{x_i, \mathrm{ad}}u$, with u unipotent in its centralizer. Relation (2.3) then implies $\mathbf{e} \in E_{t_{x_i}}$ and $\mathrm{Ad}(u)\mathbf{e} = \mathbf{e}$. To see this, take the commuting semisimple and unipotent parts of its adjoint action on the eigenvector $\mathbf{e}$.

For a k -point the result is the contrapositive of Proposition 3.4. The condition of reaching the open orbit is constructible: it is the image of the incidence condition that an aligning conjugator sends the holonomy into $t_{x_i, \mathrm{ad}}$ times the centralizer's unipotent variety, and sends $\mathbf{e}$ into $E_{t_{x_i}}^{\mathrm{op}}$. Since R is of finite type over the algebraically closed field k , a nonempty constructible subset contains a closed k -point. Thus no geometric point can reach the open orbit either. $\square$

The global assertion is now reduced to a geometric statement about the nonzero finite-type algebra R : its specializations cannot lie in these boundaries at all the selected places simultaneously. The following sections develop the local tests and the cohomological bound used to prove this statement.

4 A Frobenius-compatible realization of the Iwahori category

The local tests used below require an equivalence of enhanced monoidal categories: a triangulated equivalence alone does not identify their derived traces. This section constructs the required enhancement from Bezrukavnikov’s equivalence. We then identify the central Satake objects, their Weil structures, and the tensor interchanges that will enter the comparison with hyperspecial level.

Fix a closed point of degree d among those of Lemma 3.1, and put $r = q^d$. Let I be a split Iwahori subgroup and let $\mathcal{H} = \mathcal{H}_{I,I}$ be the category of I -equivariant constructible complexes on the affine flag variety, with finite Schubert support and coefficients in $k = \overline{\mathbb{Q}}_\ell$. It is an idempotent-complete stable category under convolution, denoted by $*$, and carries the monoidal automorphism Fr^* . All constructible categories in this section are over geometric constants. Weil structures will be specified separately. Write

$$\widetilde{\mathcal{N}} \longrightarrow \mathfrak{g}, \quad \mathcal{Z} = \widetilde{\mathcal{N}} \times_{\mathfrak{g}}^{\mathbf{R}} \widetilde{\mathcal{N}}, \quad \mathcal{C} = \mathrm{Coh}^L(\mathcal{Z}).$$

The first map is the Springer resolution followed by the inclusion of the nilpotent cone. The convolution unit of $\mathcal{C}$ is $\Delta_* \mathcal{O}_{\widetilde{\mathcal{N}}}$. For $V \in \mathrm{Rep}(L)$, set

$$D(V) = \Delta_*(\mathcal{O}_{\widetilde{\mathcal{N}}} \otimes V).$$

The diagonal representation factor gives $D(V)$ its ordinary interchange with any coherent convolution kernel. We call this the *standard interchange*. Dilation of the nilpotent coordinate by r induces the monoidal automorphism r_* of $\mathcal{C}$; equivalently, this is pullback under dilation by r^{-1} . The constant representation factor gives a preferred isomorphism $r_* D(V) \simeq D(V)$.

Let $Z(V)$ denote the central Satake kernel with its split Weil structure. We use the central construction with the scope stated in [Ope26, Section 2, “Central and Wakimoto kernels”]: its tensor and central identities hold in the homotopy category, and, for a fixed label, interchange in the other kernel is a natural equivalence of enhanced exact functors. Projection to a hyperspecial parahoric gives the Satake kernel. Write Ω for the finite group of length-zero elements of the extended affine Weyl group. It indexes the connected components of the affine flag variety; semisimplicity of G makes it finite. The choice of the test places ensures $|\Omega| \mid d$.

Theorem 4.1. *There is an enhanced monoidal equivalence $\Theta : \mathcal{H} \simeq \mathcal{C}$, together with a monoidal intertwining of Fr^{d*} and r_* , having the following properties. There are identifications*

$$\Theta Z(V) \simeq D(V) \quad (V \in \mathrm{Rep}(L)),$$

natural for ordinary representation morphisms, and an element $z \in Z(L)(k)$ such that the transported Weil isomorphism on $D(V)$ is its preferred isomorphism multiplied by $V(z)$. Under these identifications, the tensor maps between the $Z(V)$ become constant L -equivariant representation maps. On $\mathrm{Rep}(A) \subset \mathrm{Rep}(L)$ they can be chosen to be the ordinary tensor maps. Finally, the central interchange of $Z(V)$ with $Z(W)$ becomes the standard interchange whenever at least one of V, W is neutral.

Here and throughout the paper, “enhanced monoidal” means associative monoidal; no symmetry of the Iwahori convolution category is intended. The last assertion of the theorem concerns the indicated interchanges as morphisms. It does not assert that the entire central functor has been identified in an enhanced Drinfeld center.

After checking the Frobenius normalization through the monodromic construction, we make its intertwiner monoidal. Equivariant localization reduces its tensor defect to scalars on standard objects; these scalars depend only on the connected components, and taking the d th power removes

the resulting finite-group cocycle. Purity then lifts the full convolution diagram to the enhanced categories. A separate weight-filtration argument recovers the mixed central objects themselves. Polynomial degree and symmetry at hyperspecial level then determine the required Weil and tensor data.

4.1 The triangulated input

Bezrukavnikov's theorem gives a monoidal equivalence

$$\Phi : \mathrm{Ho}(\mathcal{H}) \xrightarrow{\sim} \mathrm{Ho}(\mathcal{C}). \quad (4.1)$$

We use [Bez16, Theorem 1 and Section 10], in the formulation of [BZCHN24, Theorems 1.13 and 2.17]. The latter's published Remark 2.18 explicitly states the equivalence only at the level of homotopy categories. The passage to an enhancement in Theorem 4.1 is therefore part of our argument. There is also a scalar-action convention to distinguish: [BZCHN24, Section 1.6.3, note 8, and Section 2.4.1] lets the scalar q act on points by $N \mapsto q^{-1}N$. Its notation q_* consequently denotes pullback by $N \mapsto qN$, whereas our q_* denotes pushforward by that map. For the Frobenius comparison we therefore give the generator calculation underlying [Bez16, Proposition 53], with the geometric Frobenius and scalar maps specified explicitly.

These results apply to the split root datum of G , in arbitrary residue characteristic different from ℓ . On finite Schubert supports the equivariant categories are computed with finite-type jet quotients of I . The orbit stabilizers are connected, so this description gives precisely the constructible category used above. We choose the naming of the convolution factors and the identification of the dual root datum compatibly in (4.1).

Lemma 4.2 (Frobenius normalization). *The equivalence (4.1) admits a natural isomorphism of exact functors*

$$\Phi \mathrm{Fr}^* \simeq (\mu_q)_* \Phi, \quad \mu_q(N) = qN.$$

Here Fr^* has the split Weil normalization in which geometric Frobenius acts on $k(-1)$ by q .

Proof. We track the construction on the monodromic categories before passing to Iwahori equivariance. The coherent functors in that construction are specified by the following data: representation twists, Wakimoto line bundles, the lowest-weight arrows, the tautological nilpotent endomorphism, and the two Cartan-coordinate actions. These are the data of [Bez16, Sections 4.1.1–4.1.5 and 4.4.1]. Corollary 18 of that paper is a uniqueness statement for the *additive category of free equivariant coherent sheaves* on its homogeneous-coordinate scheme: the indicated tensor functor, endomorphisms, and lowest-weight arrows determine the functor on all its morphisms. We first compare these data, and then explain the extension to the full derived categories.

Give the Wakimoto and central objects their split Weil structures. The representation maps and lowest-weight arrows are Weil maps. Here we use the lowest-weight terminology of [Bez16, Section 4.1.3]; these are the extremal arrows in the earlier construction. Their Weil structures use the top-cell normalization, and its compatibility with tensor products is the calculation in [AB09, Section 3.5, equations (17)–(18)]. If M is either a nearby-cycle logarithm or a torus-monodromy logarithm, its Tate-valued form is a Weil morphism $M : \mathcal{F} \rightarrow \mathcal{F}(-1)$. After forgetting the Tate twist, the chosen Weil map $w : \mathrm{Fr}^* \mathcal{F} \rightarrow \mathcal{F}$ therefore gives

$$w \mathrm{Fr}^*(M) w^{-1} = q^{-1}M. \quad (4.2)$$

For nearby cycles this is precisely the second identity in [AB09, Section 3.5, equation (17)]; for torus monodromy it follows from the geometric Frobenius action q^{-1} on the tame cocharacter space

$X_*(T) \otimes k(1)$. The same calculation applies to both the left and right torus actions. It also applies to the pro-unipotent objects, by passage to their finite monodromy quotients.

Under the preferred identification of a pushed-forward free bundle with the original free bundle, $(\mu_q)_*$ sends a matrix of polynomial functions $a(N)$ to $a(q^{-1}N)$. Thus it fixes the representation and lowest-weight maps and multiplies both the tautological endomorphism and each linear Cartan-coordinate map by q^{-1} . This is exactly (4.2). Corollary 18 consequently gives a natural comparison on the entire free coherent category. Its application in [Bez16, Section 4.4.1] uses two Wakimoto actions and two Cartan actions; the comparison respects the diagonal representation identification and equality of the two tautological endomorphisms. It therefore descends to the same Steinberg homogeneous-coordinate scheme used there. This is a comparison of actions, natural also in the object acted upon. Indeed, the representation and Wakimoto actions are convolution functors, and their Weil comparisons come from natural pullback and nearby-cycle comparisons. Corollary 18 is applied in Section 4.4.1 with its target the category of endofunctors of the monodromic category. Thus, writing $\mathcal{A}_P$ for the action of a free coherent label P , the comparison has the form

$$\mathrm{Fr}^* \mathcal{A}_P (\mathrm{Fr}^*)^{-1} \simeq \mathcal{A}_{(\mu_q)_* P},$$

natural in all morphisms of P and of the acted-on object.

This comparison extends to perfect complexes by the construction in [Bez16, Section 4.4.2]. More explicitly, that construction twists a finite free complex by sufficiently anti-dominant Wakimoto objects on the left and sufficiently dominant ones on the right, and evaluates it on a finite complex of free-monodromic tilting objects. Its value is the total complex of the resulting bicomplex. Frobenius preserves the tilting subcategory, and the comparison just constructed commutes with every differential and with both Wakimoto twists. It hence gives an isomorphism of these total complexes. The canonical homotopy comparisons for different choices of these twists commute with this isomorphism. The acyclic subcategory and the idempotents used in the localization of that section are preserved by dilation, so the comparison descends to the perfect-category action. Evaluating on the anti-spherical tilting object $\widehat{\Xi}$ gives the comparison for the functor denoted Φ_{perf} in [Bez16, Sections 5–6]. To choose its Weil isomorphism, recall that $\widehat{\Xi} = \widehat{T}_{w_0}$ is the unique indecomposable free-monodromic tilting object with the specified Schubert support and free-standard normalization [Bez16, Proposition 11(b) and Section 5]. Frobenius preserves the split stratification and those filtrations, so $\mathrm{Fr}^* \widehat{\Xi} \simeq \widehat{\Xi}$. Choose such an isomorphism, and obtain the one-sided version by projection. Since the action comparison is natural in the object acted upon, this choice gives a comparison natural in every perfect label.

The extension from perfect to bounded coherent objects is determined by representability, rather than by choices of cones. Write $\Upsilon(M)$ for the coherent object representing $P \mapsto \mathrm{Hom}(\Phi_{\mathrm{perf}} P, M)$, as in [Bez16, Proposition 32(a) and Sections 8.2–9.1]. Here P is a coherent perfect object, M is constructible and monodromic, and Φ_{perf} runs from coherent to constructible objects, in the direction opposite to (4.1). The comparison on perfect objects gives, naturally in P and M ,

$$\begin{aligned} \mathrm{Hom}(P, \Upsilon(\mathrm{Fr}^* M)) &\simeq \mathrm{Hom}((\mathrm{Fr}^*)^{-1} \Phi_{\mathrm{perf}} P, M) \\ &\simeq \mathrm{Hom}(\Phi_{\mathrm{perf}} (\mu_q)_*^{-1} P, M) \\ &\simeq \mathrm{Hom}((\mu_q)_*^{-1} P, \Upsilon M) \\ &\simeq \mathrm{Hom}(P, (\mu_q)_* \Upsilon M). \end{aligned}$$

The uniqueness assertion in Proposition 32(a) gives the required natural isomorphism on all objects. Its construction is compatible with shifts and exact triangles. The same construction works on the formal completion: dilation preserves the monodromy-adic filtration, and the comparison commutes

with all its quotient maps. It is therefore compatible with the passage between completed and finite-monodromy categories in Section 9.2.

It remains to pass from the completed monodromic category to I -equivariance. We make this passage in a derived category of modules. In particular, boundedness of an object will not mean boundedness of its equivariant mapping complex.

Put

$$R = \text{Sym}(\mathfrak{t}_L \oplus \mathfrak{t}_R), \quad K = R \otimes \Lambda((\mathfrak{t}_L \oplus \mathfrak{t}_R)[1]), \quad d\epsilon_x = x.$$

The two actions of R are the left and right logarithms of monodromy; on the coherent side they are the two Cartan-coordinate actions. On completed categories we may instead use the completed ring and the base change of K ; the zero-monodromy argument below will justify agreement on the bounded locus in question. All module objects below are internal to the stated monodromic or coherent category, with these actions of R .

First, the completed monodromic comparison has a concrete R -linear enhancement on which to form such modules. Write $\widehat{\mathcal{T}}$ for the additive category of free-monodromic tiltings. Proposition 7(a) and Remark 8 of [Bez16] give generation by these objects and

$$\text{Hom}^j(T, T') = 0 \quad (j \neq 0).$$

Negative vanishing also follows directly from the perverse heart. Their images under the completed coherent equivalence are coherent sheaves, by Corollary 25 as used in Section 9.3 of that paper, and have the same vanishing. For either full dg subcategory on these objects, truncation of its mapping complexes in degrees at most zero, followed by passage to degree-zero cohomology, is a zigzag of dg equivalences. Truncation respects composition and the two degree-zero R -actions. The additive comparison thus identifies these dg subcategories and their stable idempotent-complete envelopes. Those envelopes are the completed monodromic and coherent categories. This construction also carries the natural Frobenius comparison already obtained above: on tiltings its components and naturality identities are actual degree-zero maps. No uniqueness assertion for arbitrary enhancements is being used.

We record explicitly the derived module convention. For a monodromic perverse heart $\mathcal{P}$, take complexes with compatible strict K -action and invert maps which are quasi-isomorphisms on the underlying complexes. Denote the resulting stable category by $\mathbf{E}_K(\mathcal{P})$, restricting to objects with bounded cohomology in $\mathcal{P}$. Its mapping complexes are derived module mapping complexes. Equivalently, it is the category of homotopy-coherent internal K -modules in the derived category of $\text{Ind}(\mathcal{P})$ with that boundedness condition. Here is a description of this equivalence which also fixes the mapping complexes. The free-module monad is

$$T_K(M) = K \otimes_R M.$$

It preserves quasi-isomorphisms because K is finite free as a graded R -module. Work first in the Grothendieck category $\text{Ind}(\mathcal{P})$, allowing unbounded resolutions. After localization the forgetful functor is conservative, since weak equivalences were defined on underlying complexes. It preserves geometric realizations: underlying mapping cones compute cofibers, and sums and filtered colimits are exact in this Grothendieck category. The free-module adjunction consequently identifies its monad with the derived functor T_K above. Its augmented simplicial free-module resolution has the extra-degeneracy contraction after forgetting the action. The resulting bar comparison identifies the localization with coherent modules for this monad. Thus strictification is performed after resolving in the ind-category; it does not require derived maps to be strict chain maps on a previously chosen

bounded representative. Applying the free-module adjunction to the bar resolution gives, for every pair of modules, the mapping complex

$$\mathrm{RHom}_K(M, N) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{RHom}(T_K^n UM, UN),$$

where U forgets the action, and the faces and degeneracies use its unit, multiplication, and the two module structures. Composition is the composition of these derived module maps.

This description does not impose finite length on a resolution. For clarity, it nevertheless gives the same bounded derived quotient as the one with complexes in $\mathcal{P}$. On a fixed finite Schubert support the ordinary heart is noetherian. Since K is nonpositive and finite over R , a strict module with bounded cohomology in $\mathcal{P}$ admits a bounded-above representative whose terms lie in $\mathcal{P}$: first truncate above the cohomological range, and choose finite subobjects representing its highest nonzero cohomology, close them under the finitely many exterior K -operations, and proceed downwards, adjoining finite preimages of the kernels to be killed. Noetherianity keeps each such kernel finite. The required finite preimages exist because $\mathrm{Ind}(\mathcal{P})$ is locally noetherian: a noetherian subobject of a quotient is covered by the images of noetherian subobjects of the source, and finitely many of those images already cover it. Every chosen subobject is already R -stable because the monodromy action of R is central and natural on $\mathcal{P}$. In each fixed degree only finitely many preceding steps contribute, since K has finite amplitude. If N' is the resulting bounded-above strict module and a is below its cohomological range, replace it by $N'/\tau_{\leq a-1}N'$. The good upper truncation is a dg K -submodule because K is nonpositive; this quotient is bounded and quasi-isomorphic to N' , with at most one extra lower term. For roofs and homotopies choose a also below the term bounds of the bounded endpoints. The maps and homotopies then factor through the quotient. This proves that their localization is computed in the bounded subcategory. Alternatively, the normalized bar terms use $\overline{K} = K/R$ and $(\overline{K}[1])^{\otimes_{R^n}}$; their strictly decreasing degrees explain why the bar construction is degreewise finite on a bounded representative. Its infinitely many lengths remain essential in the derived mapping complex.

Now form these internal modules in the completed monodromic category. For such a module, $x = d\epsilon_x$ acts nullhomotopically on the underlying object, and hence acts zero on every perverse cohomology object, for both Cartan actions. A completed perverse object with zero monodromy equals its monodromy coinvariants. Those coinvariants are finite objects by the definition of the completed category in [Bez16, Section 3.1]. It therefore belongs to the ordinary monodromic heart. Boundedness and the fully faithful inclusion of the ordinary monodromic category into its completion put the underlying complex in the ordinary derived category. Conversely such complexes embed in the completion. The bar terms in the bar formula are obtained from the underlying objects by finitely many sums, shifts, and cones, using the finite filtration of each tensor power of K . Thus this inclusion identifies the full mapping complexes of their K -modules. Thus the completed model has reduced to the ordinary derived module quotient.

At the level of homotopy categories this quotient is exactly $D(\mathcal{P}_{\mathrm{eq}})$ in [Bez16, Section 9.3.1, Lemma 44(a)]. That lemma identifies it with the torus-equivariant derived category. Apply it to the two torus actions on the double I^0 -monodromic category, using finite-type jet quotients on every finite Schubert support. We obtain the Iwahori category $\mathrm{Ho}(\mathcal{H})$. This is an exact realization, so it identifies every H^j of the derived mapping complex with the corresponding $\mathrm{Hom}(-, -[j])$. We have used the derived quotient in Lemma 44(a), not strict chain Hom between finite tilting complexes.

On the coherent side, the same construction gives bounded coherent modules on the derived zero fiber

$$(\widetilde{\mathfrak{g}} \times_{\mathfrak{g}} \widetilde{\mathfrak{g}}) \times_{(t^*)^2}^{\mathbf{R}} \{0\} \simeq \widetilde{\mathcal{N}} \times_{\mathfrak{g}}^{\mathbf{R}} \widetilde{\mathcal{N}} = \mathcal{Z}.$$

Indeed, on an affine coherent chart with coordinate algebra A , its algebra is $A \otimes_R K$, and derived modules over this algebra are precisely internal K -modules. Its underlying A -complex has bounded

coherent cohomology exactly on the required coherent locus. The comparison of mappings is the full bar totalization, and these descriptions glue equivariantly. Formal completion has no effect on this locus because both Cartan coordinates act zero on cohomology. The enhanced comparison of the completed ambient categories therefore gives the required equivalence of these derived module models.

The ordinary monoidal comparison is compatible with this localization. Before imposing equivariance, convolution preserves $\widehat{\mathcal{T}}$, and its comparison is the one in [Bez16, Proposition 7(b) and Section 10.2]. After imposing the outer K -actions, convolution has a further closed degree-minus-one operator: it is the difference of the right nullhomotopy on the first factor and the left nullhomotopy on the second. Their differentials agree. These operators give the middle algebra $\Lambda(\mathfrak{t}[1])$, and convolution is computed by

$$(M, N) \longmapsto (M \star N) \otimes_{\Lambda(\mathfrak{t}[1])}^{\mathbf{L}} k.$$

Here $\star$ is the derived completed monodromic convolution with its inherited coherent K -actions, computed after replacing the inputs by compatible resolutions; the outer left and right actions are retained. In particular, one must not replace its middle operator by the difference of unresolved zero homotopies. Compute the displayed tensor product with its full bar resolution. For three factors the two orders of contraction are the two iterated realizations of one bisimplicial bar object; their Fubini identification gives the associativity comparison, and the same construction handles any number of factors. On the constructible side this is the torus pushforward calculation of [Bez16, Lemma 51], whose proof uses Lemma 44(a),(b). On the coherent side it is the derived fiber-product calculation for convolution. To pass from free modules to bounded modules, let $M_s \rightarrow M$ and $N_s \rightarrow N$ be finite skeleta of their normalized free-module bars, and let C_s be their contracted completed monodromic convolution. For a pair of induced free K -modules, the difference of the two middle exterior generators is a closed member of a free exterior basis. Their completed convolution is therefore free over the middle algebra $\Lambda(\mathfrak{t}[1])$, and its contraction has a finite representative. Finite cones give the same conclusion for C_s . Lemma 51 applied to these finite complexes gives natural identifications

$$\mathrm{real}_{\mathrm{eq}}(C_s) \simeq \mathrm{real}_{\mathrm{eq}}(M_s) * \mathrm{real}_{\mathrm{eq}}(N_s) \longrightarrow G := \mathrm{real}_{\mathrm{eq}}(M) * \mathrm{real}_{\mathrm{eq}}(N),$$

where the last map is the geometric convolution of the augmentations.

Form $C = \mathrm{colim}_s C_s$ in the derived category of K -modules over $\mathrm{Ind}(\mathcal{P})$ on the constructible side. This is a derived module colimit, with its full mapping complexes. Normalized bar length shifts the upper perverse bound towards minus infinity: its terms contain $\overline{K}[1]$, and the middle contraction is a derived tensor product over a nonpositive algebra. The latter preserves upper cohomological bounds. Together with the uniform finite amplitude of completed monodromic convolution on the fixed supports, this gives numbers $b_s \rightarrow -\infty$ such that

$$\mathrm{Cone}(C_s \longrightarrow C_t) \in D^{\leq b_s}(\mathrm{Ind}(\mathcal{P})) \quad (t \geq s).$$

The same bound holds with C in place of C_t , since filtered colimits are exact. All these complexes have a common upper bound. Consequently, for every fixed a , the module truncation $\tau_{\geq a}C$ is represented by $\tau_{\geq a}C_s$ for sufficiently large s . It has bounded cohomology in the finite heart $\mathcal{P}$, so that the ordinary equivariant realization of Lemma 44(a) applies to it.

Use the fixed smooth shift in Lemma 44(b) to match the perverse indices of the module and equivariant categories. The canonical maps from C_s to $\tau_{\geq a}C$, together with the geometric augmentation above, give, for sufficiently large s , the natural comparison

$$\mathrm{real}_{\mathrm{eq}}(\tau_{\geq a}C) \simeq \tau_{\geq a}G.$$

Indeed, the first map becomes an isomorphism after $\tau_{\geq a}$ by the stabilization just proved. The geometric augmentation has a cone with upper bound tending to minus infinity by Lemma 44(b) and the finite amplitude of geometric convolution. Applying the fixed lower perverse truncation therefore identifies both sides with the truncation of $\text{real}_{\text{eq}}(C_s)$. Geometric convolution on finite Schubert supports is bounded. Varying a below its lower bound now proves that C itself has bounded cohomology in $\mathcal{P}$ and that $\text{real}_{\text{eq}}(C) \simeq G$. This proves boundedness in the constructible model before using the coherent comparison. Naturality and associativity follow by applying these augmentation spans to each finite diagram in the common multibar construction.

We compare with the coherent calculation separately. The completed ambient tilting comparison respects convolution at the enhanced level: convolution preserves tiltings, and the same degree-zero truncation of their multimorphism complexes transports the monoidal identities of Section 10.2. It therefore carries each finite C_s , its outer actions, and its augmentation to the corresponding finite coherent bar calculation. Write F for this completed comparison. The cones of $C_s \rightarrow C$ are now bounded modules. Their perverse cohomology has zero Cartan monodromy and is supported on a fixed finite Schubert union, so its simple constituents belong to a fixed finite set. The images under F of these simples have a common upper coherent bound B . Finite extensions and bounded Postnikov towers imply

$$F(\text{Cone}(C_s \rightarrow C)) \in D_{\text{coh}}^{\leq b_s + B}.$$

The same finite-simple argument applies to the bounded input cones of $M_s \rightarrow M$ and $N_s \rightarrow N$, so their images under F have coherent upper bounds tending to minus infinity. The finite free-module calculation therefore gives bounded coherent augmentation spans

$$F(C) \longleftarrow F(C_s) \simeq F(M_s) *_{\text{coh}} F(N_s) \longrightarrow F(M) *_{\text{coh}} F(N).$$

The left cones tend to minus infinity by the displayed estimate. The right cones do so as well: coherent convolution has a uniform upper cohomological amplitude on the fixed supports, by finite Tor amplitude over the smooth middle Springer resolution and the final proper projection. Applying any fixed lower coherent truncation to a sufficiently long finite span identifies its two ends. Both ends are bounded coherent objects, giving the required natural identification $F(C) \simeq F(M) *_{\text{coh}} F(N)$. The same finite-diagram argument preserves associativity, the outer actions, and the structural maps. No unbounded coherent bar or compactness assertion in an ind-coherent category is used. The geometric truncations were perverse truncations, whereas these last spans use the ordinary coherent t -structure; the ambient equivalence was not assumed to identify the two. Throughout, mapping complexes remain the full derived bar totalizations. Consequently this computation restricts to the stated categories. It is the derived-localization construction of the ordinary equivalence and monoidal structure in Sections 9.3 and 10.3, including their maps, rather than an identification of generator objects alone. We use this realization for Φ in (4.1).

Finally define σ_q on R and K by

$$x \longmapsto qx, \quad \epsilon_x \longmapsto q\epsilon_x$$

in both Cartan directions. Restriction along this map changes the monodromy and its nullhomotopy by the same scalar. It acts on every term, face, and degeneracy of the module bar construction, so the natural comparison on completed tiltings induces a natural comparison on the entire derived module category and all its mapping complexes. The realization in Lemma 44(a) respects this operation: on a trivial torus torsor, pullback by the q -power map replaces tame monodromy T by T^q , and hence replaces its logarithm by q times that logarithm. The equivariant smooth descent in that proof uses the same isogeny on the acting torus and therefore glues this comparison. On the coherent

derived zero fiber, restriction along σ_q is $(\mu_q)_*$, including its action on the derived coordinates. We have proved the claimed natural exact comparison, with its action on every shifted Hom group. This agrees with (4.2): that formula conjugates a pulled-back morphism by a chosen Weil identification, whereas restriction along σ_q describes the canonical monodromy of the pulled-back object. These are the respective q^{-1} and q descriptions of the same comparison.

There is a useful full-mapping check. In rank one for the torus group, the completed double-monodromic category is modeled by $A = k[[x]]$, with $R = k[x_L, x_R]$ acting diagonally. Thus $A \otimes_R K$ is quasi-isomorphic to $\Lambda(\eta)$, where $|\eta| = -1$ and $\eta = \epsilon_R - \epsilon_L$. The unit is the augmentation module. Resolve its underlying A -module by $P = A \otimes \Lambda(e)$, $de = x$, with both ϵ_L and ϵ_R acting by e . Then $P \otimes_A P = A \otimes \Lambda(e_1, e_2)$, and the middle operator is $e_1 - e_2$. It acts freely on the exterior factor generated by $e_2 - e_1$; the derived middle contraction removes this factor and returns P . This verifies the convolution unit as well as the need to retain the actions on the derived ambient product. The semifree resolution of its augmentation module has generators p_n of degree $-2n$ and differential $dp_n = \eta p_{n-1}$. It gives

$$\mathrm{Ext}_{\Lambda(\eta)}^\bullet(k, k) = k[u], \quad |u| = 2.$$

For $\eta \mapsto q\eta$, the comparison on this resolution sends p_n to $q^n p_n$ in the restricted module. Pushforward therefore sends u to qu , agreeing with geometric Frobenius on $H^2(BT, k)$. The infinite polynomial tail is retained; no bound on the chain Hom of a finite representative has been imposed. No monoidality of the Frobenius comparison is asserted at this stage. $\square$

Lemma 4.3. *At the level of objects, $\Phi Z(V) \simeq D(V)$ for every $V \in \mathrm{Rep}(L)$.*

Proof. Let I^0 be the pro-unipotent radical of I . Forgetting from I -equivariance to I^0 -equivariance fits into the diagram of [BZCHN24, Theorem 2.17]. On the coherent side its counterpart is direct image under the closed affine inclusion

$$i : \widetilde{\mathcal{N}} \times_{\mathfrak{g}}^{\mathbf{R}} \widetilde{\mathcal{N}} \longrightarrow \widetilde{\mathfrak{g}} \times_{\mathfrak{g}} \widetilde{\mathcal{N}},$$

where $\widetilde{\mathfrak{g}}$ is the Grothendieck–Springer resolution. The representation action in Bezrukavnikov’s construction identifies convolution by the central sheaf with tensoring by V on this one-sided-monodromic category; see [Bez16, Sections 2.2.2, 3.5, 4, and 9–10]. In particular, applying that action to the forgotten unit identifies the direct image of $\Phi Z(V)$ with $i_* D(V)$. The compatibility of the central action with forgetting equivariance is also expressed in the pro-monodromic formulation of [Bez16, Proposition 13].

Direct image under i is exact for the ordinary coherent t -structures and detects their cohomology sheaves. It follows that $\Phi Z(V)$ is a sheaf, since $i_* D(V)$ is a sheaf. On these hearts i_* is fully faithful: it is restriction of scalars along the quotient defining the closed immersion. The displayed identification after i_* therefore lifts to an isomorphism $\Phi Z(V) \simeq D(V)$. If the forgetful diagram is written with a common shift, the shift cancels by applying the same diagram first to the unit. This argument identifies objects and does not require an enhanced assertion about their central structures. $\square$

4.2 Pure generators and equivariant localization

Normalize every Schubert intersection complex to weight zero, using the fixed square root of q . A *word* will mean a convolution of finitely many such normalized IC complexes, including the empty word 1_I . We regard words as formal labels together with their evaluation in $\mathcal{H}$; thus a concatenation has a specified convolution evaluation. The IC complexes generate $\mathcal{H}$ under finite

cones and retracts, by the orbit filtration and recollement. Consequently the words form a set of monoidal generators.

Let $T = I/I^0$ and put $S_T = H^\bullet(BT, k)$. This graded polynomial ring acts on morphisms by left equivariance. The following freeness property will let us verify identities after inverting its nonzero homogeneous elements.

Lemma 4.4. *For two IC words P, Q , the group $\mathrm{Hom}_{\mathcal{H}}^j(P, Q)$ is pure of weight j . Their total graded Hom is a finitely generated, torsion-free S_T -module. The same assertions hold when either word is a concatenation of several prescribed words.*

Proof. The goodness theorem for split Schubert varieties and their convolutions gives very pure stalks, with the normalization just specified; see [dCHL18, Theorem 2.2.1 and Corollary 2.2.3]. Verdier duality gives the same degree–weight assertion for costalks. For the inclusion j_O of a smooth orbit, passing from point costalks to the ordinary fibers of $j_O^!Q$ removes the common shift and twist contributed by the orbit dimension. Thus both $H^a((j_O^*P)_x)$ and $H^b((j_O^!Q)_x)$ have weights a and b , respectively.

The stabilizer of $x \in O$ has torus quotient T and a cohomologically trivial unipotent radical. Its equivariant cohomology is therefore S_T , compatibly with the left action. The local terms of the orbit filtration of $\mathrm{RHom}(P, Q)$ have a coefficient-cohomology spectral sequence with terms

$$\mathrm{Hom}(H^a((j_O^*P)_x), H^b((j_O^!Q)_x)) \otimes H^c(BT, k). \quad (4.3)$$

Their total degree and weight are both $b - a + c$. Every differential changes total degree by one and commutes with Frobenius, so it vanishes. The local graded Hom consequently has a finite filtration whose quotients are finite free graded S_T -modules. The orbit spectral sequence has the same weight property and also degenerates. There are finitely many orbits on the supports in question, so its resulting module filtration is finite. Extensions of free S_T -modules are torsion free (in fact they split as S_T -modules). This proves both conclusions. Concatenating words merely gives another convolution of IC complexes, so the final assertion is included in the same argument. $\square$

Let Δ_w and ∇_w be the standard and costandard complexes for the orbit indexed by an extended affine Weyl element w . Their normalizations can be chosen consistently with convolution; they are invertible objects, and length-additive convolution gives the corresponding standard or costandard. The standard objects also generate $\mathcal{H}$. Write K_T for the graded localization of S_T at all its nonzero homogeneous elements. It is a graded field, in the sense that every nonzero homogeneous element is invertible.

Lemma 4.5. *The localization of graded Hom spaces from S_T to K_T is compatible with convolution. In the localized category, $\Delta_w \simeq \nabla_w$; distinct standard objects are orthogonal; and every object is a finite direct sum of shifts of standard objects. Within each connected component, any two standards are linked, before localization, by a chain of nonzero homogeneous morphisms between standards.*

Proof. First consider a standard object Δ_w . Its invertibility identifies each of the left and right actions of $S_T = \mathrm{End}^\bullet(1_I)$ with its full graded endomorphism ring. The two identifications differ by a graded automorphism of S_T . Thus every nonzero homogeneous element acting on the right becomes invertible after localization for the left action. Standard generation extends this property to all objects, since the property of a natural transformation being invertible is preserved under cones and retracts. Interchanging left and right gives the converse. It follows that convolution in either variable preserves the maps being inverted. Hence it descends to the localized category.

For an affine simple reflection s , the relevant Schubert variety is $\mathbb{P}^1$ with two strata. The cone of $\Delta_s \rightarrow \nabla_s$ is supported at the closed point. To identify its character also for an affine simple

reflection, write the corresponding affine root as $\alpha + n\delta$. Conjugation by a constant torus element $t \in T$ sends the root coordinate ct_0^n to $\alpha(t)ct_0^n$, where t_0 is the loop parameter. Thus the tangent character of the opposite chart at the closed point is α or $-\alpha$; its finite-root part is nonzero. After restriction to T , the punctured chart is the punctured line for that character. The equivariant localization triangle for its zero section computes the closed-point term by the Euler complex. Applying $\mathrm{RHom}(1_I, -)$ gives, up to the standard normalization shifts and twists,

$$\mathrm{Cone}(S_T[-2](-1) \xrightarrow{\alpha} S_T).$$

Here the shift and twist make the Euler map of degree zero; its cohomology is $S_T/(\alpha)$, up to the same normalization. The remaining factors of the closed-point stabilizer are unipotent, so restriction to T does not change its equivariant cohomology. Since $\alpha \neq 0$ in characteristic zero, this complex becomes zero over K_T and has nonzero cohomology before localization. The unit generates the category supported at the closed point, so its Hom detects the cone there. Hence the cone vanishes after localization, giving $\Delta_s \simeq \nabla_s$ there. A length-zero orbit is already closed, and length-additive convolution now proves the same assertion for all w .

The usual adjunction calculation gives $\mathrm{Hom}^\bullet(\Delta_w, \nabla_v) = 0$ for $w \neq v$ and identifies the endomorphism ring for $w = v$ with S_T . After localization the standards are consequently orthogonal and have endomorphism ring K_T . Finite cones and retracts of such objects split into finite sums of their shifts: homogeneous matrices over a graded field reduce to their rank normal form. Standard generation gives the assertion for every object.

Finally, adjunction gives $\mathrm{Hom}^\bullet(1_I, \nabla_s) = 0$, whereas the nonzero Euler complex above gives a nonzero homogeneous morphism from 1_I to a shift of Δ_s . Convolution by an invertible standard preserves nonzero morphisms. Applying this observation successively along a reduced expression links Δ_ω to every standard in the component $\omega \in \Omega$. Reversing a chain when necessary links any two standards in that component. $\square$

4.3 Making Frobenius monoidal

We next improve the Frobenius intertwiner in (4.1) on the graded diagram of IC words. This is the only step where the divisibility condition on d is used. It removes a possible scalar tensor defect between different components.

Lemma 4.6. *If $|\Omega| \mid d$, the exact-functor intertwiner between $\Phi \mathrm{Fr}^{d*}$ and $r_*\Phi$ can be chosen to respect convolution and the unit on the complete graded Hom diagram of IC words.*

Proof. Choose the intertwiner for one Frobenius and normalize its value on the unit, whose degree-zero endomorphisms are k . Compare its value on $P * Q$ with the convolution of its values on P and Q , using the monoidal structures of Φ , Fr^* , and q_* . Transporting this discrepancy through the equivalences gives a natural degree-zero automorphism

$$\delta_{P,Q} : P * Q \longrightarrow P * Q$$

commuting with shifts. If $P = \Delta_w$ and $Q = \Delta_v$, their convolution is invertible, so $\delta_{P,Q}$ is a scalar. Naturality for the nonzero homogeneous maps in Lemma 4.5, and invertibility of the other factor, shows that this scalar depends only on the components of w and v . Denote it by $c(\omega, \omega')$.

The natural transformation δ extends to the localized category. There all objects split into sums of shifted standards, so naturality forces its value on any two objects supported in components ω, ω' to be the same scalar $c(\omega, \omega')$. In particular this holds for IC words. By Lemma 4.4, localization is

injective on the endomorphisms of their convolution. The scalar equality therefore already holds before localization.

Associativity of convolution gives

$$c(\omega, \omega')c(\omega\omega', \omega'') = c(\omega', \omega'')c(\omega, \omega'\omega''), \quad c(1, \omega) = c(\omega, 1) = 1.$$

Thus c is a normalized multiplicative 2-cocycle on Ω . Frobenius preserves components and acts trivially on the scalar field. Iterating the intertwiner d times consequently changes this defect to c^d . Positive-degree cohomology of a finite group is annihilated by its order, by restriction and corestriction to the trivial subgroup. Hence the class of c^d in $H^2(\Omega, k^\times)$ is zero. Rescaling the iterated intertwiner on each component by a normalized 1-cochain removes its defect. This rescaling is defined on the entire homotopy category: apply the chosen scalar to every object in the indicated open-and-closed component, and take direct sums over the finite component support of an object. It remains a natural intertwiner of exact functors on all objects and all their morphisms. The resulting intertwiner respects all graded morphisms between words, their tensor products, and the nullary operation specifying the unit. $\square$

4.4 Purity and the enhanced diagram

We recall the precise formality statement needed to pass from the graded diagram to an enhancement. It also explains why possible nonsemisimplicity of pure Weil modules causes no obstruction. For a complex with automorphism, an eigenvalue has weight j if its complex absolute values are $r^{j/2}$, with the chosen coefficient realization. All the eigenvalues used below are algebraic Weil numbers supplied by the preceding purity argument.

Lemma 4.7. *In the derived category of complexes over k with one automorphism, consider the full subcategory consisting of complexes with bounded below, degreewise finite-dimensional cohomology, pure of weight j in degree j . Its mapping spaces are discrete, and cohomology is an equivalence from this subcategory to the category of graded pure modules. This equivalence is symmetric monoidal for tensor over k with the diagonal automorphism. Consequently, a diagram of mapping complexes of this kind, including all its multilinear composition and tensor operations, is determined by its cohomology diagram.*

Proof. A complex with one automorphism is an object of the derived category of $R = k[u, u^{-1}]$ -modules. The ring R is hereditary. Every finite-dimensional R -module has a resolution by finitely generated projectives of length at most one, so it is compact in the derived category. It follows that a complex as in the statement splits into its shifted cohomology modules. One can construct this splitting by lifting each cohomology module with a length-one projective resolution and taking the direct sum; boundedness below and compactness justify the same construction when the complex is unbounded above.

Modules of distinct weights have disjoint u -spectra, and hence have neither Hom nor Ext between them. For pure modules M_i, N_j of weights i, j , respectively, the terms contributing to a positive homotopy group of a mapping space are

$$\mathrm{Hom}_{D(R)}(M_i[-i], N_j[-j-n]) = \mathrm{Ext}_R^{i-j-n}(M_i, N_j), \quad n > 0.$$

They vanish if $i \neq j$ by the weight distinction, and if $i = j$ because the Ext degree is negative. In degree zero, the only surviving terms are the ordinary maps $M_i \rightarrow N_i$. The compactness just noted permits the same calculation for the direct sums of cohomology pieces. Thus cohomology is fully faithful on mapping spaces and is evidently essentially surjective. Equal-weight modules may have

nonzero Ext_R^1 ; this does not affect the calculation, since those groups do not occur in these mapping spaces.

The Künneth isomorphism over k preserves the automorphism and adds both weights and degrees. Since the complexes are bounded below, only finitely many terms contribute to each degree of a tensor product. Cohomology is therefore a symmetric monoidal equivalence on this subcategory. Apply it to each mapping complex of a diagram and each tensor product serving as the source of a composition operation. All these sources remain in the same subcategory, with discrete mapping spaces. The operations and their coherent identities are consequently determined by their cohomology operations and identities. $\square$

Proposition 4.8. *The equivalence on the graded diagram of IC words, with the Frobenius intertwiner from Lemma 4.6, lifts to an enhanced monoidal equivalence $\Theta : \mathcal{H} \simeq \mathcal{C}$ intertwining Fr^{d*} and r_* . It induces the prescribed graded Hom identifications on the words.*

Proof. Transport the Weil isomorphism of each normalized IC object across Φ , and lift that isomorphism to the enhancement of $\mathcal{C}$. This requires only the lift of an isomorphism of objects: an action of the freely generated group $\mathbb{Z}$ imposes no additional relation. Give each formal word the convolution of these chosen lifts. Lemma 4.6 says that the resulting Frobenius actions on the graded Hom spaces are exactly those transported through Φ .

For a list of input words $P_1, \dots, P_m$ and an output word Q , use the mapping complex

$$\text{RHom}(P_1 * \dots * P_m, Q)$$

as the multimorphism complex. The chosen Weil structures give it the conjugation automorphism. Its cohomology is pure in its degree by Lemma 4.4, on both sides of (4.1). Composition, tensor product, and the unit are equivariant operations on these complexes, since the Frobenius automorphisms on the original enhanced categories are monoidal. Lemma 4.7 identifies the two enhanced diagrams from their identified graded diagrams, simultaneously with all these operations. The formal-word indexing ensures that every tensor product being represented is itself among the colors of this diagram.

Forget the automorphism in this identified enriched multicategory and take its stable idempotent-complete envelope. The relevant universal property concerns the full enhanced mapping complexes: restriction identifies exact functors out of the envelope with enriched functors on the generator category. Applying this property in each variable extends the multilinear tensor operations and all their coherence data. Applying it to the already identified diagrams with automorphism extends the Frobenius actions and their monoidal intertwining as well. Thus this passage uses an equivalence of enriched multicategories, not merely agreement of triangulated functors on generator objects. The IC objects generate the constructible side; their images generate the coherent side because Φ is an equivalence. These envelopes are therefore exactly $\mathcal{H}$ and $\mathcal{C}$. $\square$

We have constructed the enhanced equivalence needed for traces. It agrees with Φ on the pure generator diagram, but an identification on generators does not by itself specify its value on every mixed perverse object. Central Satake kernels are mixed. We therefore compare their weight filtrations before normalizing their tensor and Weil structures.

4.5 Central objects and their normalization

Write $\eta_V : r_* D(V) \xrightarrow{\sim} D(V)$ for the preferred Weil identification, and write ρ_V for the action of L on V .

Lemma 4.9 (Comparison on mixed central objects). *For every $V \in \text{Rep } L$, the enhanced equivalence constructed in Proposition 4.8 satisfies*

$$\Theta(Z(V)) \simeq \Phi(Z(V)) \simeq D(V)$$

as geometric objects. The first isomorphism can be chosen to respect the transported Weil structures and the identifications of pure weight pieces supplied by the comparison on IC generators.

Proof. Transport the perverse t -structure to the coherent category by Φ . Use for Φ the globally natural exact-functor intertwiner constructed in Lemma 4.6. It transports the Weil structures on every term and arrow of a weight filtration, even though its monoidality was needed only on the IC-word diagram. The functor Θ is t -exact for this structure: it matches the simple perverse objects, and every perverse object under consideration has finite length. The same argument applies to its inverse. The central sheaf $Z(V)$ is mixed perverse, by its nearby-cycles construction, and its finite weight filtration has pure perverse quotients

$$P_j = \text{gr}_j^W Z(V).$$

These statements can be checked on a finite Schubert support. The weight filtration carries the equivariance because it is functorial and compatible with smooth pullback.

By geometric semisimplicity for pure perverse sheaves [BBD82, Théorème 5.3.8], each P_j is geometrically a direct sum of IC complexes with Weil multiplicity spaces. Geometric semisimplicity also holds in the equivariant perverse heart. Indeed, forgetting equivariance is fully faithful on this heart for the connected equivariance groups in use. For a morphism of underlying perverse sheaves, the two pullbacks through the equivariance isomorphisms agree if they agree at the identity: connectedness supplies H^0 of the acting group, and the vanishing of negative perverse Hom groups excludes higher terms in degree zero. This can be checked on the finite-type quotients defining the equivariant category. Ordinary splitting idempotents therefore split equivariantly as well. The multiplicity spaces need not have semisimple Frobenius.

The comparison on IC generators identifies the images of the P_j , together with their Weil structures and graded morphisms. We extend these identifications through the weight filtration. Suppose they have already been extended through weight $j - 1$, and put

$$B = W_{j-1}Z(V), \quad A_- = W_{j-2}Z(V).$$

The next step is an extension of P_j by B . Use Φ to identify the target heart with the source heart. The induction hypothesis and the chosen pure-piece identification match the two images of B and P_j , respectively. The two extension classes then lie in the same space $\text{Ext}^1(P_j, B)^F$. The exact sequence $0 \rightarrow A_- \rightarrow B \rightarrow P_{j-1} \rightarrow 0$ gives

$$\ker(\text{Ext}^1(P_j, B) \longrightarrow \text{Ext}^1(P_j, P_{j-1})) = \text{im}(\text{Ext}^1(P_j, A_-) \longrightarrow \text{Ext}^1(P_j, B)). \quad (4.4)$$

All Ext groups here are geometric groups with their Frobenius action. By Lemma 4.4, $\text{Ext}^1(P_j, P_i)$ has weight $1 + i - j$. For $i \leq j - 2$ this weight is negative. Applying the long exact sequences through the filtration of A_- shows that $\text{Ext}^1(P_j, A_-)$ has only negative weights. Consequently the map in Equation (4.4) is injective on Frobenius-invariant classes. Its target records the extension $W_j Z(V)/W_{j-2} Z(V)$ between the two consecutive pure pieces. Its connecting morphism is a degree-one morphism between sums of IC complexes with Weil multiplicities, so the projected classes agree by the comparison of Θ and Φ on these graded Hom groups. Thus the two full extension classes agree.

It remains to choose an isomorphism of these extensions compatible with Frobenius. With the endpoint identifications fixed, the choices form a torsor under

$$H = \text{Hom}(P_j, B).$$

The filtration of B shows that all weights of H are among the negative weights $i - j$, $i \leq j - 1$. Hence $F - 1$ is invertible on H , where F denotes Frobenius. If an isomorphism f has Frobenius defect $\delta = F(f) - f$, translating it in this torsor by $-(F - 1)^{-1}\delta$ gives a Frobenius-compatible isomorphism. This proves the induction, including its compatibility with the chosen pure-piece identifications. Generalized weight spaces suffice throughout. Finally, Lemma 4.3 identifies $\Phi(Z(V))$ with $D(V)$ geometrically. $\square$

The comparison of objects leaves freedom in their Weil maps, tensor maps, and central interchanges. The next proposition makes that freedom explicit. A morphism between diagonal representation kernels will be called constant if it is induced by an L -linear map of their representation factors. A representation is neutral if it factors through $A = L_{\text{ad}}$.

Proposition 4.10 (Weil and tensor normalization). *There are identifications $\Theta(Z(V)) \simeq D(V)$, natural and additive in $V \in \text{Rep } L$, and an element $z \in Z(L)$ with the following properties. Representation morphisms become their ordinary maps on diagonal representation kernels, and the transported Weil map on $D(V)$ is*

$$\rho_V(z) \eta_V : r_* D(V) \longrightarrow D(V). \quad (4.5)$$

The transported tensor maps are constant; on neutral representations they can be taken to be the ordinary diagonal convolution identifications. The half-braid of $Z(V)$ with $Z(W)$ becomes the ordinary interchange whenever either V or W is neutral.

Proof. Semisimplicity of $\text{Rep } L$ allows us to choose the object identifications first on irreducibles and then extend them using the ordinary multiplicity spaces. They are then additive and natural in representation morphisms.

For an irreducible V , compare the transported Weil map with η_V . Their ratio lies in the units of

$$E_V = \text{End}^0(D(V)) = (k[\mathcal{N}] \otimes \text{End}_k(V))^L. \quad (4.6)$$

To obtain this equality, compute degree-zero Hom in the standard coherent heart on the ordinary diagonal and use $\Gamma(\widetilde{\mathcal{N}}, \mathcal{O}) = k[\mathcal{N}]$. Thus the ratio is an equivariant polynomial matrix $a_V(N)$. Its value at $N = 0$ is a nonzero scalar c_V , by irreducibility.

The graded algebra E_V is finite-dimensional. For example, evaluation at a regular nilpotent is injective on equivariant polynomial maps: equivariance determines the values on its dense orbit, and regularity determines the map on $\mathcal{N}$. Since $(E_V)_0 = k$, its ideal of positive polynomial degree is nilpotent. Under η_V , the scaling action on this algebra is

$$\tau(f)(N) = f(r^{-1}N).$$

Changing the object identification by $b \in E_V^\times$ changes a_V to $b a_V \tau(b)^{-1}$. Suppose the first positive-degree term of a_V/c_V has degree m , so that

$$a_V/c_V = 1 + a_m + \text{terms of higher degree}.$$

Taking $b = 1 + b_m$ changes its degree- m term to $a_m + (1 - r^{-m})b_m$. The choice

$$b_m = \frac{a_m}{r^{-m} - 1}$$

removes that term. The denominator is nonzero because $r > 1$. Successive corrections terminate in the finite-dimensional graded algebra and put the Weil map in the form $c_V \eta_V$. Extend these choices naturally over the irreducible decompositions.

Let V, W be irreducible and let

$$J_{V,W} : D(V) * D(W) \xrightarrow{\sim} D(V \otimes W)$$

be the transported tensor map. Evaluation at $N = 0$ is an invertible L -linear map. Compatibility with the Weil structures therefore gives

$$c_X = c_V c_W \quad \text{for every irreducible } X \subset V \otimes W. \quad (4.7)$$

The Weil maps on the source and target of $J_{V,W}$ consequently have the same scalar. Their compatibility now says $J_{V,W}(r^{-1}N) = J_{V,W}(N)$, so $J_{V,W}$ is constant. Additivity gives the assertion for arbitrary representations. Together with the unit normalization, the scalars in Equation (4.7) define a tensor automorphism of the identity functor of $\text{Rep } L$. Such automorphisms are exactly the elements of $Z(L)$, giving the element z in Equation (4.5). In particular, this Weil map is ordinary on neutral representations.

We next determine the interchange of two central labels. For irreducible V, W , Weil compatibility makes the transported map

$$h_{V,W} : D(V) * D(W) \longrightarrow D(W) * D(V)$$

constant, by the same scaling argument. Compose it with the inverse ordinary flip and denote the resulting L -linear endomorphism of $V \otimes W$ by $T_{V,W}$. The polynomial matrix

$$N \longmapsto d\rho_W(N)$$

defines a degree-zero endomorphism of $D(W)$. It is induced from the base $\mathfrak{g}$, so convolving it with $D(V)$ simply tensors the matrix with id_V . Naturality of the half-braid in its other kernel variable gives

$$[T_{V,W}, \text{id}_V \otimes d\rho_W(N)] = 0 \quad (N \in \mathcal{N}).$$

Nilpotents span the semisimple Lie algebra $\mathfrak{g}$. Schur's lemma therefore gives $T_{V,W} \in \text{End}_k(V) \otimes k \text{id}_W$; its L -equivariance then makes it scalar on the irreducible V . Write

$$h_{V,W} = \beta_{V,W} \text{flip}_{V,W}.$$

The hexagon and the central tensor identities imply multiplicativity on tensor constituents in each variable. For example, if $X \subset W \otimes U$ is irreducible, then $\beta_{V,X} = \beta_{V,W} \beta_{V,U}$. These equalities are unaffected by the constant tensor identifications, since both sides of each hexagon are scalar multiples of the same ordinary interchange. With either irreducible label fixed, the scalars in the other variable thus define a tensor automorphism of the identity on $\text{Rep } L$, hence the action of a central element. They are trivial on neutral representations. Additivity now proves the assertion about $h_{V,W}$ whenever either label is neutral.

It remains to normalize the constant tensor maps on $\text{Rep } A$. We first show that their tensor structure respects the ordinary symmetry: for neutral V, W ,

$$D(\text{flip}_{V,W}) \circ J_{V,W} = J_{W,V} \circ h_{V,W}. \quad (4.8)$$

Here $h_{V,W}$ is already the ordinary flip. Let K be hyperspecial, and let $M \in \mathcal{H}_{I,K}$ and $J \in \mathcal{H}_{K,I}$ be the unshifted projection kernels. Their convolution $C = M * J$ is the nonzero unshifted closed-flag

kernel. Projection compatibility for central sheaves identifies the projected functor, including its interchanges, with Satake; see [Ope26, Section 2, Theorem “Central and Wakimoto kernels”]. For neutral labels the Satake parity is trivial. The difference between the two sides of (4.8) therefore vanishes after right convolution by M , and hence after right convolution by C .

This difference is a constant map. Under Θ , its convolution with C is the tensor product of its underlying linear map with the nonzero coherent kernel $\Theta(C)$. A nonzero linear map over k remains nonzero after tensoring with a nonzero complex. The original difference must therefore vanish. The constant tensor maps define a symmetric tensor autoequivalence of $\text{Rep } A$ whose underlying k -linear functor is the identity.

By ordinary Tannaka duality over the algebraically closed field k , this autoequivalence comes from an automorphism of A , up to tensor isomorphism. It fixes every irreducible isomorphism class. Its outer automorphism is consequently trivial by the description of representations through the root datum, so the automorphism is inner. The autoequivalence is tensor isomorphic to the identity. Using this tensor isomorphism to adjust the identifications on neutral labels makes their tensor maps ordinary. Its components are constant L -linear maps, so all preceding Weil and interchange normalizations are preserved. $\square$

Only the interchanges and tensor identities supplied in the homotopy category have been used in this proposition. For a fixed central label, naturality in the other kernel variable is the natural equivalence of exact enhanced functors furnished by [Ope26, Section 2, Theorem “Central and Wakimoto kernels”]. These statements specify the central compatibility needed below without requiring a lift of Z to an infinity-categorical center.

Proof of Theorem 4.1. Proposition 4.8 supplies the enhanced monoidal equivalence and its Frobenius intertwining. Lemma 4.9 identifies the central objects under this equivalence, and Proposition 4.10 gives all the stated Weil, tensor, and interchange normalizations. $\square$

5 Horizontal traces and spherical transfer

We now pass from the enhanced Iwahori category to a coherent model of its Frobenius trace. The point of the comparison is to retain maps between representation labels: the global support of Proposition 2.6 uses matrix entries as well as characters. We will identify the hyperspecial trace as a retract and compare its matrix operations and homogeneous Satake morphisms inside this model.

Fix one of the places of Lemma 3.1, of degree d , and put $r = q^d$. Write I and K for Iwahori and hyperspecial level. Let $\mathcal{H}_{P,Q}$ be the category of kernels from level Q to level P , with the convolution convention of Section 4. The Frobenius twist in the one-factor trace is $\Phi = \text{Fr}^{d*}$. Brackets $[J]$ denote the image of a kernel in its horizontal trace. All the coends below are derived.

5.1 Mapping complexes and level changes

The following description also fixes the rotation convention used in the comparison.

For a small rigid category $\mathcal{C}$, we form its presentable twisted trace as

$$\text{Ind}(\mathcal{C}) \otimes_{\text{Ind}(\mathcal{C} \otimes \mathcal{C}^{\text{rev}})} \text{Ind}(\mathcal{C}),$$

where the regular right action and twisted left action are

$$B \cdot (P, Q) = Q * B * P, \quad (P, Q) \cdot B = \Phi P * B * Q.$$

By its *horizontal trace* we mean the small idempotent-complete category of compact objects in this tensor product. The object $[J]$ is the image of $\mathbf{1} \otimes J$.

Lemma 5.1. *For a rigid enhanced kernel category with monoidal automorphism Φ , the images of kernels compactly generate its presentable trace and generate its horizontal trace under finite colimits and retracts. Moreover,*

$$\mathrm{Map}([J], [J']) \simeq \int^T \mathrm{Map}(\Phi T * J, J' * T). \quad (5.1)$$

The formula applies to the matrix category with levels I, K . Each corner trace embeds fully faithfully in the matrix trace. At a closed point of degree d , the cyclic trace of its d geometric factors is identified with the one-factor trace twisted by Fr^{d} . This identification respects level transfers and full-cycle slides.*

Proof. Use the relative tensor product just defined, and put $\mathcal{D} = \mathcal{C} \otimes \mathcal{C}^{\mathrm{rev}}$. The right module functor $\mathcal{D} \rightarrow \mathcal{C}$ sending (P, Q) to $Q * P$ preserves compact objects. Its right adjoint is therefore continuous; rigidity makes this adjoint a module functor. On the unit it has value

$$\int^T (T^!, T), \quad \mathbf{1} \longrightarrow T^! * T$$

with the indicated duality convention. Indeed, writing the adjoint as the coend with coefficients $\mathrm{Map}(Q * P, \mathbf{1})$ and then applying Yoneda gives this expression. Base change to the twisted left module proves (5.1), including compactness. Generation follows from generation of the relative tensor product by elementary tensors.

A representative with coend label T acts by inserting coevaluation, rotating T from last to first, applying the representing map, and contracting by evaluation. Explicitly, if f has coend label T and g has coend label U , their composite has label UT and representative

$$\Phi U \Phi T * J \xrightarrow{\Phi U * f} \Phi U * J' * T \xrightarrow{g * T} J'' * UT.$$

This follows from balancing and the two duality triangles. These are also the formulas for the path functor of [Ope26, Proposition 2.5, “Coherent path rotation”].

For the two levels use the sum of their units. Extend the cyclic path functor by zero on paths whose endpoint levels differ, and use rotation with level change on the diagonal blocks. The matrix category is rigid: its generators are invertible standard kernels, Satake kernels, and the adjoint projection kernels. Off-diagonal generation follows from the parahoric orbit stratification, using minimal-length Iwahori lifts of partial-flag cells and left–right exchange. Projection kernels have their usual adjoints with the required shifts and Tate twists, which belong to these stable categories; rigidity uses those adjoints. If both endpoints in (5.1) lie in the aa corner, consider the bifunctor $\mathrm{Map}(\Phi T' * J, J' * T)$. For pure matrix blocks, convolution composability and equality of the source and target blocks force both T' and T into the aa corner. This remains true for the derived coend. The underlying linear category is the finite direct sum of its matrix blocks, with zero Hom complexes between different blocks. By enriched additivity one may compute its coend on pure-block objects; a higher bar chain cannot change blocks, because the intervening Hom complex would be zero. Mixed direct sums thus introduce no additional relations. The coend and its composition are the corner calculation, proving full faithfulness.

For the closed-point assertion put $F_0 = \mathrm{Fr}^*$. In the d -fold bar construction number indices cyclically, so that the left action of $(P_i, Q_i)_i$ on $(J_i)_i$ has i th entry $F_0(P_{i-1}) * J_i * Q_i$. Balancing first in the Q_i contracts regular modules and replaces right and left entries L_i, J_i by $W_i = J_i * L_i$. The remaining balance identifies right multiplication by P on W_i with left multiplication by $F_0 P$ on W_{i+1} . Contracting successively for $i = d, d-1, \dots, 2$ leaves the twist F_0^d and the label

$$F_0^{d-1} W_2 * \dots * F_0 W_d * W_1.$$

For $d = 1$ this expression is W_1 . These contractions are just associativity of the relative tensor product and $\mathcal{C} \otimes_{\mathcal{C}} (-) \simeq (-)$.

It is useful to see the same calculation on maps. Put J, J' in the first geometric factor and units elsewhere. The contributing integrand is

$$\mathrm{Map}(F_0 T_d * J, J' * T_1) \otimes \bigotimes_{i=2}^d \mathrm{Map}(F_0 T_{i-1}, T_i).$$

Successive coends remove the intermediate T_i by Yoneda, giving (5.1) with twist F_0^d . The calculation holds also between different rational levels: all factors in a cycle use the same level, and each T_i belongs to the same prescribed off-diagonal block. External labels suffice by the orbit filtration and equivariant Künneth. Composition contracts the intermediate maps in the same order, so the calculation respects composition, including complex symmetry signs. Weil maps on all geometric factors combine to the Weil map for F_0^d . Thus projection transfers and full-cycle slides become exactly their one-factor versions. Finally rotation moves all factors into one, so these one-factor images generate the fixed-corner trace. $\square$

For a Weil kernel T with identification $\Phi T \rightarrow T$, write χ_T for the endomorphism of the trace unit represented by this map in the coend. More generally, if T has an interchange with J , the composite $\Phi T * J \rightarrow T * J \rightarrow J * T$ defines its character operator χ_T on $[J]$. Thus $[T]$ denotes a trace object, whereas χ_T denotes an operator.

Lemma 5.2. *The cohomology of $\mathrm{End}([1_I])$ is concentrated in degree zero and is the affine Hecke algebra with parameter r . Its unshifted standard Weil classes are the characteristic-function generators in the normalization of [Ope26, Proposition 2.6, “Hecke action and transfer”].*

Proof. The endomorphisms of the trace unit compute the vertical trace. Apply the standard-orbit semiorthogonal decomposition of the kernel category. Hochschild homology with Frobenius is additive for this decomposition; equivalently, the directed morphisms between standard strata exclude mixed-stratum Hochschild chains. Each stratum contributes the twisted trace of the formal equivariant point algebra $H^\bullet(BT)$. Frobenius multiplies its linear generators by r , so its twisted diagonal intersection is k . The unshifted standard Weil class represents the generator of this summand. One first works on finite downward-closed supports and then takes their union.

We can verify the relations directly in this universal trace. For an affine simple reflection s , let Δ_s be its unshifted open rank-one kernel. The convolution fiber over relative position 1 is a projective line with one point removed, whereas over relative position s it is a projective line with two points removed. They are respectively $\mathbb{A}^1$ and $\mathbb{G}_m$. Write $\mathrm{cl}(T)$ for the class of a Weil kernel in its Grothendieck group. The two-stratum filtration of convolution gives the identity

$$\begin{aligned} \mathrm{cl}(\Delta_s * \Delta_s) &= \mathrm{cl}(\mathbf{1}) \mathrm{cl}(R\Gamma_c(\mathbb{A}^1, k)) \\ &\quad + \mathrm{cl}(\Delta_s) \mathrm{cl}(R\Gamma_c(\mathbb{G}_m, k)). \end{aligned}$$

The assignment $\mathrm{cl}(T) \mapsto \chi_T$ is additive: on a cone presentation the off-diagonal entries disappear by dinaturality, and the shifted diagonal contributes the negative class. The map is multiplicative by the composition rule in Lemma 5.1. The two coefficient complexes have Weil traces r and $r - 1$, respectively. Thus the standard trace class $H_s = \chi_{\Delta_s}$ satisfies $H_s^2 = (r - 1)H_s + r$. For length-additive products, the open convolution correspondence gives $\Delta_w * \Delta_{w'} \simeq \Delta_{ww'}$, yielding the usual length-additive relation, including length-zero elements. These relations and the standard-class basis identify the algebra. The local character formulas in [Ope26, Proposition 2.6] show that applying the path functor gives the usual Hecke operators. No faithfulness of that global functor is needed for the universal algebra calculation. $\square$

5.2 The coherent trace and its representation bundles

Write $\widetilde{\mathcal{N}} \rightarrow \mathfrak{g}$ for the Springer resolution followed by the inclusion of the nilpotent cone. Define the derived equation space

$$Y_N = \{(s, N) \in L \times \mathfrak{g} : \mathrm{Ad}(s)N = r^{-1}N\}. \quad (5.2)$$

Its flag version consists of (s, N, B) with B a Borel subgroup, $s \in B$, $N \in \mathfrak{n}_B$, and the displayed equation imposed derivedly in $\mathfrak{n}_B$. Let P_N be the direct image of its structure sheaf on Y_N . For a representation V of L , put $S(V) = \mathrm{Sat}(V)$, and retain the notation $Z(V)$ for the central Iwahori kernel.

Proposition 5.3. *The equivalence of Theorem 4.1 identifies the Iwahori horizontal trace with $\mathrm{Coh}^L(Y_N)$, carrying $[\mathbf{1}_I]$ to P_N and $[Z(V)]$ to $P_N \otimes V$. Sheaf duality $\mathbb{D} = \mathrm{RHom}(-, \mathcal{O})$ is a duality on this coherent model, and $\mathbb{D}P_N \simeq P_N$.*

Let $z \in Z(L)$ be the element of Theorem 4.1. Full-cycle slide on the representation factor is $V(zs)$. Hyperspecial transfer identifies $[S(V)]$ with $P_N^S \otimes V$, where P_N^S is a retract of P_N . If s_{sph} denotes spherical holonomy, then

$$s_{\mathrm{sph}} = zs. \quad (5.3)$$

For neutral labels this comparison respects representation morphisms, homogeneous spherical-kernel morphisms, and matrix operations obtained from slides by composition, contraction, and tensoring by neutral spectators. It also respects invariant holonomy functions coming from arbitrary representations of L .

We prove the coherent assertion first and the transfer assertion in the next subsection. The distinction matters: the coherent trace theorem provides the ambient category, whereas the transfer calculation identifies the spherical retract and its maps.

Proof of the coherent assertion. The coherent trace theorem [BZCHN24, Theorems 1.19(2)–(3) and 3.23] applies to the derived Steinberg category in (4.1) with automorphism r_* . Here r_* means pushforward by the literal dilation $N \mapsto rN$. The scaling action of [BZCHN24, Section 1.6.3] has weight -1 on points, so its parameter for this functor is r^{-1} . It gives Coh of the twisted loops in $[\widehat{\mathcal{N}}/L]$, and sends the unit to the coherent Springer object. Here $\widehat{\mathcal{N}}$ denotes the formal completion of $\mathfrak{g}$ along $\mathcal{N}$. With our rotation convention, $r_* = (r^{-1})^*$ and the loop equation is (5.2). Formal completion imposes no additional condition: positive-degree invariant polynomials vanish on the degree-zero cohomology of this equation space, because r is not a root of unity. Thus every test point already satisfies the formal nilpotence condition. This also agrees with [BZCHN24, Proposition 4.3]. The singular-support condition is vacuous for this Springer version of the trace [BZCHN24, Remark 4.14]. We keep the equations derived in the presentation and in the flag space.

For the map comparison use the cyclic pull-push description of the trace [BZCHN24, Theorem 3.23]. Set

$$X_1 = [\widetilde{\mathcal{N}}/L], \quad Y_1 = [\widehat{\mathcal{N}}/L], \quad l = r^{-1}.$$

A kernel on $X_1 \times_{Y_1} X_1$ is pulled back along the graph which closes its endpoints and then pushed to the twisted loops in Y_1 . We use ordinary kernel convolution, with diagonal pullback and tensor product. In the equivalent $!$ normalization, convolution inserts $\omega_{X_1}^{-1}$ at the middle diagonal; tensoring a kernel with the second-factor ω_{X_1} compensates for this insertion. At the closing graph the compensation is the pullback of that same factor, using its natural transport under l . Thus both normalizations give the stated formula for the unit and the same representation transport.

In first-to-second kernel order, closure goes from x to lx and the base loop is an arrow $y \rightarrow ly$. A representation bundle is pulled from BL at the start, so the projection formula gives $P_N \otimes V$. To calculate rotation explicitly, a two-segment path has entries and arrows

$$x, x_1, lx, \quad y \xrightarrow{\alpha} y_1 \xrightarrow{\beta} ly.$$

After rotation these become

$$l^{-1}x_1, x, x_1, \quad l^{-1}\beta, \alpha.$$

The arrow $l^{-1}\beta$ identifies the new start with the old start. Moving a representation once across the cut, with the other segment the unit, therefore acts by the loop arrow, namely $V(s)$. The Weil identification of $Z(V)$ contributes $V(z)$, giving $V(zs)$. If the representation is a spectator, its transport back to the start uses the same further arrow $l^{-1}\beta$ after rotation. Consequently a coend representative tensored with a spectator and crossed back by standard interchange gives precisely the original map tensored with the bundle from BL . This calculation uses arrows of the fiber products and therefore holds for derived fiber products as well.

Finally the twisted loop and its flag version are Gorenstein with trivial dualizing complex in stack normalization. In the tangent complex of the graph intersection with the diagonal, determinants cancel in pairs. On the affine equation presentation the vector directions and their equations cancel, leaving virtual dimension $\dim L$ before quotient by L . On the flag presentation, the remaining group-flag incidence has trivial equivariant canonical line: its flag tangent and Borel-subgroup tangent have product determinant equal to that of $\text{Lie } L$. These descriptions also verify the trivializations directly. Proper duality for the flag projection now gives $\mathbb{D}P_N \simeq P_N$. $\square$

5.3 The spherical retract and its maps

Let $M \in \mathcal{H}_{I,K}$ and $J \in \mathcal{H}_{K,I}$ be the unshifted projection kernels, and write $C = MJ$. They are independent coend labels; we do not identify them with each other's shifted rigid duals. The transfers have coend labels M and J :

$$i_V : [S(V)] \longrightarrow [Z(V)], \quad l_V : [Z(V)] \longrightarrow [S(V)].$$

They use the Weil structures and the projection interchanges $MS(V) \simeq Z(V)M$ and $JZ(V) \simeq S(V)J$.

Completion of the proof of Proposition 5.3. The composition rule of Lemma 5.1 gives

$$l_V i_V = c_K \text{id}, \quad i_V l_V = \chi_C, \quad c_K = \#(K/I)(\mathbb{F}_{q^d}). \quad (5.4)$$

To check this identity in the universal trace, the composite JM is flag cohomology on the hyperspecial unit. Geometrically it splits into even shifts of that unit: its odd extension groups vanish. Naturality of the projection interchange makes its interchange with $S(V)$ the unit interchange on these shifted summands. Dinaturality of the coend removes off-diagonal Weil entries, so the character is the Weil trace of flag cohomology, namely c_K . The reverse composite $MJ = C$ is the unshifted closed-flag kernel. These are exactly the closed-flag calculations in [Ope26, Proposition 2.6, ‘‘Hecke action and transfer’’]; their proof uses composition and the projection identities, so it gives the identity in the universal trace before applying the global path functor. The multiple-projection compatibility from fusion ensures that crossing a product through a level change is the successive crossing of its factors. Thus χ_C/c_K is an idempotent. On P_N it is the spherical idempotent of Lemma 5.2; denote its image by P_N^S .

We next identify this idempotent on representation labels. For irreducible V , the object

$$Z(V)C \simeq MS(V)J$$

is a shifted simple perverse sheaf. Indeed one projection forgets equivariance, while the other pulls the Satake intersection complex back along the smooth flag fibration with connected fibers. Consequently the actual interchange with C , transported through Θ , is scalar times standard interchange. With standard interchange the preceding coherent calculation makes χ_C equal to its value on P_N tensored with V . This map is nonzero: the hyperspecial unit embeds by (5.4), is nonzero by its spherical corner calculation, and tensoring with a nonzero representation preserves nonzero maps. Both the actual and standard operators divided by c_K are idempotents. A nonzero scalar multiple of a nonzero idempotent is idempotent only when the scalar is one. Hence the interchange with C is standard, and the hyperspecial retract on every label is

$$[S(V)] \simeq P_N^S \otimes V. \quad (5.5)$$

Transfer commutes with full slide. Explicitly, composing slide with i_V in either order gives coend labels $MS(V)$ and $Z(V)M$; the projection interchange identifies the representatives, including their Weil maps. Thus the slide on (5.5) is $V(zs)$. Naturality in ordinary representation maps identifies those maps with the usual constant representation maps tensored with P_N^S .

The same argument identifies character functions, including those of nonneutral representations. A character with spherical coend label $S(V)$ transfers to the label $MS(V)J = Z(V)C$. Its character is the character of $D(V)$ with Weil factor $V(z)$, followed by the closed-flag operation. Dividing by c_K to identify the retract gives the invariant function $\text{Tr}_V(zs)$. This proves (5.3) for the spherical holonomy coordinate.

It remains to compare higher morphisms with representation spectators. This step will later determine the action of the entire spherical loop algebra, including its degree-two coordinates. The retraction of i_V is $c_K^{-1}l_V$. Accordingly, transfer on an arbitrary map $f : [S(U)] \rightarrow [S(V)]$ is the map between retracts

$$f \mapsto c_K^{-1}i_V f l_U : [Z(U)] \longrightarrow [Z(V)].$$

Consider a map between neutral labels represented in (5.1) by $\Phi T * S(U) \rightarrow S(V) * T$. Compose with i_V, l_U and divide by c_K . Its transferred representative, with coend label MTJ , is

$$\begin{aligned} \Phi(MTJ)Z(U) &\longrightarrow \Phi M \Phi T S(U)J \\ &\longrightarrow \Phi M S(V)TJ \longrightarrow Z(V)MTJ. \end{aligned} \quad (5.6)$$

Tensor the original representative on the right with $S(R_1)$ and cross this spectator back through T . Tensor and projection compatibility identify the transferred representative with (5.6) tensored with $Z(R_1)$, crossed back through MTJ . In fact that crossing is exactly the successive J, T, M crossing, using $S(R_1)$ at the intermediate level.

For an actual homogeneous spherical morphism we take T to be the unit. This includes positive cohomological degrees: the spherical morphism is the middle arrow of (5.6), and no image of that morphism under Z is used. For a slide on a label $S(W)$, take $T = S(W)$: its representative is the identity, followed by the Weil identification; evaluation naturality and the duality triangles give this description. A character of an arbitrary representation W has the same coend label $S(W)$ and its Weil map as representative, but both endpoint labels are the unit. It therefore admits the same comparison with a neutral spectator even if W is not neutral. These are the only coend labels required for the asserted operations. On the spherical side the spectator crossing is ordinary representation interchange. On the coherent side it is also standard: MTJ is respectively C or

$Z(W)C$, the crossing with C was just proved standard, and the crossing with $Z(W)$ is standard for a neutral spectator by Theorem 4.1. The endpoint tensor maps on neutral representations have the ordinary normalization from that theorem. The coherent cut calculation therefore identifies spectator tensoring with tensoring by the actual representation bundle. Compositions and equivariant contractions preserve this identification. This proves the remaining assertion without an enhanced Drinfeld-center hypothesis. $\square$

6 Coherent local tests after regrading

The trace comparison gives an Iwahori Springer object and a spherical retract in a coherent category. For the global argument we need a model in which the spherical retract is the structure sheaf and the other tests are direct images from explicit flag spaces. We obtain that model by linear Koszul duality and a change of cohomological degree. We will also identify the ring action on the spherical retract, including its representation-valued matrix operations.

6.1 The slice and its equation algebras

We now describe the coherent trace near one of the semisimple classes chosen in Lemma 3.1. Our aim is to replace its Springer object by flag objects over the positive resonance space. This requires both a Koszul duality and a change of cohomological grading; we establish the two operations, including their effect on morphisms, before applying them to the flags.

Fix the selected element $t \in L$ and the corresponding scalar r . Write

$$H = Z_L(t), \quad E_+ = \ker(\mathrm{Ad}(t) - r), \quad W = \ker(\mathrm{Ad}(t) - r^{-1}).$$

Here H is connected. The diagonal cocharacter $m : \mathbb{G}_m \rightarrow H$ of the chosen triple is central in H ; its weights on E_+ and W are respectively 2 and -2 . An L -invariant nondegenerate form on $\mathfrak{g}$ identifies W^* with E_+ . These facts, and the finiteness of the H -orbits in either resonance space, are the resonance assertions recalled in Lemma 3.2; for W one applies the same assertion to t^{-1} . Denote by $z \in Z(L)$ the coordinate correction in (5.3).

Lemma 6.1 (Semisimple slice). *There is an affine saturated H -invariant open neighborhood $U_H \subset H$ of the identity class such that the conjugation map*

$$L \times^H (tU_H) \longrightarrow L, \quad [g, tu] \longmapsto gtug^{-1}, \quad (6.1)$$

is strongly étale over its image. More precisely, it is the pullback of the induced étale map $U_H//H \rightarrow L//L$ along the adjoint quotient. The analogous statement holds with t replaced by $z^{-1}t$. The neighborhood can be chosen so that the transverse conjugation operator and all nonresonant blocks of the vector equations are invertible. It contains the entire unipotent locus of H . The identity point of $U_H//H$ is the only point above the selected semisimple class; the corresponding fiber in U_H still contains every unipotent element.

Proof. The transverse differential of (6.1) is $1 - \mathrm{Ad}(tu)$ on $\mathfrak{g}/\mathrm{Lie} H$. It is invertible at $u = 1$ and, more generally, at every unipotent $u \in H$: on a nonidentity t -eigenspace the operator $\mathrm{Ad}(u)$ is unipotent, so $\mathrm{Ad}(tu)$ cannot have eigenvalue 1. The same argument applies to each nonresonant block of $\mathrm{Ad}(tu) - r$ and of $\mathrm{Ad}(z^{-1}tu) - r^{-1}$. Their determinants are conjugation-invariant regular functions on H , nonzero on the unipotent locus. Inverting them therefore gives a saturated affine neighborhood of the identity class.

At $[1, t]$ the map (6.1) identifies the closed orbit and its stabilizer with those of t . Luna's fundamental lemma [Lun73, fundamental lemma] therefore makes it strongly étale after saturated shrinking around that orbit. This is the form of the semisimple slice that we use. There are only finitely many semisimple H -classes of u for which tu is conjugate to t , as is seen from the finite Weyl orbit of t in a maximal torus. Invariant functions remove the classes other than $u = 1$. Every shrinking just made contains the identity class in $H//H$, and consequently retains all unipotent elements. Multiplying by the central element z^{-1} changes neither the differential nor the stabilizers, and gives the second assertion. $\square$

We allow further saturated shrinkings of U_H around the identity class. Set $A_0 = \mathcal{O}(U_H)$. By the coordinate comparison (5.3), the spherical coordinate is tu , whereas the coherent Iwahori coordinate is $s = z^{-1}tu$. Pullback along $U_H//H$ replaces L -equivariance by H -equivariance through (6.1). All these base changes are flat and have cohomological degree zero. They consequently commute with cohomology and with the Hom comparisons between bounded coherent objects used below. On an affine presentation, this follows by taking a degreewise finite free resolution: a fixed Hom degree against a bounded target uses only finitely many terms. Taking rational reductive invariants preserves the comparison. For the Ind-extensions the same assertion holds with compact source, by filtered colimits.

Eliminating the nonresonant vector coordinates together with their Koszul generators in (5.2) gives the algebra

$$R_- = A_0 \otimes \text{Sym}(W^* \oplus W^*[1]), \quad d\theta = (\text{Ad}(u) - 1)N. \quad (6.2)$$

The equation notation means that, for $\alpha \in W^*$, $d\theta_\alpha = \langle \alpha, (\text{Ad}(u) - 1)N \rangle$, where N is the universal vector in W . The vector coordinates have degree 0, the generators θ have degree -1 , and both have m -weight 2. The elimination is a quasi-isomorphism of dg algebras: an invertible coefficient block first changes its equation generators so that their differentials are the corresponding coordinates, and then removes these contractible pairs.

The spherical trace calculation of [Ope26, Proposition 2.7, ‘‘Spectral form of a Frobenius trace’’] gives, by the same elimination,

$$B_{\text{sph}}^0 = A_0 \otimes \text{Sym}(E_+^*[-2] \oplus E_+^*[-1]), \quad d\eta = (\text{Ad}(u) - 1)e. \quad (6.3)$$

Its two sorts of generators have degrees 2 and 1, respectively, and both have m -weight -2 . The superscript 0 records that the grading has not yet been changed.

We use the following grading convention. For an even-weight rational module $M = \bigoplus_w M_w$, its *shear* places M_w^d in degree $d + w$. All algebra weights here are even, so this operation preserves the signs in the differential and the tensor symmetry. For a module in the odd $m(-1)$ -sector, place M_w^d in degree $d + w - 1$ instead. Thus the change of degree is even in each sector. Since every rational module is the direct sum of its weight spaces, these rules give equivalences of the corresponding graded module categories. On the even sector the equivalence is compatible with tensor products. We will always apply the Koszul functor first and shear its output.

Define the ordinary affine scheme

$$Y = \{(u, e) \in U_H \times E_+ : \text{Ad}(u)e = e\}. \quad (6.4)$$

The sheared version of (6.3) is its equation Koszul algebra: its vector coordinates have degree 0 and its equation generators have degree -1 .

Lemma 6.2 (The equation schemes). *The algebra R_- and the shear of B_{sph}^0 have cohomology only in degree zero. Their classical schemes are complete intersections of dimension $\dim H$.*

Proof. Let V be either W or E_+ . For an H -orbit $\mathcal{Q} \subset V$, the pairs (u, v) with $v \in \mathcal{Q}$ and $uv = v$ have dimension

$$\dim \mathcal{Q} + \dim Z_H(v) = \dim H.$$

Restricting u to U_H leaves a nonempty open part of every stabilizer, since $1 \in U_H$. There are finitely many orbits by [Ope26, Lemma 3.6, “Resonance orbits”], so the full solution space has dimension $\dim H$. It is cut out by $\dim V$ equations in the smooth affine scheme $U_H \times V$, of dimension $\dim H + \dim V$. The equations therefore have their expected codimension. In a regular ring an ideal generated by this many elements at that codimension is a complete-intersection ideal; its displayed generators form a regular sequence locally. The equation Koszul complex has no negative cohomology, as asserted. $\square$

6.2 Covariant Koszul duality and the grading

The algebras just obtained involve the mutually dual vector spaces W and E_+ . The following covariant form of Koszul duality passes from the first to the second. Its full faithfulness uses the positive weights of the coordinates of R_- ; we include the argument because it also specifies which unbounded module categories occur in the construction.

Proposition 6.3. *For the algebra R_- in (6.2), rational internal derived Hom from the augmentation A_0 followed by shear defines an exact fully faithful functor*

$$\mathcal{K} : \mathrm{Coh}^H(R_-) \longrightarrow \mathrm{Coh}^H(Y). \quad (6.5)$$

Before shear the functor is A_0 -linear and commutes with tensoring by finite-dimensional H -representations. It extends fully faithfully to the Ind-categories of these coherent categories.

Proof. We first compute the endomorphism algebra of the augmentation and show that coherent inputs have bounded coherent shears. We then prove full faithfulness by comparing an object with its Koszul reconstruction. The difference is detected away from the zero section, where the positive grading makes its maps into coherent objects vanish.

Work initially in the unbounded derived category of rational H -equivariant dg modules. Free modules with finite-dimensional representation coefficients are compact generators. Indeed their mapping groups are rational invariant tests on cohomology; invariants are exact in characteristic zero, commute with direct sums, and these tests detect every nonzero rational representation. Semifree resolutions made from such free modules consequently compute the derived functors below.

Write x for the vector coordinates on W and $J_u = \mathrm{Ad}(u)|_W - 1$. A semifree resolution of A_0 is

$$K' = R_- \otimes \mathrm{Sym}(W^*[1] \oplus W^*[2]), \quad dp = x, \quad dl = \theta - J_u p. \quad (6.6)$$

Here p and l have degrees -1 and -2 and both have m -weight 2. The displayed formula is written in vector coordinates, so $d\theta = J_u x$ and $d^2 l = 0$. Adjoin the p ’s before the l ’s to see semifreeness. Replacing θ by $\theta - J_u p$ separates the pairs (x, p) and $(\theta - J_u p, l)$. Each pair is contractible; for the second pair this uses characteristic zero, or equivalently the usual contraction with divided polynomial powers. Thus the augmentation $K' \rightarrow A_0$ is a quasi-isomorphism.

Let G_0 denote rational internal derived Hom from A_0 , computed using K' . Its acting algebra is

$$B^0 = A_0 \otimes \mathrm{Sym}(W[-1] \oplus W[-2]), \quad d\eta = J_u^t \beta, \quad (6.7)$$

where η has degree 1 and β degree 2, both of m -weight -2 . To verify the algebra structure, let the generators act on K' by the derivative operators in p and l . Choosing the sign of the derivative

in p gives precisely the differential in (6.7). These operators graded commute. Composition with $K' \rightarrow A_0$ identifies the resulting map from B^0 to $\underline{\mathrm{RHom}}_{R_-}(A_0, A_0)$ with the polynomial dual of the powers of l and the exterior dual of the powers of p . Using factorials for the polynomial dual gives an isomorphism of complexes, and proves that the map is a quasi-isomorphism of algebras. We identify the opposite algebra with B^0 using its graded commutativity when expressing the Hom functor as a functor to left modules.

After shear, the generators β and η have degrees 0 and -1 . Under $W^* = E_+$ the transpose equation is the fixed-vector equation for u^{-1} on E_+ . Multiplication by the invertible operator $-\mathrm{Ad}(u)$ changes it to the fixed-vector equation for u . Consequently the shear B of B^0 is an equation algebra for Y , and Lemma 6.2 identifies it with $\mathcal{O}(Y)$. This identifies its equation scheme with that of B_{sph}^0 after shear. It does not yet compare the spherical algebra action with the action constructed by G_0 ; that comparison will require the representation-valued transfer maps.

We next check coherence, including the absence of a completion in this computation. A coherent R_- -module has a finite cohomology filtration, so it suffices to take a finite module M over $H^0(R_-)$ concentrated in degree zero. In a fixed cochain degree of $\mathrm{Hom}_{R_-}(K', M)$ only finitely many powers of l occur. The complex is, as a total graded module, finite over

$$A_0[x, \beta] = A_0 \otimes \mathrm{Sym}(W^*) \otimes \mathrm{Sym}(W[-2]);$$

the finite exterior factor comes from p . Its differential is linear over this ring. Noetherianity therefore makes its cohomology finite over the same ring. Multiplication by every x is null-homotopic on this Hom complex: on the source K' it is the commutator of the differential with multiplication by the corresponding p . Thus x acts trivially on cohomology, which is finite over $A_0[\beta]$. After shear all these remaining algebra generators have degree zero. Choosing finitely many homogeneous generators of cohomology shows that only finitely many sheared cohomological degrees occur, and that each is coherent. The finite filtration proves the claim for every coherent input. Rational weight decompositions, rather than products of weight spaces, are used throughout.

It remains to show that G_0 preserves all maps between coherent objects. In the unbounded rational module categories it has a left adjoint L_0 , given by tensoring with the augmentation Koszul bimodule. The unit $N \rightarrow G_0 L_0 N$ is an isomorphism for $N = B^0 \otimes V$, with V a finite-dimensional representation, by (6.7); it is therefore an isomorphism for all perfect B^0 -modules.

Let C_x be the finite Koszul module on the coordinates x over R_- . It is perfect. Each of its bounded coherent cohomology modules is killed by the ideal (x) and is therefore a finite A_0 -module. Since U_H is smooth, such a module has a finite resolution by equivariant projective A_0 -modules, each a summand of some $A_0 \otimes V$. To see this equivariantly, choose a finite set of module generators and its finite-dimensional H -span to construct successive equivariant free covers. Smoothness bounds the projective dimension; at the last step an ordinary splitting can be averaged. The Hom space from a finitely presented equivariant module consists of rational vectors, as is checked from a finite presentation, so averaging gives an equivariant splitting. The cohomology filtration now puts C_x in the thick subcategory generated by the modules $A_0 \otimes V$. It follows that

$$L_0 G_0 C_x \simeq C_x, \quad G_0 C_x \in \mathrm{Perf}^H(B^0). \quad (6.8)$$

For every B^0 -module N , applying L_0 induces an isomorphism

$$\mathrm{Hom}_{B^0, H}^\bullet(G_0 C_x, N) \xrightarrow{\sim} \mathrm{Hom}_{R_-, H}^\bullet(C_x, L_0 N). \quad (6.9)$$

Indeed both sides commute with colimits in N , by compactness of $G_0 C_x$ and C_x , and the assertion holds on the perfect free generators by the unit calculation. The same argument holds with any finite-dimensional representation tensored onto C_x . Applying (6.9) to $N = G_0 M$ and using adjunction

shows that the counit

$$L_0 G_0 M \longrightarrow M \quad (6.10)$$

induces an isomorphism on all such tests. Write D_M for its cone. The representation tests detect rational internal Hom, so $\underline{\mathrm{RHom}}_{R_-}(C_x, D_M) = 0$. Koszul self-duality, up to an invertible representation and a shift, gives

$$C_x \otimes_{R_-}^{\mathbf{L}} D_M = 0. \quad (6.11)$$

We explain the consequence of (6.11) for maps into a coherent object M' . Forget H -equivariance temporarily while retaining the central m -grading. If $x_1, \dots, x_n$ are linear coordinates, the finite Koszul module on $x_1^a, \dots, x_n^a$ is obtained by successive triangles from the Koszul module on the x_i 's. Its tensor product with D_M therefore also vanishes. The usual Koszul description of local cohomology as the colimit over these powers shows that the local cohomology of D_M on (x) vanishes. Equivalently, D_M is the totalization of its finite Čech localization complex, whose terms are localizations at nonempty products of the x_i . Each term is a module on which at least one coordinate x_i is invertible.

For such a coordinate x , every m -equivariant map from an $R_-[x^{-1}]$ -module to M' vanishes in the derived category. It suffices to check the free localized modules with every weight shift, since these generate the unbounded localized module category under colimits and derived Hom into M' carries colimits to limits. Fix a weight a for the free generator. The localization is the telescope whose j th free generator represents x^{-j} and has weight $a - 2j$. Derived Hom from this telescope is the derived inverse limit of the corresponding weight complexes of M' . The cohomology of M' is bounded below in m -weight, uniformly over its finitely many nonzero cohomological degrees: its cohomology is finite over an algebra generated over weight-zero A_0 in weight 2. Those weight complexes are therefore acyclic for all sufficiently large j . Their derived inverse limit is zero. Applying this to the finite Čech complex gives

$$\mathrm{Hom}_{R_-,m}^{\bullet}(D_M, M') = 0. \quad (6.12)$$

This is the graded completion argument: although the counit need not be an isomorphism on all unbounded modules, its difference is invisible to bounded coherent targets with these positive weights. If $W = 0$, the Koszul module is R_- itself and the cone is already zero, so no localization is needed.

The vanishing (6.12) also holds with H -equivariance. We spell out why averaging is valid even for the unbounded resolutions just used. Choose equivariant semifree resolutions; after forgetting equivariance they are still semifree. A nonequivariant null-homotopy of an equivariant map can be averaged using the invariant integral on $\mathcal{O}(H)$. For each vector v , its H -span is finite-dimensional, and the H -span of the image of that finite-dimensional space under the homotopy is again finite-dimensional. Hence evaluating the conjugated homotopy on v gives a regular finite-dimensional matrix-valued function on H , to which the integral applies. The resulting pointwise definition preserves module linearity, since each conjugated homotopy is module-linear. It also preserves the homotopy identity and is H -equivariant. Thus forgetting equivariance is injective on the mapping groups in question.

Apply this vanishing to the counit triangle. For coherent M, M' it gives

$$\mathrm{Hom}_{R_-,H}^{\bullet}(M, M') \simeq \mathrm{Hom}_{R_-,H}^{\bullet}(L_0 G_0 M, M') \simeq \mathrm{Hom}_{B^0, H}^{\bullet}(G_0 M, G_0 M').$$

Shear is an equivalence of the graded module calculi, so the coherence already proved yields (6.5) and its full faithfulness. The displayed resolution is A_0 -linear and commutes with constant representation tensors, proving the remaining compatibility before shear. Finally, Ind-extension of a fully faithful exact functor between these small coherent categories is fully faithful. This last assertion concerns their Ind-categories; it does not identify them with an unbounded quasi-coherent category. $\square$

6.3 The image of the flag object

We apply Proposition 6.3 to the Springer object P_N of the coherent trace. The first step is to identify its flag space over the slice; the second is a calculation with the same augmentation resolution.

The components of the t -fixed flag variety are indexed by finitely many types b . Each is an H -flag variety. Over its component put

$$U_b = \{(B, u) : B \text{ has type } b, u \in U_H \cap B_H\}, \quad B_H = B \cap H. \quad (6.13)$$

Here B denotes a Borel subgroup of L and B_H a Borel subgroup of H . For such a B , let

$$W_B = W \cap \mathfrak{n}_B, \quad E_{+,B} = E_+ \cap \mathfrak{n}_B.$$

These form vector bundles on the component, since they are the fixed t -eigensubbundles of the universal nilradical bundle. They are preserved by $u \in B_H$.

Lemma 6.4. *Over U_H , the flag space defining P_N is the disjoint union, over b , of the derived schemes*

$$\{(B, u, N) : (B, u) \in U_b, N \in W_B, (\text{Ad}(u) - 1)N = 0 \text{ in } W_B\}. \quad (6.14)$$

The base U_b is smooth of dimension $\dim H$. All derived structure in (6.14) is in its displayed vector equation.

Proof. A Borel containing tu contains its semisimple part $(tu)_{\text{ss}}$. The transverse invertibility in Lemma 6.1 places the identity component of its centralizer inside H : on $\mathfrak{g}/\text{Lie } H$, the semisimple part of $\text{Ad}(tu)$ has no fixed vector, as follows by Jordan decomposition from invertibility of $1 - \text{Ad}(tu)$. Choose in the Borel a maximal torus containing $(tu)_{\text{ss}}$. It lies in this connected centralizer and is a maximal torus of H . Since t is central in connected H , it belongs to that torus. Thus every Borel containing tu also contains t and u . The same holds for $z^{-1}tu$, since the center of L belongs to every Borel. Set-theoretically the flag incidence is therefore the union of the U_b 's.

This identification is scheme-theoretic. Before imposing the vector equation, the incidence is the group Grothendieck resolution $L \times^B B$, a smooth scheme. Pulling it back along the strongly étale map of Lemma 6.1 gives another smooth scheme. The induction presentation identifies it, by smooth H -torsor descent, with the induced incidence over U_H . Each U_b is itself smooth of dimension $\dim H$: it is the corresponding open part of the group Grothendieck resolution for H . Its inclusion in this incidence is closed, and the set-theoretic identification above exhausts the incidence. The smooth reduced structures therefore coincide. In particular the flag condition has introduced no derived excess.

Inside the nilradical, decompose the vector equation into its t -eigenbundles. The nonresonant blocks are invertible, so removing their coordinate-equation pairs leaves exactly (6.14), with the equation valued in W_B . $\square$

Let $\tilde{Y}_b$ denote the derived scheme

$$\tilde{Y}_b = \{(B, u, e) : (B, u) \in U_b, e \in E_{+,B}, (\text{Ad}(u) - 1)e = 0 \text{ in } E_{+,B}\}. \quad (6.15)$$

The last equation is imposed derivedly, even though the ambient Y is classical. Projection defines a proper map $\pi_b : \tilde{Y}_b \rightarrow Y$. Set

$$P_b = R\pi_{b*} \mathcal{O}_{\tilde{Y}_b}, \quad P = \bigoplus_b P_b. \quad (6.16)$$

Proposition 6.5. *The functor $\mathcal{K}$ sends P_N to P , without a shift or an equivariant line twist. It sends $\mathbb{D}P_N$ to the same object, using the self-duality of P_N in the coherent trace model.*

Proof. We compute first over U_b . Use $R_{-,B}$ for the analogue of R_- with W_B in place of W , and B_B^0 for its Koszul dual. Restriction of coordinate functions gives $(R_-)_{U_b} \rightarrow R_{-,B}$, whereas the inclusion $W_B \subset W$ gives $B_B^0 \rightarrow (B^0)_{U_b}$. Both are maps of dg algebras: u preserves W_B , so the equations restrict along the same inclusion as the coordinates. Restricting (6.6) and taking Hom shows that the transform of the flag summand before direct image is

$$(B^0)_{U_b} \otimes_{B_B^0}^{\mathbf{L}} \underline{\mathrm{RHom}}_{R_{-,B}}(\mathcal{O}_{U_b}, R_{-,B}). \quad (6.17)$$

Indeed in the free Hom complex the coefficient terms involving x and θ only involve the generators attached to W_B . The remaining derivative generators give extension of scalars. The factorial identification of the dual polynomial powers in (6.6) makes this an identification of modules with their B^0 -actions, not merely of cohomology groups. Locally split the vector-bundle inclusion $W_B \subset W$. To construct $(B^0)_{U_b}$ from B_B^0 , first adjoin the complementary closed polynomial generators β , and then the complementary exterior generators η . The latter have differentials linear in the already present β 's. This is a semifree, hence K-flat, extension; no invariant splitting of W_B is needed.

The last factor in (6.17) is particularly simple, and we record its shifts explicitly. Locally over U_b write $C = \mathcal{O}_{U_b}$, $F = W_B$, $n = \mathrm{rk} F$, and $S = \mathrm{Sym}_C(F^*)$. As a dg S -algebra, $R_{-,B} = S \otimes_C \mathrm{Sym}(F^*[1])$ is the Koszul complex on its vector equations. Its exterior-product pairing gives

$$\underline{\mathrm{RHom}}_S(R_{-,B}, S) \simeq R_{-,B} \otimes_C \det(F)[-n]. \quad (6.18)$$

This is Koszul self-duality and does not require the vector equations to be a regular sequence. On the other hand the coordinate sequence cutting out $x = 0$ in S is regular, so

$$\underline{\mathrm{RHom}}_S(C, S) \simeq C \otimes_C \det(F)[-n]. \quad (6.19)$$

Derived coinduction adjunction and (6.18) give

$$\begin{aligned} \underline{\mathrm{RHom}}_{R_{-,B}}(C, R_{-,B}) \otimes_C \det(F)[-n] &\simeq \underline{\mathrm{RHom}}_{R_{-,B}}(C, \underline{\mathrm{RHom}}_S(R_{-,B}, S)) \\ &\simeq \underline{\mathrm{RHom}}_S(C, S) \simeq C \otimes_C \det(F)[-n]. \end{aligned}$$

Canceling this identical invertible factor gives

$$\underline{\mathrm{RHom}}_{R_{-,B}}(C, R_{-,B}) \simeq C. \quad (6.20)$$

The pairings and adjunction are intrinsic to the vector bundle, so these local formulas glue equivariantly. They cancel both the cohomological shift and the determinant character. In particular the sole cohomology in (6.20) has degree and m -weight zero.

After shear, B_B^0 becomes a connective equation algebra: its generators β have degree zero and weight -2 , and its generators η have degree -1 . The shear of (6.20) still has only C in degree zero. The standard t-structure for this connective algebra therefore identifies it with its H^0 -module. Weight considerations force every β to act by zero, because C has only weight zero. Thus it is precisely the zero-section module C over the sheared algebra, including its module structure.

The K-flat extension in (6.17) now sets the generators from W_B equal to zero. Its remaining coordinates are dual to W/W_B , so its vector space is the annihilator $W_B^\perp \subset W^* = E_+$. The induced differential is the transpose equation on this annihilator. As before, changing u^{-1} to u is an invertible change of equation generators. Finally,

$$W_B^\perp = E_+ \cap \mathfrak{n}_B = E_{+,B}. \quad (6.21)$$

To verify this, the invariant form pairs different t -eigenspaces only when their eigenvalues are inverse. An element of E_+ therefore annihilates W_B exactly when it annihilates all of $\mathfrak{n}_B$. Since $\mathfrak{n}_B^\perp = \text{Lie } B$, this says it belongs to $E_+ \cap \text{Lie } B$. The action of t on $\text{Lie } B/\mathfrak{n}_B$ is trivial, whereas its eigenvalue on E_+ is $r \neq 1$; hence this intersection is already in $\mathfrak{n}_B$. The algebra remaining in (6.17) is consequently the derived equation algebra of (6.15).

It remains to commute this calculation with direct image. Each free term of the augmentation resolution satisfies the projection formula. For a bounded input model only finitely many resolution powers contribute to a fixed un-regraded Hom degree. The proper flag projections have bounded cohomological dimension, so these termwise projection formulas give the derived Hom comparison before shear. Shear also commutes with this direct image: m acts trivially on U_b , the complexes are direct sums by weight, and quasi-coherent pushforward commutes with these sums. Alternatively, a finite affine cover of the flag base computes pushforward by a Čech complex of bounded length, for which the weightwise assertion is immediate. Thus direct image of the algebra just calculated is exactly $R\pi_{b*}\mathcal{O}_{\tilde{Y}_b}$, with no shift. Summing over the disjoint flag components proves the first assertion. The second follows from $\mathbb{D}P_N \simeq P_N$ in the coherent trace model. $\square$

6.4 Recovering the spherical summand

The Koszul calculation has identified the Iwahori test object P . We next identify its spherical retract. This requires comparing endomorphisms with representation coefficients, since invariant endomorphisms alone would not determine the module action needed by the global support. In particular, B_{sph}^0 comes from the spherical trace, whereas B^0 acts through the independent Koszul construction. Their common equation scheme does not identify these actions. We first reconstruct the internal algebra of the retract, then prove that the retract is an equivariantly trivial line.

Recall the spherical retract P_N^S from Proposition 5.3, and define

$$L_S^0 = G_0(\mathbb{D}P_N^S), \quad L_S = \text{sh}(L_S^0).$$

Here sh is the change of degree by m -weight described above. Duality enters because a covariant global trace functional is represented in the form $\text{RHom}(\mathbb{D}[J], F_N)$, for a representing Ind-object F_N constructed in (7.5): applying $\mathbb{D}$ reverses the test morphism, and the first variable of RHom reverses it back. Thus the tests to which we apply G_0 are the duals of the trace objects. Since $\mathbb{D}P_N = P_N$, the object L_S is a retract of $P = \bigoplus_b P_b$. Figure 2 records this retract and the one that we obtain by identifying its spherical term.

Lemma 6.6 (Polynomial representatives on the slice). *Put $C = \mathcal{O}(L)^L$ and $C' = \mathcal{O}(U_H)^H$, with the map $C \rightarrow C'$ given by the spherical coordinate $s_{\text{sph}} = tu$. Write $B_{\text{sph}} = \text{sh } H^\bullet(B_{\text{sph}}^0)$ for the classical spherical equation algebra, retaining polynomial degree as the original cohomological degree divided by two. For neutral representations V, V' the restriction and quotient map*

$$(C' \otimes_C (\mathcal{O}(L) \otimes \text{Sym}(\mathfrak{g}^*) \otimes V^* \otimes V'))^L \longrightarrow (B_{\text{sph}} \otimes V^* \otimes V')^H \quad (6.22)$$

is surjective and preserves polynomial degree. Every resulting class can be expressed, over C' , by homogeneous Satake morphisms, slides on neutral representations, and equivariant contractions. The transferred expressions respect tensoring by neutral spectators. When there is no Lie-variable factor, transfer identifies the holonomy matrix map with the same matrix map on the coherent side.

Proof. Consider the ordinary polynomial equation algebra

$$S_r = \frac{\mathcal{O}(L) \otimes \text{Sym}(\mathfrak{g}^*)}{(\text{Ad}(s_{\text{sph}})e - re)}.$$

The strong slice identifies the base-changed holonomy presentation with $L \times^H (tU_H)$. Adding the representation $\mathfrak{g}$ gives its induced vector bundle. Removing the nonresonant vector coordinates in the equation leaves precisely the E_+ equation algebra. Consequently

$$[\mathrm{Spec}(C' \otimes_C S_r)/L] \simeq [\mathrm{Spec} B_{\mathrm{sph}}/H].$$

The map from the polynomial algebra to S_r is a surjection, homogeneous in the Lie-variable degree. After the flat base change, exactness of reductive invariants and the induced presentation prove (6.22), separately in each polynomial degree. This explains both the equivariant lift from the slice and the extension of polynomials from E_+ to the full Lie algebra.

Shrink so that the image of $\mathrm{Spec} C'$ lies in the invariant open of Lemma 3.1. Its central-torsor property gives

$$C' \otimes_C \mathcal{O}(L) \simeq C' \otimes_{\mathcal{O}(A)^A} \mathcal{O}(A).$$

Thus neutral matrix coefficients generate the required holonomy functions over C' . Before base change one may equivalently use neutral matrix coefficients together with full invariant holonomy functions. Denominators from further invariant shrinking are absorbed into C' .

Expand a lifted equivariant matrix polynomial into these holonomy coefficients and polynomial Lie coefficients. Retain the finite representation coefficients in each term, take tensor products, and contract by ordinary equivariant maps, evaluations and coevaluations. Exactness of invariants ensures that these contractions also span the equivariant maps. The Lie-polynomial terms are actual homogeneous morphisms in derived Satake. A matrix coefficient attached to a neutral representation W is realized by first introducing $W^* \otimes W$ in such a Satake morphism, then sliding W , and evaluating. Every term is a cycle of the spherical Koszul model, since it uses no odd equation generator. There are no further cohomology classes to represent: after shear the equation algebra is classical by Lemma 6.2.

Proposition 5.3 compares exactly these operations and their tensor products with neutral spectators. In degree zero in the Lie variables they consist solely of holonomy matrices and ordinary representation maps; hence their transferred maps are the same holonomy matrices. Subsequent duality transposes those maps. $\square$

Proposition 6.7. *After shrinking the saturated slice U_H around the identity class, there are an isomorphism*

$$L_S \simeq \mathcal{O}_Y \tag{6.23}$$

and an H -equivariant isomorphism of graded algebras over $\mathcal{O}(U_H)$

$$\mathrm{sh} H^\bullet(B_{\mathrm{sph}}^0) \xrightarrow{\sim} \underline{\mathrm{End}}_Y^\bullet(L_S) = \mathcal{O}(Y). \tag{6.24}$$

The isomorphism is induced by spherical transfer, duality, and G_0 . It respects the representation-valued operations of Proposition 5.3. As an isomorphism between the two equation models over U_H , it preserves the open orbit and its boundary over every unipotent $u \in U_H$.

Proof. We first reconstruct the internal endomorphism algebra without assuming that L_S is a line. The full corner embedding of Lemma 5.1, the retract (5.5), and the spherical loop calculation of [Ope26, Proposition 2.7, “Spectral form of a Frobenius trace”] give

$$(H^\bullet(B_{\mathrm{sph}}^0) \otimes V^* \otimes V')^H \simeq \mathrm{Hom}_{B^0, H}^\bullet(L_S^0 \otimes V'^*, L_S^0 \otimes V^*) \tag{6.25}$$

for neutral L -representations V, V' , restricted to H . The order on the right is reversed by $\mathbb{D}$; duality retains the displayed Hom degree. To justify base change in this formula, first make the comparison

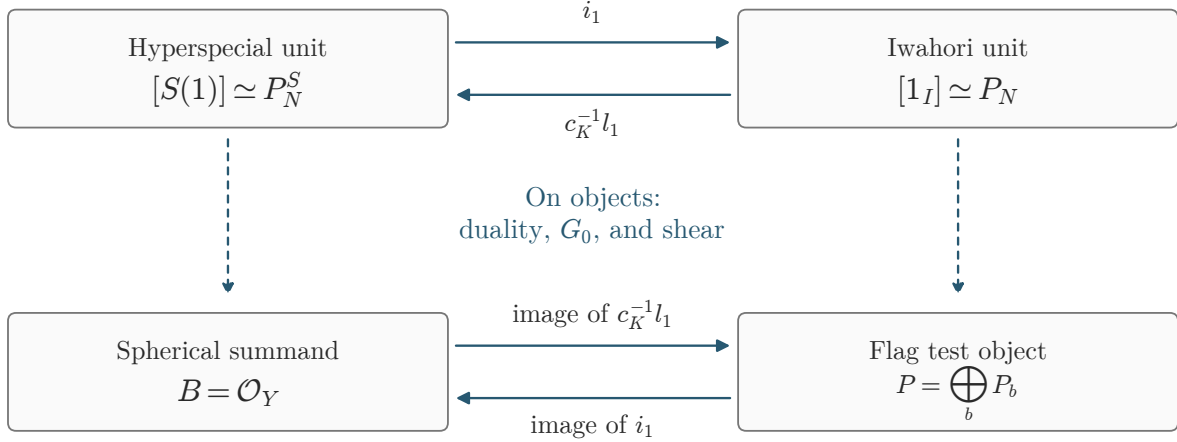


Figure 2: The hyperspecial unit is a retract of the Iwahori unit. After coherent duality, the Koszul functor G_0 , and shear, this becomes the retract $B = \mathcal{O}_Y$ of the flag object P . The dashed arrows record the objects corresponding under these operations; duality reverses the solid arrows. The comparison of representation-valued maps is (6.25), and the resulting internal algebra comparison is (6.24).

on the two retracts in the trace. Their invariant coordinate rings agree by (5.3). The étale extension is flat and therefore tensors the cohomology of their mapping complexes. On the coherent side these complexes can be computed after pullback to the slice, using degreewise finite free resolutions against bounded targets and exact reductive invariants. The same comparison with compact source passes to Ind-extensions by filtered colimits. Finally G_0 is $\mathcal{O}(U_H)$ -linear and respects representation tensors before shear; duality takes constant tensors to their duals. These facts give (6.25) with the stated coefficients.

To recover the internal algebra, we need naturality of this comparison for representation-valued maps, not just its displayed vector-space isomorphisms. Lemma 6.6 supplies the required representatives and proves their spectator compatibility. Its surjection is homogeneous for the original polynomial degree, so it applies to every cohomological degree in (6.25) before the shear as well.

In particular the comparison takes matrix maps with coefficients in $\mathcal{O}(U_H)$ to the same matrix maps, transposed by duality. This includes constant H -maps between restrictions of L -representations: the strong slice expresses them as equivariant matrix functions on the induced L -presentation. Every representation of $H/Z(L)$ occurs as a summand of a restricted representation of A . Indeed matrix coefficients for the closed subgroup $H/Z(L) \subset A$ are quotients of those for A , and complete reducibility extracts their irreducible constituents. The endomorphisms at issue have trivial $Z(L)$ -sector. We may therefore use all these representation summands in (6.25).

Testing a rational H -module against all finite-dimensional representations determines it. Applying this observation to (6.25) identifies $H^\bullet(B_{\text{sph}}^0)$ with the H -internal endomorphism cohomology of L_S^0 . To recover multiplication, tensor the first tested map with the representation coefficients of the second and compose. The spectator comparison proves that this is exactly the tensor pairing defining internal composition. Coefficients from $\mathcal{O}(U_H)$ give its scalar action by the same argument. Duality may reverse multiplication, which is harmless for the commutative spherical ring. Thus this is an algebra comparison over $\mathcal{O}(U_H)$, not merely a vector-space comparison of invariant Hom's.

After shear it gives (6.24), apart from the line assertion. Endomorphism degree changes by its m -weight. Only even weight sectors occur: the flag formula for P , and hence its retract L_S , has this parity. There is no completed internal Hom hidden in this step. After shear we have coherent modules on the affine scheme Y with reductive equivariance; bounded coherent targets and resolutions finite in each degree compute the ordinary internal endomorphism cohomology with its rational H -action.

We have proved that the internal endomorphism cohomology of L_S is commutative and concentrated in degree zero. We next use the complete intersection to deduce that L_S is a line. Let

$$i : Y \hookrightarrow U_H \times E_+$$

be its regular equation embedding. The ambient scheme is smooth, so i_*L_S is perfect there and $Li^*i_*L_S$ is perfect on Y . The bimodule cohomology filtration of the derived tensor square of $\mathcal{O}_Y$ over the ambient ring has regular-sequence Tor terms. Tensoring it with L_S gives a finite filtration whose quotients are L_S tensored with exterior powers of the equation space, shifted by $[j]$, for $0 \leq j \leq \dim E_+$; its last quotient is the unshifted L_S . Locally these exterior factors are free. Positive self-Ext vanishing splits off the last quotient: the successive obstruction groups have strictly positive degree. Hence L_S is locally a summand of a perfect complex, and is perfect.

A minimal presentation over a local ring of a nonzero perfect complex with two occupied degrees would give positive self-Ext: in the top degree of the Hom complex, the last differential has entries in the maximal ideal, so its cokernel remains nonzero modulo that ideal. Thus L_S is locally a vector bundle in a single degree. Commutativity of its endomorphism algebra forces its rank to be at most one. The scheme Y is connected, since m contracts it to the zero section over connected U_H . The nonzero retract L_S therefore has rank one throughout, with a locally constant shift.

We now remove the shift and trivialize the line equivariantly. Take u in a maximal torus of H , regular both for H and for tu in L , and remove the further resonances where $\text{Ad}(u) - 1$ is singular on E_+ . On this dense torus locus the flag formula for P is an ordinary etale cover. Its fibers have trivial stabilizer-torus action and lie in degree zero. The retract L_S consequently has zero shift and trivial torus character there. Along the torus with $e = 0$ this character is locally constant, so the fiber at $(1, 0)$ has trivial torus character. Since H is connected, its one-dimensional character is then trivial.

Reductivity now gives an invariant section generating at $(1, 0)$. The affine quotient of Y is the quotient of U_H , since the m -weights contract all e -directions. Every orbit over the identity class has $(1, 0)$ in its closure up to the unique closed orbit in that fiber. Thus an invariant section generating at $(1, 0)$ generates over this entire quotient fiber. Its nongenerating locus is closed and invariant, and its image in the affine quotient is closed. Shrinking the saturated neighborhood removes that image. The section trivializes L_S , proving (6.23) and the last equality in (6.24).

It remains to verify the geometric compatibility of the ring comparison. It need not fix the E_+ coordinates literally, but it is H -equivariant over U_H . At $u = 1$ its induced automorphism preserves the unique open H -orbit in E_+ and hence the boundary. Suppose u is unipotent and e lies in that open orbit. The stabilizer H_e is reductive, so u can be contracted to 1 inside H_e . If the image of (u, e) under the coordinate comparison had its vector in the boundary, equivariance and closedness of the boundary would force the image of $(1, e)$ into the boundary as well, a contradiction. Applying the same argument to the inverse gives the asserted preservation in both directions. $\square$

6.5 The local test available to global cohomology

We collect the constructions in the form used in the remaining sections. In the statement, the equation embedding has conormal bundle obtained from the finite H -representation E_+^* ; this is the

source of the uniform finite filtrations used below.

Theorem 6.8. *Let t , $H = Z_L(t)$ and m be associated with a selected place as in Lemmas 3.1 and 3.2. After an etale base change on invariant holonomy functions and a saturated shrinking about the identity class, the Iwahori trace has a fully faithful coherent realization. Its composite with coherent duality, the covariant Koszul functor G_0 , and the change of degree by m -weight is contravariant and has the following properties.*

Its unit corresponds to

$$P = \bigoplus_b P_b \in \mathrm{Coh}^H(Y), \quad Y = \{(u, e) \in U_H \times E_+ : \mathrm{Ad}(u)e = e\},$$

where the P_b are the derived flag direct images in (6.16). The scheme Y is a classical complete intersection in the smooth affine scheme $U_H \times E_+$. The hyperspecial unit is the retract $B = \mathcal{O}_Y$ of P under (6.23). The unsheared representation comparisons are (6.25); their internal algebra and scalar actions are (6.24). Before duality the slide on a label V is the matrix $V(tu)$.

The ordinary H -equivariant algebra $\mathrm{End}(P)^{\mathrm{op}}$ is the affine Hecke algebra with parameter r , pulled back along the etale map on its center. It has no higher cohomology, and its spherical corner is $\mathrm{End}(B) = \mathcal{O}(U_H)^H$.

All these comparisons respect the matrix operations, homogeneous spherical maps, and neutral spectators of Proposition 5.3. They permit tests by every finite representation of $H/Z(L)$. In the even sectors used below, a test of m -weight w changes cohomological degree from b to $b + w$; the other sectors use the stated parity adjustment. The ring comparisons preserve the open orbit and boundary over unipotent u . The covariant coherent and Koszul realizations extend fully faithfully to Ind-completions; duality is applied to the compact test objects.

The same statements hold for finite products of selected places, with external test objects, separate parity sectors, and the sum of the m -weights.

Proof. The assertions at one place have been proved above and in Proposition 5.3. On the chosen slice $s = z^{-1}tu$, so the slide there is $V(zs) = V(tu)$. Tensoring with a representation commutes with the Koszul functor before shear; duality replaces the representation by its dual. An m -weight w coefficient is shifted by $[-w]$ under shear, which gives the stated degree change for its Hom test. The ring and the flag objects have even weights. On an odd sector the additional subtraction of one in the degree convention preserves even shifts, as in the definition of shear. The covariant Ind-extensions are fully faithful because they are the colimit extensions of fully faithful functors on the coherent compact objects. Duality remains on these compact test objects.

For the endomorphism assertion, start with Lemma 5.2, make the flat etale base change, and apply the fully faithful Koszul comparison and duality. Duality reverses composition. Equivariant maps have m -weight zero, so shearing does not change their degree. This proves the stated identification with $\mathrm{End}(P)^{\mathrm{op}}$. The spherical corner is $\mathcal{O}(Y)^H$ by (6.23); its equality with $\mathcal{O}(U_H)^H$ follows from the nonzero m -weights of the vector coordinates.

For products, coherent Hom's of external objects satisfy Künneth: use affine presentations, bounded coherent targets, resolutions with finite terms in each degree, and exact reductive invariants. The constructible kernel products have the same equivariant Künneth description, also obtainable from their orbit filtrations. The cut and coend calculations tensor together. Thus the stable idempotent-complete subcategory generated by external objects is fully faithfully realized in coherent objects on the product. This subcategory suffices for all subsequent tests; no assertion that every coherent object is an external product is needed. Shearing adds the weights from the different factors and keeps their parity sectors separately. This proves the product statement and its Ind-extension. $\square$

7 A simultaneous lower bound

The local models of Section 6 change a cohomological degree by a weight of the cocharacters m_i . We now prove that, after localizing at the Hecke character of f , these changed degrees are nonnegative. The bound will hold for every finite collection of the places chosen in Lemma 3.1, with the same lower bound zero. This uniformity permits the simultaneous tests in Section 8.

Fix distinct places $x_1, \dots, x_n$ from that sequence, with $n \geq 1$, and put

$$T_I = \{x_1, \dots, x_n\}, \quad U' = U \setminus T_I, \quad \Gamma' = W_{U'}, \quad d_i = \deg(x_i).$$

At the places of T_I we use Iwahori level; elsewhere we retain the full level D . Let K_I be the fixed-place path functor at this level. As before, $y_0 \in U \setminus T_I$ is the auxiliary place, and C is its full spherical Hecke algebra. Write $\mathfrak{m}_C$ for the maximal ideal of C determined by f . All absolute values in this section are taken in one fixed complex realization.

For neutral representations $V_i \in \text{Rep}(L)$, set

$$N_b(V) = H^b K_I(\boxtimes_{i=1}^n Z(V_i)), \quad V = \boxtimes_{i=1}^n V_i. \quad (7.1)$$

There is one nontrivial label at the selected geometric point of each Frobenius cycle and a unit label at its other points. If D_* is divisible by all the d_i , *complete Frobenius through degree D_** means the product of the full-cycle slides, with their Weil structures, raised in the i th factor to the power D_*/d_i .

The numerical comparison behind the bound is already visible in the local model. Retain its centralizer H_i and cocharacter m_i at x_i , and let $W_i \subset V_i^*$ be an irreducible H_i -summand in the even $m_i(-1)$ -sector. If w_i is its m_i -weight and $w = \sum_i w_i$, duality and the change of grading replace degree b by $b + w$. At the slice-center value $[u_i] = [1]$ for every i , the complete-Frobenius eigenvalues on this dual-label test have modulus $q^{-D_* w/2}$. We will prove the arithmetic inequality

$$|\alpha| \leq q^{D_* b/2}$$

after quotienting $N_b(V)$ by $\mathfrak{m}_C$ and localizing at the away Hecke values of f . A nonzero test satisfying both bounds must have $b + w \geq 0$. The next two subsections establish the arithmetic inequality by simultaneous degeneration and constituent purity; we then apply it to the coherent tests and pass to ordinary quasi-coherent complexes.

7.1 Disjoint Hecke actions and simultaneous degeneration

The path functor and moving-leg cohomology carry Hecke operations in two apparently different descriptions. Their agreement is needed before we use finite generation.

Lemma 7.1. *At a place carrying a unit path label, the spherical trace action on K_I agrees with the usual disjoint Hecke action on compact shtuka cohomology, with the Satake conventions fixed in Section 2. This agreement holds with arbitrary labels and full or parahoric levels at the other places.*

Proof. Both actions use the same modification correspondence and the Weil-stalk trace at the auxiliary place. One can check the normalization after temporarily imposing Iwahori level there. For a closed simple-wall kernel, the parallel-wall calculation of [Ope26, Proposition 2.6], applied at position the identity, gives pullback followed by summation over Frobenius-fixed rank-one flag refinements. This is the ordinary Hecke correspondence; subtracting the identity gives the simple generator. A length-zero kernel gives the graph of a unique transport, with coefficient one for its unshifted constant sheaf. These kernels generate the Iwahori Hecke algebra.

Transfer between Iwahori and hyperspecial level is similarly pullback and summation over Frobenius-fixed full flags, with the index normalization in [Ope26, Proposition 2.6]. All other modification labels are carried unchanged through these diagrams. Since there is no leg at the auxiliary place, the Frobenius-fixed flags are finite *étale* in families; a further finite level there trivializes them. Thus the calculation is compatible with the other levels and with compact direct image. $\square$

To compare all the central labels in (7.1) at once, we place their degenerations over a single rational point of a second curve. The following elementary choice of map is useful.

Lemma 7.2. *For disjoint finite sets $T_I, \text{supp}(D), \{y_0\} \subset |X|$, there is a finite generically separable morphism*

$$\pi : X \longrightarrow \mathbb{P}_{\mathbb{F}_q}^1$$

unramified over $0, 1, \infty$, such that T_I lies above 0 , $\text{supp}(D)$ lies above ∞ , and y_0 lies above 1 .

Proof. Take high powers of an ample line bundle and choose sections s, t with prescribed first jets at the indicated points. Prescribe simple zeros of s at T_I , simple zeros of t at $\text{supp}(D)$, and simple zeros of $s - t$ at y_0 , keeping the other necessary values nonzero. These conditions can be imposed over the residue fields, and force s, t , and $s - t$ to be nonzero sections. We seek a pair with no common zero and with each of these three sections having only simple zeros. Then $[s : t]$ is the required morphism.

Here is the finite-field existence argument. At any fixed finite set of additional closed points, Riemann–Roch makes the jet conditions independent once the degree of the line bundle is large. The local probability of satisfying the conditions is positive, including over residue fields with two elements: one can choose the two first jets with nonzero determinant. At a point of degree l outside the prescribed set, a forbidden double zero or common zero has probability $O(q^{-2l})$, with a constant independent of the line-bundle degree. Indeed, for a line bundle of degree m the dimension bound

$$h^0(\mathcal{L}(-2x)) \leq \max(0, m - 2l + 1)$$

gives this estimate for double zeros, and the analogous bound for $\mathcal{L}(-x)$ gives it for common zeros. The fixed jet prescriptions affect only the constant. If the required order of vanishing is impossible for a nonzero section, its probability is zero. Since the number of degree- l points is $O(q^l)$, the sum of the excluded tail probabilities tends to zero. The product of the positive local probabilities therefore has positive limit. A suitable pair exists. Its simple zero at any point of T_I also ensures generic separability. $\square$

Proposition 7.3. *For any finite T_I and neutral labels V_i , the groups $N_b(V)$ are finitely generated C -modules. There is a finite set of additional places which can be omitted from the away Hecke actions such that the following comparison holds. For a sufficiently divisible D_* , $N_b(V)$ identifies, compatibly with C , the remaining away Hecke actions, and complete Frobenius, with a direct summand of moving-leg cohomology at the same level over a geometric generic tuple in $(U')^n$. On the latter group complete Frobenius acts through a tuple in $(\Gamma')^n$ whose degree is D_* in every factor.*

Proof. We use the one-leg case of the characteristic-zero parahoric comparison of [Sal23, Theorem 5.2(1)]. Its setting allows a smooth affine group scheme over a curve with quasi-split reductive generic fiber, parahoric reduction at the degenerating point, and arbitrary full level away from that point, encoded by dilatation. For one moving Satake leg, with a stationary unit label omitted, the canonical comparison is an isomorphism

$$p_! \Psi \mathcal{S} \xrightarrow{\sim} \Psi p_! \mathcal{S}. \quad (7.2)$$

Here Ψ is the full trait nearby-cycle functor and $p_!$ is the filtered limit over Harder–Narasimhan bounds. The assertion is made on this limit itself: the fusion and partial-Frobenius constructions of [Sal23, Proposition 3.5, Theorem 3.8, and Lemma 4.2] apply to the resulting ind-constructible complexes. Creation of a dual pair of Satake legs, fusion of nearby cycles, and annihilation construct an inverse, whose two compositions with the canonical map are the tensor evaluation–coevaluation identities [Sal23, Lemma 4.6, Corollary 4.7, and proof of Theorem 5.2]. Thus this characteristic-zero comparison is established directly on the Harder–Narasimhan limit.

We use the canonical map in (7.2) for all equivariance statements. It respects the full local Galois action and the Weil descent data; its inverse consequently does too. This does not require separate local Weil equivariance of every auxiliary fusion map used to construct the inverse [Sal23, paragraph following Lemma 4.6 and Corollary 6.1]. All coefficients and labels can be taken over a sufficiently large finite extension E of $\mathbb{Q}_\ell$.

Choose π as in Lemma 7.2. Enlarge D to $\pi^*(m_\infty\infty)$ for m_∞ large enough. We will recover the original level by finite level-group invariants. Let $\mathcal{G}$ be the smooth group scheme on X with Iwahori fibers at T_I and hyperspecial fibers elsewhere, and take its Weil restriction to $\mathbb{P}^1$. This is smooth affine with connected geometric fibers and quasi-split semisimple generic fiber: restriction of a split Borel becomes a product of Borels over a separable closure. It is parahoric also at the branch places: under finite separable extension of complete discretely valued fields, the building and facet identification, with rescaled valuation, identifies the Weil restriction of a parahoric model with the corresponding parahoric model. This is the Weil-restriction property of Bruhat–Tits models; see [Kal21, Appendix F, Fact F.1], which allows extensions that are not tamely ramified. Connectedness can also be checked on the local fibers, where the additional congruence groups are geometrically unipotent. The full level $m_\infty\infty$ can be encoded by dilating this model at the identity along that divisor, as in [Sal23, Section 2]. This leaves the model at 0 unchanged. Thus the hypotheses of (7.2) hold, with degeneration at 0 and the enlarged full level at ∞ .

Over the strict local curve at 0, the cover $X \rightarrow \mathbb{P}^1$ splits into its geometric branches. On the branches selected by the x_i we place the labels V_i ; on the other branches we place the unit. To apply the global theorem, first induce this external tensor label from the connected dual group to its semidirect product with the finite splitting group. This amounts to taking its finite family of translates, with permutation descent. After restriction to the strict local curve, the original external tensor is a summand. Naturality of the canonical map (7.2) permits projection onto this summand, although that summand need not descend separately over the global good open. Increasing D_* makes Frobenius preserve every selected summand. The relative Satake normalization splits as the product of the normalizations on the branches; any common base shift or twist in a convention is compensated on both sides. Neutral labels have no component-parity correction.

For completeness, the integral comparison cone in this particular application also vanishes. Choose an $\mathcal{O}_E$ -Satake lattice in the induced label, and keep the full level at ∞ separate from the parahoric model. Let $\mathcal{C}_{\mathcal{O}_E}$ be the cone of (7.2) for that lattice, before taking finite level-group invariants or rational summands. Reduction modulo the uniformizer ϖ and extension of coefficients to E commute with this comparison: on its source, compute constructible trait nearby cycles and compact direct image on finite-type charts and then pass to the filtered Harder–Narasimhan limit; on its target, nearby cycles on the curve base itself are the geometric generic fiber. Apply the torsion case of [EX24, Theorem 2.2] with a stationary unit leg and constant coefficient on its restricted-shtuka factor at 0. Writing $k_E = \mathcal{O}_E/(\varpi)$, it gives

$$\mathcal{C}_{\mathcal{O}_E} \otimes_{\mathcal{O}_E}^L k_E = 0.$$

Multiplication by ϖ on this cone is consequently invertible, so

$$\mathcal{C}_{\mathcal{O}_E} \simeq \mathcal{C}_{\mathcal{O}_E} \otimes_{\mathcal{O}_E} E \simeq \mathcal{C}_E = 0,$$

where the last equality is the independent characteristic-zero comparison just proved. This argument applies to the actual one-leg Satake coefficient used here. It makes no completion assertion about general Hecke-finite modules. In the remainder of the proof all invariants and selected summands are taken over E , where they preserve the comparison isomorphism.

The resulting generic shtuka cohomology is the moving-leg cohomology on X along the branch tuple. Indeed torsors for the Weil restriction are equivalent to torsors on the finite cover. This can be checked étale locally on the base: over a strict henselian local base a smooth torsor on the finite cover trivializes *étale* locally. The equivalence works with arbitrary base schemes and commutes with the Frobenius pullback in the shtuka diagram. Modification data agree away from the branch locus and, near 0, by the split description. Level structures agree by definition. Consequently these identifications commute with an exhaustion by finite-type opens.

On the special fiber, the local-model morphism for shtukas is smooth after sufficient truncation on each bounded modification. Nearby cycles therefore pull back from the Beilinson–Drinfeld Grassmannian for the parahoric group scheme. A spherical modification degenerating to an Iwahori place gives the central sheaf $Z(V_i)$, with unit flag label, by its nearby-cycle construction [Ope26, Theorem 2.3]. If Z is defined using unipotent nearby cycles, retain that summand. The Künneth isomorphism for nearby cycles over a trait, on bounded supports, tensors these calculations across the split branches. Although the trait has diagonal inertia on the product, the factorwise unipotent projections are still defined. They are stable under a sufficiently divisible Frobenius power. This gives precisely the central coefficients in (7.1), with its path normalization, as a summand of the comparison.

The special-fiber Frobenius in this construction is the product of the slides in (7.1), by cyclicity of the path functor [Ope26, Proposition 2.5, Coherent path rotation]. A change of local Frobenius lift does not change its spectrum. On the unipotent nearby-cycle summand, inertia acts through tame logarithmic monodromy N , and Frobenius conjugates N by a nontrivial power of q or its inverse. Multiplying a Frobenius lift by $\exp(cN)$ thus conjugates it by another exponential of N . The same argument works separately on the factors and commutes with disjoint Hecke operations. The branches give the same strict local curves as the coordinates used earlier to construct the path functor.

We may omit from the away Hecke actions all places over 0 and ∞ . Every remaining Hecke correspondence is disjoint from the degeneration; it is finite *étale* after the usual level refinement and carries the modification coefficient along unchanged. Hence (7.2) is equivariant for it, in particular for C at y_0 . Taking finite level-group invariants at ∞ now recovers the original level D . These invariants are exact and commute with the comparison.

To make the passage from the branch tuple to independent moving legs explicit, let ξ be a geometric generic point of the trait at 0, and let $\alpha : \xi \rightarrow (U')^n$ be its selected branch tuple. This tuple need not be generic in the product. Choose a geometric generic specialization $\bar{\eta}_n \rightarrow \alpha$ in $(U')^n$. The specialization isomorphism of [Xue20b, Theorem 4.2.3] identifies the fiber of each ind-smooth cohomology sheaf at α with its fiber at $\bar{\eta}_n$. On the latter fiber, [Xue20a, Theorem 2] applies: its restriction excluding nonzero Frobenius graphs is satisfied at this generic point. This proves finite generation over C .

These identifications also retain the actions. For any finite collection of away Hecke operations, restrict the leg positions to the open avoiding their places. Both α and $\bar{\eta}_n$ lie in this open, and specialization is natural for the Hecke morphisms there. Taking the filtered union handles all

the away operations, and hence their quotients and localizations. By the product Weil action in [Xue20b, Proposition 5.0.4], recalled in Lemma 2.1, the local Weil action pulled back along α is the product of its coordinate branch actions. Because π is unramified over 0, after D_* fixes the branches the i th action is a local lift of $\text{Frob}_{x_i}^{D_*/d_i}$. Its degree is therefore D_* . Changing the paths used in specialization conjugates these lifts and leaves their degrees unchanged. This proves every assertion. $\square$

7.2 The constituents after imposing Hecke values

Proposition 7.3 does not by itself transfer a weight bound from ordinary fibers to nearby cycles. We obtain that bound by first identifying the possible irreducible Weil constituents after imposing the Hecke values of f .

Fix b and work at the Iwahori level above. In the regular module M_b^{reg} of Section 2, first quotient by $\mathfrak{m}_C$ and then localize at the joint neutral spherical Hecke values of f outside the finite exceptional set of Proposition 7.3. Denote the result by M' . Each of its A -isotypic pieces is finite dimensional. On each such piece the localization simply retains the corresponding joint primary summand. The neutral algebra at a place is $k[A]^A$, acting by the Hecke identity. Define the finite-dimensional $(\Gamma')^n$ -module

$$D_b = \left(M' \otimes \bigotimes_{i=1}^n V_i \right)^A.$$

By (2.1) and exactness of A -invariants, D_b is the generic moving-leg cohomology with labels V_i after the same C -quotient and away Hecke localization. Proposition 7.3 therefore identifies the corresponding reduction of $N_b(V)$ with a direct summand of D_b , equivariantly for the complete Frobenius given there by a tuple of local Weil lifts. It suffices to control the constituents of D_b to bound the spectrum on that summand.

Proposition 7.4. *Every irreducible constituent of D_b occurs among the constituents of*

$$\boxtimes_{i=1}^n (V_i \circ \sigma_{\nu, \text{ad}}) \big|_{(\Gamma')^n}. \quad (7.3)$$

This assertion concerns occurrence of constituents and does not prescribe their multiplicities.

Proof. Let $\mathcal{B}_{\Gamma'}$ be the coordinate algebra of $\text{Hom}(\Gamma', A)$ used in Section 2, with Γ' regarded as an abstract group. It acts on M' . Choose finitely many coefficient vectors of a basis of D_b , take their A -spans, and let $M'' \subset M'$ be the $\mathcal{B}_{\Gamma'}$ -module they generate. Put

$$R' = \mathcal{B}_{\Gamma'} / \text{Ann}(M'').$$

This algebra is of finite type. To see this, choose finitely many representations whose matrix entries generate $k[A]$. Their matrix entries, as the Weil element varies, carry the chosen generating vectors into a fixed finite list of A -isotypics. Those isotypics are finite dimensional after the C -quotient. Thus the span of these entries in R' is finite dimensional: vanishing of an operator is tested on the generating vectors and their A -spans, since all coordinate operators commute. Finitely many entries therefore generate R' .

At any k -point of R' , the universal matrices specialize to a homomorphism $b' : \Gamma' \rightarrow A(k)$. This homomorphism is continuous and defined over a finite extension of $\mathbb{Q}_\ell$. Indeed the entries are finite-dimensional linear evaluations of their actions on the generating spaces. Testing those actions by (2.1) uses only a fixed finite list of representation labels. The resulting finite-dimensional quotients carry the continuous partial Weil actions of Lemma 2.1. The chosen primary summands

are determined, on these finitely many spaces, by finitely many Hecke equations. Enlarging the coefficient field accommodates these equations and the finitely many coordinates of the k -point.

For every away place used in the localization, the Hecke identity shows that

$$[b'(\mathrm{Frob}_y)]^{\mathrm{ss}} = [\sigma_{\nu,\mathrm{ad}}(\mathrm{Frob}_y)]^{\mathrm{ss}}.$$

Indeed a power of each prescribed class-function difference kills the generators of M'' , hence all of M'' , and vanishes at every point of R' . These Frobenius eigenvalues are ℓ -adic units. The geometric image of b' is compact by continuity. In a faithful matrix representation, the powers of one such Frobenius also have compact closure. If $h \in \Gamma'$ has degree one and this Frobenius has degree d , then $b'(h)^d$ differs from its image by an element of the compact normal geometric image. Its powers, and therefore all powers of $b'(h)$, have compact closure. The degree splitting now extends b' continuously to $\pi_1(U')$.

Chebotarev and the ordinary character criterion imply that, in every algebraic representation, b' and $\sigma_{\nu,\mathrm{ad}}$ have the same semisimplification on $\pi_1(U')$. The latter representation is semisimple because the parameter has reductive Zariski closure; the same remains true on the dense Weil group. Thus the assertion of the proposition holds after specialization at every k -point of R' .

It remains to pass from these pointwise statements to D_b . Let J be the annihilator in the ordinary group algebra $k[(\Gamma')^n]$ of the semisimple representation (7.3). The universal matrices over R' give an action on $\bigotimes_i V_i$. At every k -point, J kills each successive composition factor, so J^r acts by zero there for an integer r bounded by the dimension of that tensor product. Its matrix entries therefore lie in the nilradical of R' . Since R' is noetherian, a further power of J acts by zero over R' itself, and hence on D_b . The quotient $k[(\Gamma')^n]/J$ is the finite-dimensional semisimple image algebra of (7.3). Its simple modules are precisely the constituents appearing there, which proves the claim. $\square$

Proposition 7.5. *Quotient $N_b(V)$ by $\mathfrak{m}_C$ and localize at the neutral spherical Hecke values of f outside the finite exceptional set in Proposition 7.3. On the resulting finite-dimensional space, every eigenvalue α of complete Frobenius through a sufficiently divisible degree D_* satisfies*

$$|\alpha| \leq q^{D_* b/2}. \quad (7.4)$$

Proof. Every irreducible constituent of each factor of (7.3) is lisse on all of U , is defined over a finite extension of $\mathbb{Q}_\ell$, and has algebraic Frobenius eigenvalues. The constituent argument in [Ope26, Lemma 6.4] therefore gives a single pure weight for that constituent at every point of U . More explicitly, its determinant is made finite order by an algebraic constant-field twist; purity of the twisted irreducible sheaf then shows that every eigenvalue of the original constituent has the same exponent, independent of the place. This argument permits arbitrary ramification outside U .

Choose pairwise disjoint closed points for the moving legs in U' , and take complete Frobenius through a common multiple of their degrees. By (2.1), smooth transport, and Lemma 2.2, D_b is a subquotient to which the compact-cohomology weight bound applies. Thus the sum of the pure weights in every occurring external-tensor constituent is at most b . The Hecke quotient and localization can be performed at a geometric generic point; this argument only transports their resulting Weil subquotient and does not require Hecke modifications at a colliding leg.

The same constituents extend across all the points of T_I , by Proposition 7.4. Consequently the tuple of local Frobenius lifts in Proposition 7.3 has, on each constituent, the same sum weight as the disjoint good tuple. This gives (7.4) for the nearby-cycle comparison and its direct summand $N_b(V)$. $\square$

7.3 Central localization and the change of grading

We now apply this spectral inequality to the coherent models. Exact representability on the compact trace categories, with the duality of Section 5, gives an Ind-object F_N characterized by

$$K_I(J) = \mathrm{RHom}(\mathbb{D}[J], F_N). \quad (7.5)$$

This is the same representability argument as in [Ope26, proof of Proposition 2.7], applied to the finite tensor product of the compact categories in Proposition 5.3. All Hom complexes are taken equivariantly. Pull F_N to the chosen *étale* slices at x_i . We retain the sectors with trivial action of $Z(L)^n$, so the acting group becomes

$$H^\natural = \prod_{i=1}^n H_i/Z(L).$$

We also retain the sectors with trivial action of every $m_i(-1)$. The flag and spherical objects in Theorem 6.8 belong to these sectors.

Let $\mathcal{Z}$ be the tensor product of C , the neutral spherical algebras at all other places of $U \setminus T_I$, and the slice-center rings $k[U_{H_i}]^{H_i}$. An infinite tensor product here means the filtered union of its finite tensor products. Its character is given by the Hecke values of f and by the identity classes $u_i = 1$ on the slices; denote the corresponding maximal ideal by $\mathfrak{m}$.

Lemma 7.6. *The algebra $\mathcal{Z}$ acts on the representing Ind-object by commuting dg endomorphisms. Pullback to the slices and localization at $\mathfrak{m}$ commute with mapping from compact objects. In particular, the latter operation computes ordinary flat localization on the cohomology of the mapping complexes.*

Proof. The slice-center rings act centrally on the coherent models. For the away factors, include any finite disjoint list of additional places in the path functor with spherical labels, then restrict to the unit label at those places in the horizontal trace. The endomorphisms of these units carry the ordinary spherical algebra action computed by the loop model in [Ope26, Proposition 2.7]. They therefore act on the functor on the remaining trace compacts and hence on F_N .

These actions commute and are compatible as the finite list grows. Indeed insertion of further unit labels compares the external products of the balanced bar constructions defining the traces; their unit-insertion identities give the required coherence. Thus they define an action of the filtered union $\mathcal{Z}$ itself. Lemma 7.1 identifies these actions on mapping cohomology with the disjoint Hecke actions already used above. The *étale* comparisons respect these scalars by Section 6. Finally localization is tensoring with the flat algebra $\mathcal{Z}_{\mathfrak{m}}$, and a compact source commutes with the colimits computing this tensor product. $\square$

Apply this localization and the Koszul functors and cohomological change of grading of Theorem 6.8, extended to Ind-categories. The order of localization and these functors is immaterial. Denote the resulting object by

$$F^{\mathrm{ind}} \in \mathrm{Ind} \left(\mathrm{Coh}^{H^\natural} \left(\prod_{i=1}^n Y_i \right) \right).$$

Set $P = \boxtimes_i P_i$ and $B = \mathcal{O}_{\prod_i Y_i}$. By (6.23), B is the selected spherical summand of P . The change of grading uses the sum of the m_i -weights.

Theorem 7.7 (Simultaneous lower bound). *For every finite collection of the chosen places and every finite algebraic representation W of $H^\natural$, one has*

$$\mathrm{RHom}(P \otimes W, F^{\mathrm{ind}}) \in D^{\geq 0}(k). \quad (7.6)$$

The lower bound zero is independent of the collection of places and of W .

Proof. It suffices to consider irreducible W . Representations in the discarded parity sectors give zero. Every remaining W occurs in the restriction of V^* for some external tensor of neutral L -representations V_i , by the restriction statement of Theorem 6.8. Before the change of grading, (7.5) and the local comparison identify its mapping groups with the W -summand of (7.1), pulled to the slices and localized. Here $[Z(V_i)]$ corresponds to $P_{N,i} \otimes V_i$ and $\mathbb{D}P_{N,i} = P_{N,i}$. If m_i has weight w_i on W , put $w = \sum_i w_i$. A class in the original cohomological degree b has degree $b + w$ after the change of grading: the changed source with label W is $(P \otimes W)[-w]$.

The original degree- b mapping group is finitely generated over $\mathcal{Z}_{\mathfrak{m}}$. Before the slice extension this follows from Proposition 7.3, since a finite set of C -generators also generates over the larger central algebra. The *étale* extension preserves finite generation because we retain its slice-center scalars; taking a representation summand preserves it as well. We may therefore prove vanishing by reduction modulo $\mathfrak{m}$ and Nakayama's lemma. No noetherian hypothesis on the infinite tensor product $\mathcal{Z}$ is needed here.

The bound (7.4) persists on this reduction. First quotient by $\mathfrak{m}_C$ and localize at the away places used in Proposition 7.5; then perform the remaining central reductions and base changes. These operations commute with complete Frobenius. On the slice, the Frobenius action is the matrix action on the coherent representation labels, as in Section 6.

More precisely, before duality its action on the i th label is

$$V_i((t_i u_i)^{D_*/d_i}). \quad (7.7)$$

Dualizing an endomorphism means that on V_i^* one uses the representation matrix with inverse group argument, acting on the Hom complex by precomposition. This operator preserves each H_i -summand, since $t_i u_i \in H_i$. Its characteristic polynomial has coefficients in the slice-center ring, and the same polynomial identity holds on the mapping groups. At the central value $[u_i] = [1]$, every eigenvalue on the W -summand consequently has absolute value

$$q^{-wD_*/2},$$

because the absolute-value direction of t_i is $m_i/2$, with base q^{d_i} , by Lemma 3.1 and Section 3. Unipotent u_i does not change these eigenvalues.

If $b + w < 0$, this absolute value is strictly larger than $q^{bD_*/2}$, contrary to (7.4), unless the reduced mapping group is zero. Nakayama's lemma then makes the original localized group zero. This proves (7.6). Although the sufficiently divisible integer D_* may depend on the finite collection and the labels, the inequality $b + w \geq 0$ does not, giving the asserted uniformity. $\square$

7.4 Passage to ordinary quasi-coherent complexes

Let F^{qc} be the image of F^{ind} under the natural functor from Ind-coherent objects to ordinary equivariant quasi-coherent complexes on $\prod_i Y_i$. The next proposition specifies exactly which mapping complexes survive this passage.

Proposition 7.8. *The complex F^{qc} is concentrated in degrees at least zero. If Q belongs to the thick subcategory generated by the objects $P \otimes W$, for finite representations W of $H^{\natural}$, then the natural comparison is an isomorphism:*

$$\text{RHom}(Q, F^{\text{ind}}) \xrightarrow{\sim} \text{RHom}(Q, F^{\text{qc}}). \quad (7.8)$$

Proof. For a perfect source, the comparison is automatic. Apply Theorem 7.7 to the summand B of P and all its representation twists. Since reductive invariants are exact, these tests detect the ordinary cohomology of F^{qc} . They show that it lies in degrees at least zero.

Fix Q as in the statement. The left side of (7.8), even after replacing Q by $Q \otimes W$ for arbitrary finite W , has a uniform lower bound: start with Theorem 7.7 and use the finitely many cones, shifts, and retracts constructing Q . The right side also has a uniform lower bound, since Q is bounded coherent and $F^{\text{qc}} \in D^{\geq 0}$.

Let i be the regular equation embedding of $\prod_i Y_i$ into the smooth ambient space from Section 6. Its conormal bundle is the structure sheaf tensored with a finite representation. The complex Li^*i_*Q is perfect, because i_*Q is bounded coherent on that smooth ambient space. The finite Postnikov filtration of the self-intersection bimodule gives a filtration of Li^*i_*Q whose final quotient is Q and whose other quotients are

$$Q \otimes W_j[j], \quad j > 0,$$

for the exterior powers W_j of the conormal representation. This uses the regular-sequence formula for the Tor groups; it requires no splitting of the self-intersection bimodule.

For a source T , let $E(T)$ be the cone of the comparison (7.8). Suppose some $E(Q \otimes W)$ is nonzero. The uniform lower bounds give a least integer h in which one of these cones has nonzero cohomology. Choose such a W and apply the preceding filtration to $Q \otimes W$. Contravariance gives

$$E(T[j]) = E(T)[-j],$$

so all its positive- j pieces have cohomology only in degrees at least $h + j$. The final unshifted quotient therefore supplies an injection

$$H^h E(Q \otimes W) \hookrightarrow H^h E(Li^*i_*(Q \otimes W)).$$

The target is zero because its source is perfect. This contradiction proves the comparison. $\square$

In particular, the canonical map $H^0(F^{\text{qc}}) \rightarrow F^{\text{qc}}$ is available, and maps from finite constructions using the flag and spherical summands of P can be tested in ordinary quasi-coherent complexes. Both facts will be used in Section 8.

8 Simultaneous boundary tests and completion of the proof

We now compare the two consequences of a failure of enhancement. By Proposition 3.4, such a failure confines the finite type support $\text{Spec } R$ to the boundary of the open orbit at every selected place. We shall construct a coherent test at each place whose map to the spherical summand vanishes on every derived fiber over this support. Noetherianity then makes a finite product of these maps vanish on the whole support. The cuspidal vector f , however, survives every such product, as we shall see from a simple quotient of its Iwahori Hecke module.

Throughout this section, assume that the occurring excursion character ν has no enhancement as in Theorem 1.2. Retain the nonzero eigenvector f and its cyclic support algebra R from Proposition 2.6, together with the infinite sequence of places x_i from Lemma 3.1. At x_i write

$$r_i = q^{\deg x_i}, \quad H_i = Z_L(t_i), \quad E_i = \ker(\text{Ad}(t_i) - r_i), \quad H_i^{\natural} = H_i/Z(L).$$

Let m_i be the cocharacter of the prescribed triple. The local models of Theorem 6.8 are

$$Y_i = \{(u, e) \in U_{H_i} \times E_i : \text{Ad}(u)e = e\}, \quad B_i = \mathcal{O}(Y_i), \quad P_i = \bigoplus_b P_{i,b}.$$

Here b runs through the components of the t_i -fixed flag variety, and $P_{i,b}$ is the derived incidence pushforward in (6.16). The spherical summand of P_i is B_i . For a finite initial segment of length n , put

$$Y = \prod_{i=1}^n Y_i, \quad B = \bigotimes_{i=1}^n B_i, \quad P = \boxtimes_{i=1}^n P_i, \quad H^\natural = \prod_{i=1}^n H_i^\natural.$$

We use F_n^{ind} and F_n^{qc} for the localized, sheared objects constructed in Section 7. Thus $F_n^{\text{qc}} \in D^{\geq 0} \text{QCoh}^{H^\natural}(Y)$, and Proposition 7.8 compares mapping complexes from the thick closure of P , including its representation twists.

8.1 The cyclic support on the local models

The local equation $\text{Ad}(u)e = e$ describes a pair consisting of a residual holonomy and a raising vector. The global relation (2.3) gives such a pair at every chosen Frobenius. We first make precise how the cyclic module generated by f appears on the product of these local spaces.

Proposition 8.1. *There are morphisms*

$$\lambda_i : [\text{Spec } R/A] \longrightarrow [Y_i/H_i^\natural] \quad (8.1)$$

whose field-valued images have unipotent residual holonomy and a vector in E_i outside its open H_i -orbit. For every $n \geq 1$, the product morphism $\lambda = (\lambda_1, \dots, \lambda_n)$ is affine. Write R_n for the $H^\natural$ -equivariant B -algebra representing $\lambda_ \mathcal{O}$ on Y . There is an equivariant B -module map*

$$R_n \longrightarrow H^0(F_n^{\text{qc}}) \quad (8.2)$$

that sends 1 to the class of f in the spherical summand. In particular, the composite

$$B \longrightarrow R_n \longrightarrow H^0(F_n^{\text{qc}}) \longrightarrow F_n^{\text{qc}} \quad (8.3)$$

is the section defined by f .

Proof. Evaluation on the slices. At x_i , evaluate the universal homomorphism $\mathbf{b}$ at $\gamma_i = \text{Frob}_{x_i}$ and retain the universal vector $\mathbf{e}$. Equation (2.3) gives

$$\text{Ad}(\mathbf{b}(\gamma_i))\mathbf{e} = r_i \mathbf{e}.$$

The invariant functions of $\mathbf{b}(\gamma_i)$ on R are exactly their values on $t_{i,\text{ad}}$. Consequently this evaluation factors through the fiber of the adjoint conjugation quotient at the specified class.

Use the fixed point of the strongly étale slice base over this class. The central-torsor assertion of Lemma 3.1 identifies the resulting adjoint presentation with the presentation using full holonomy on the chosen sheet. The slice equivalence and elimination of the nonresonant equations therefore identify the evaluation with a map to $[Y_i/H_i^\natural]$. Here we also use the ring identification (6.24): on points, its inverse converts the spherical equation coordinates into the coordinates of Y_i . The local comparison preserves the open orbit and its boundary.

The point fixed in this construction is a point of the *quotient* $U_{H_i} // H_i$. The group coordinate u itself can vary throughout its unipotent locus. This distinction explains why the evaluation has

unipotent residual holonomy, rather than necessarily $u = 1$. By Proposition 3.4, any k -point whose vector reaches the open orbit would give an enhancement of ν . Our standing assumption therefore proves the boundary assertion for k -points. It also proves it for all field-valued points: a nonempty inverse image of the open-orbit locus after a field extension would give a nonempty constructible locus in the finite type support and hence a k -point.

For affineness, first use the adjoint presentations, before passing to the slices. Pulling back along their affine product presentation gives

$$(A^n \times \operatorname{Spec} R)/A, \quad h \cdot (g_1, \dots, g_n, z) = (g_1 h^{-1}, \dots, g_n h^{-1}, h z). \quad (8.4)$$

One frame trivializes the diagonal action: using g_1 identifies this quotient with $A^{n-1} \times \operatorname{Spec} R$. It is thus affine, and remains so after the slice base changes. This proves the assertion about λ and defines R_n .

The cyclic module and its grading. We retain the matrix coefficients of holonomy when identifying the action on cohomology. Before shearing, the framed regular module (2.5) and the inclusion (2.7) give a map

$$(k[A^n] \otimes R)^A \longrightarrow \mathcal{M}_n, \quad 1 \longmapsto f, \quad (8.5)$$

where on the right we take the sector on which $Z(L)^n$ acts trivially. By (2.6), this is a module map for the ordinary raising-equation rings: their holonomy functions act by the transported matrices $g_i \mathbf{b}(\gamma_i) g_i^{-1}$, and their degree-two variables act by $g_i \mathbf{e} g_i^{-1}$.

Base change this map to the étale slice bases over $A//A$. We may shrink each base so that it has only its selected point over the prescribed adjoint class. The full conjugation quotient is a central torsor there and splits after this base change. Project the target to the sheet prescribed by the full Hecke character of f . On that sheet, full holonomy functions are generated by the adjoint holonomy functions together with the base functions. After nonresonant elimination these functions and the raising coordinates give the entire spherical equation ring. Thus the map respects that entire ring, although the full holonomy functions need not have preserved its image before the sheet projection. The coordinate comparison (5.3) ensures that the projection retains f . After the slice equivalence, the source is the spherical-coordinate presentation of R_n . On the target apply the flat central localization of Section 7.

We record explicitly why the target becomes $H^\bullet(F_n^{\text{qc}})$ after shearing. A test by a neutral representation $V = \boxtimes_i V_i$ of L^n computes the mapping groups from the corresponding spherical summand tensored with V^* , by the transfer (5.4) and the functional (7.5). The comparison (6.25) respects the constant $H^\natural$ -maps between the restrictions of these labels. Since every finite representation of $H^\natural$ is a summand of such a restriction, these tests determine the complete rational equivariant module. They also determine its module structure: a ring operation paired with a finite representation is the spherical matrix operation on one side and precomposition by the corresponding coherent map on the other. This is precisely the spectator compatibility in Theorem 6.8.

For a representation test W on which the m_i have weights w_i , shearing changes the unsheared cohomological degree b to $b + \sum_i w_i$, as in (7.6). The spherical object becomes B by (6.23); mapping from this perfect source agrees with mapping to the quasi-coherent realization. We therefore obtain the required identification with $H^\bullet(F_n^{\text{qc}})$, including its full ring action.

To specify the direction of that action, put

$$C_{\text{sph}} = \bigotimes_i \operatorname{sh} H^\bullet(B_{\text{sph},i}^0), \quad \alpha : C_{\text{sph}} \xrightarrow{\sim} B$$

for the product of (6.24). Let R_n^{sph} be the source algebra in spherical coordinates. The inverse point map used to define λ gives exactly

$$R_n = R_n^{\text{sph}} \otimes_{C_{\text{sph},\alpha}} B; \quad b \cdot r = \alpha^{-1}(b)r \quad \text{on the underlying module } R_n^{\text{sph}}.$$

The cohomology identification above is α -linear: the operation $c \in C_{\text{sph}}$ on the spherical module becomes precomposition by $\alpha(c)$ on the coherent test. Hence it transports the source map to a B -module map from precisely R_n .

This transport is also compatible with the grading used in the final projection. Before shear, R_n^{sph} inherits the cohomological grading of $(k[A^n] \otimes R)^A$, in which the frame functions have degree zero. The degree-zero slice base changes retain this grading, together with the separate m_i -weight gradings coming from $H^\natural$ -equivariance. On the slice a holonomy function has original degree and m_i -weight $(0, 0)$, while a linear raising coordinate in factor i has degree and weight $(2, -2)$. Thus every element of B has sheared degree zero. The isomorphism α is $H^\natural$ -equivariant and preserves the sheared grading; its weight grading also recovers the original grading. Frame coordinates in R_n can have nonzero m_i -weights and need not have sheared degree zero. Nevertheless the B -action preserves every sheared degree of the source. Reindexing the rational weight direct sums leaves its underlying ungraded B -module unchanged.

Finally project to the sectors on which all $m_i(-1)$ act trivially and to sheared degree zero. These projections are B -linear: B has trivial indicated parities and is concentrated in sheared degree zero. They retain f , which has weight and cohomological degree zero. Forgetting the grading on the source gives (8.2). The last arrow of (8.3) is the canonical truncation map $H^0(F_n^{\text{qc}}) \rightarrow F_n^{\text{qc}}$, available because the latter is concentrated in nonnegative degrees. $\square$

8.2 Flags detecting the boundary

We construct the local test at a single selected place. For the following definition and lemma, write $H = H_i$, $E_+ = E_i$, $Y = Y_i$, $t = t_i$, $r = r_i$, and $m = m_i$. A component b of the t -fixed flag variety is called *bad* if

$$E_+ \cap \mathfrak{n}_{\mathcal{B}}$$

contains no point of the open H -orbit in E_+ , for a flag $\mathcal{B}$ of type b . This condition is independent of the flag within its component, since that component is a homogeneous H -space. The notation $\mathfrak{n}_{\mathcal{B}}$ denotes the nilradical of the Borel corresponding to the flag.

Lemma 8.2. *Let $(u, e) \in Y$ be a field point with u unipotent and e in the boundary of the open H -orbit in E_+ . After extension of its residue field, there is a flag $\mathcal{B}$ of bad type such that $u \in \mathcal{B} \cap H$ and $e \in E_+ \cap \mathfrak{n}_{\mathcal{B}}$.*

Proof. Work over an algebraically closed residue field. Choose a triple for e compatible with t , write e^- for its lowering element, and denote its diagonal cocharacter by h . Both h and m lie in H , and m is central in H . If $h = m$, set $c = t h(a^{\deg x_i})^{-1}$, the semisimple factor centralizing this triple. The weight-zero to weight-two map for the triple is surjective on each $\mathfrak{sl}_2$ -summand where weight two occurs. Taking c -invariants preserves this surjectivity; these invariant weight spaces are precisely $\text{Lie}(H)$ and E_+ . The orbit tangent map $\text{Lie}(H) \rightarrow E_+$ is then surjective, placing e in the open orbit. Hence $h \neq m$.

Put $\delta = m - h$, using additive notation for the commuting cocharacters. The raising and lowering elements of the chosen triple have m -weights 2 and -2 , since they have t -eigenvalues r and r^{-1} . They have the same h -weights. Thus δ centralizes the triple. Since u centralizes e , the triple parabolic contains u ; intersecting with H gives

$$u \in P_H(h) = P_H(-\delta).$$

We describe a Borel containing the whole of u . Write $u = u_+ u_0$ using the Levi decomposition of $P_H(-\delta)$, with $u_0 \in Z_H(\delta)$. Choose a Borel $\mathcal{B}_0$ of $Z_H(\delta)$ containing u_0 , and choose a maximal torus

T in $\mathcal{B}_0$. The element t and the cocharacters h, m, δ are central in this Levi and therefore belong to T . Choose a generic rational direction ϵ in the cocharacter space of T whose signs on the roots of $Z_H(\delta)$ give $\mathcal{B}_0$. Declare a root of L positive by the first nonzero entry in the lexicographic ordering

$$(-\delta, m, \epsilon).$$

After a small generic choice of ϵ , no root is left unordered. This ordering defines a Borel $\mathcal{B}$ of L . Since m is central in H , its intersection with H is the product of $\mathcal{B}_0$ with the unipotent radical of $P_H(-\delta)$. It consequently contains both u_+ and u_0 . Moreover e has δ -weight zero and positive m -weight, so $e \in \mathfrak{n}_{\mathcal{B}}$.

It remains to prove that its type is bad. Decompose $\mathfrak{g}$ into its m -weight spaces $\mathfrak{g}_j$ and consider the polynomial

$$\Delta(x) = \prod_{j>0} \det((\text{ad } x)^j : \mathfrak{g}_{-j} \longrightarrow \mathfrak{g}_j)^j, \quad x \in E_+. \quad (8.6)$$

Choose volume forms to interpret the determinants as scalars; their vanishing loci are independent of this choice. The polynomial is nonzero at a raising element whose diagonal cocharacter is m , by $\mathfrak{sl}_2$ representation theory. Its nonvanishing is H -invariant because m is central in H , so it is nonzero throughout the open orbit.

In the following trace calculation, a cocharacter denotes its differential. Write c_{Δ} for the weight in the evaluation identity $\Delta(\text{Ad}(\delta(z))x) = z^{c_{\Delta}} \Delta(x)$. Then

$$\begin{aligned} c_{\Delta} &= \sum_{j>0} j(\text{Tr}(\text{ad } \delta | \mathfrak{g}_j) - \text{Tr}(\text{ad } \delta | \mathfrak{g}_{-j})) = \text{Tr}(\text{ad } m \text{ ad } \delta) \\ &= \text{Tr}((\text{ad } \delta)^2) > 0. \end{aligned}$$

For the second equality, the trace pairing of h with δ vanishes: δ commutes with the triple, so it pairs trivially with the bracket $[e, e^-]$ defining its neutral element. The final inequality follows because $\delta \neq 0$ is a cocharacter of the semisimple group L . On the other hand, the priority given to $-\delta$ ensures that every δ -weight in $E_+ \cap \mathfrak{n}_{\mathcal{B}}$ is nonpositive. A polynomial of the strictly positive character just computed must vanish on that subspace. It therefore contains no point of the open orbit, proving that the type of $\mathcal{B}$ is bad. $\square$

For each place, let the units of the derived incidence algebras define

$$q_i : Q_i \longrightarrow B_i, \quad Q_i = \text{fib} \left(B_i \longrightarrow \bigoplus_{b \text{ bad}} P_{i,b} \right). \quad (8.7)$$

These are maps of bounded coherent equivariant objects. They belong to the thick subcategory generated by P_i , since B_i and all the $P_{i,b}$ are summands of P_i .

Lemma 8.3. *The pullback of q_i to $\text{Spec } R$ is zero on every derived residue-field fiber.*

Proof. By Proposition 8.1, every field point of the support evaluates to a point (u, e) with u unipotent and e in the boundary. Lemma 8.2 supplies a point of one of the bad incidence spaces above (u, e) , after field extension if necessary. Evaluation at that point gives a left inverse, on the derived fiber, to the unit

$$B_i \longrightarrow \bigoplus_{b \text{ bad}} P_{i,b}.$$

One can obtain this left inverse by the natural derived base-change map followed by evaluation, or by pushing the structure-sheaf map to the chosen flag point and using adjunction. Its composition with the unit is the identity of the residue field. In the derived category of a field, the fiber arrow of a split injection is zero. Faithfully flat extension of residue fields detects zero maps between complexes of vector spaces, proving the assertion over the original residue field. $\square$

8.3 A tensor argument on noetherian support

Vanishing on residue fields does not by itself annihilate a map over R . The next lemma explains why the infinite supply of places is sufficient. It allows different maps at different places and does not require their sources to be perfect. It is a sequential variant of tensor nilpotence with parameters [Tho97, Theorem 3.8]. We give the proof using noetherian support reduction, as in the argument attributed to Neeman in [Tho97, Sections 3.6.5–3.6.7 and History 3.17].

Lemma 8.4 (Successive tensor vanishing). *Let S be a noetherian ring and let $v_i : D_i \rightarrow S$, $i \geq 1$, be maps in $D(S)$ whose sources are pseudo-coherent and bounded above. Suppose*

$$v_i \otimes_S^{\mathbf{L}} \kappa(\mathfrak{p}) = 0 \quad \text{for every } i \geq 1 \text{ and every } \mathfrak{p} \in \operatorname{Spec} S.$$

Then, for some $n \geq 1$,

$$v_1 \otimes_S^{\mathbf{L}} \cdots \otimes_S^{\mathbf{L}} v_n = 0. \tag{8.8}$$

More generally, for every bounded coherent complex K and every starting index a , there is $b \geq a$ such that $(v_a \otimes_S^{\mathbf{L}} \cdots \otimes_S^{\mathbf{L}} v_b) \otimes_S^{\mathbf{L}} \operatorname{id}_K = 0$.

Proof. For $a \leq b$, write

$$D_{a,b} = D_a \otimes_S^{\mathbf{L}} \cdots \otimes_S^{\mathbf{L}} D_b, \quad v_{a,b} = v_a \otimes_S^{\mathbf{L}} \cdots \otimes_S^{\mathbf{L}} v_b.$$

We prove the more general assertion by noetherian induction on $\operatorname{supp} K$. We first note that the asserted property, with all starting indices allowed, is preserved by triangles. In a triangle $K_1 \rightarrow K \rightarrow K_2$, choose an initial product that kills K_2 . Naturality shows that its map to K factors through K_1 . A subsequent product killing K_1 kills this factorization, again by naturality. Shifts and finite sums also preserve the property.

The assertion is immediate when $K = 0$. Otherwise let $\mathfrak{p}_1, \dots, \mathfrak{p}_s$ be the generic points of $\operatorname{supp} K$. The localized complex $K_{\mathfrak{p}_j}$ has finite-length cohomology. Finite devissage by copies of the residue field, together with the preceding triangle observation, gives a product beginning at a that kills $K_{\mathfrak{p}_j}$. Increasing the final index preserves vanishing, so choose one b that works for all j . Denote the resulting map by

$$z = v_{a,b} \otimes \operatorname{id}_K : D_{a,b} \otimes_S^{\mathbf{L}} K \longrightarrow K.$$

Its localization is zero at every $\mathfrak{p}_j$.

The source is pseudo-coherent and bounded above, while the target is bounded coherent. Accordingly, its derived Hom group commutes with localization; this is also [The26, Lemma 15.101.2(3)]. To see the finiteness needed here, resolve the source by a bounded-above complex of finite projective modules and represent the target by a bounded complex. Only finitely many terms contribute to each degree of the Hom complex, so localization commutes with that complex and its cohomology. It follows that $\operatorname{Ann}_S(z)$ is not contained in any $\mathfrak{p}_j$. Finite prime avoidance gives

$$s \in \operatorname{Ann}_S(z) \setminus \bigcup_j \mathfrak{p}_j.$$

Thus $sz = 0$, and the triangle for multiplication by s provides a factorization of z through

$$K' = \operatorname{fib}(s : K \longrightarrow K).$$

This is bounded coherent, with support in the proper closed subset $\operatorname{supp} K \cap V(s)$. By induction a subsequent product $v_{b+1,c}$ kills K' . Tensor naturality with respect to the factorization through K' now shows that $v_{a,c} \otimes \operatorname{id}_K = 0$. This proves the induction. The original assertion follows by taking $K = S$ and $a = 1$. $\square$

Apply this lemma to the maps obtained by pulling (8.7) back to $\text{Spec } R$. The ring R is noetherian by Proposition 2.6; the sources are pseudo-coherent and bounded above because they are derived pullbacks of bounded coherent objects. Lemma 8.3 verifies the residue-field hypothesis. We obtain an integer n for which the product is zero on $\text{Spec } R$.

This zero also holds equivariantly. Indeed A is linearly reductive, and we can compute these maps using bounded-above equivariant free resolutions, whose underlying complexes are K-projective. Ordinary vanishing then supplies an actual null-homotopy, which can be averaged to an equivariant one. The averaging is well defined even for the present unbounded resolutions: the orbit of an individual vector, and the span of its image under the homotopy, lie in finite-dimensional rational subrepresentations. Integration against the invariant integral on $\mathcal{O}(A)$ is therefore defined on each vector and preserves module linearity.

Set $Q_\Pi = \boxtimes_{i=1}^n Q_i$ and $q_\Pi = \boxtimes_{i=1}^n q_i$. By adjunction for the affine morphism λ , the equivariant vanishing just obtained is precisely

$$(Q_\Pi \xrightarrow{q_\Pi} B \longrightarrow R_n) = 0. \quad (8.9)$$

Composing with (8.2) and then the truncation map in (8.3), we conclude that

$$q_{\Pi}^* f = 0 \quad \text{in } H^0 \text{RHom}(Q_\Pi, F_n^{\text{qc}}). \quad (8.10)$$

Since Q_Π lies in the thick closure of P , the same vanishing holds with F_n^{ind} by Proposition 7.8.

8.4 The cuspidal class survives the same tests

We keep the finite set of places supplied by the tensor lemma. Write $\mathcal{H} = \text{End}(P)^{\text{op}}$, so that precomposition gives a left action of this ordinary algebra on $\text{Hom}(P, F_n^{\text{ind}})$. The contradiction will take place in a finite-dimensional simple quotient of the Iwahori Hecke module containing f . First we identify that module and record why its spherical part cannot meet a bad summand.

Lemma 8.5. *The complex $\text{RHom}(P, F_n^{\text{ind}})$ is concentrated in degree zero. Its degree-zero module has a simple $\mathcal{H}$ -quotient that detects f in the spherical summand and on which the local centers act by the chosen classes of the t_i . In this simple quotient, every idempotent projecting to a bad flag summand in any one factor acts by zero.*

Proof. Let $\mathcal{H}_I$ be the tensor product of the affine Iwahori Hecke algebras at the selected places, and let M_I be the space of compactly supported functions on the corresponding bundle groupoid. Before base change, the functional (7.5) on the Iwahori units is $M_I[0]$, with its ordinary Hecke action; see [Ope26, Section 2, Hecke action and transfer]. Write

$$\mathcal{Z}_I = \bigotimes_i k[L]^L, \quad \mathcal{Z}_{\text{sl}} = \bigotimes_i k[U_{H_i}]^{H_i}$$

for the local class-coordinate rings before and after the slice. The map between them includes the coordinate convention of (5.3).

Theorem 6.8 identifies the entire ordinary endomorphism algebra, and the full mapping module, as

$$\mathcal{H} \simeq \mathcal{H}_I \otimes_{\mathcal{Z}_I} \mathcal{Z}_{\text{sl}}, \quad \text{RHom}(P, F_n^{\text{ind}}) \simeq (M_I \otimes_{\mathcal{Z}_I} \mathcal{Z}_{\text{sl}})_{\mathfrak{m}}[0]. \quad (8.11)$$

The subscript denotes the central localization of Section 7, including its away Hecke factors. Indeed the slice changes the original groups by this flat scalar extension. Duality accounts for the opposite algebra; the fully faithful Koszul functor then preserves the mapping complexes. The retained

parity and central sectors contain the source P , so they do not change its mapping groups. Finally equivariant maps have m_i -weight zero, hence shearing does not change their degree. These facts prove both identities in (8.11) and the asserted concentration. If one also localizes the category in the local center, the first identity is localized in those same local center variables.

Project the original space of compactly supported functions onto its finite-dimensional cuspidal subspace at this Iwahori level. In a complex realization the automorphic inner product is positive definite, the Hecke action is closed under adjoints, and the orthogonal projection is Hecke-equivariant. The joint finite-dimensional image of the local Hecke algebras and the commuting away spherical algebras is therefore semisimple. The spherical summand contains f by level transfer. Select a simple quotient that detects f , within the simultaneous away eigenspace of its specified character. The local class coordinates act on its spherical summand by the prescribed full Hecke values. They consequently act on the whole simple quotient by those values. Via (8.11), extend its action to the entire algebra $\mathcal{H}$ by evaluating $\mathcal{Z}_{\text{sl}}$ at the selected identity classes. The image algebra is unchanged by this scalar extension, so the resulting module is still simple; in particular the new flag idempotents on the slice act on it. Its central values are precisely those used in the localization at $\mathfrak{m}$, so it remains a quotient after localization. This construction transfers back to k through the chosen complex realization and gives the required simple module, denoted by S .

Let e_{sph} be the idempotent of the spherical summand B , and let e_b project to a bad summand in one factor of P . If both idempotents act nontrivially on S , simplicity implies that $e_{\text{sph}}\mathcal{H}e_b\mathcal{H}e_{\text{sph}}$ acts nontrivially on $e_{\text{sph}}S$. For example, generate S from $e_{\text{sph}}S$, project to e_bS , and then generate back to $e_{\text{sph}}S$. Some composite from B through that bad summand and back to B must therefore act nontrivially.

Such a composite belongs to

$$\text{End}_{H^\natural}(B) = \bigotimes_{i=1}^n k[U_{H_i}]^{H_i}, \quad (8.12)$$

with the corresponding equality after any localization of these local center rings. To justify the equality, $\text{End}(B)$ before invariants is $\mathcal{O}(Y)$, and m_i acts trivially on U_{H_i} while all linear functions in e_i have strictly negative m_i -weight. Thus an invariant function is independent of the e_i . This observation also identifies its action on S with evaluation at the selected quotient classes.

Now set all residual group coordinates to $u_i = 1$ and take an open-orbit vector in the factor containing the bad summand. The support of that summand is empty there, by its definition and (6.16). Consequently every composite through it vanishes on this fiber. By (8.12), the composite is independent of the raising vectors, so its value at the selected quotient classes is zero. It acts by zero on S , contradicting the preceding paragraph. Therefore every bad idempotent acts by zero. $\square$

Proposition 8.6. *For every finite initial segment of the selected places,*

$$q_\Pi^* f \neq 0 \quad \text{in } H^0 \text{RHom}(Q_\Pi, F_n^{\text{ind}}).$$

Proof. Apply $\text{RHom}(-, F_n^{\text{ind}})$ to the exterior product of the fiber diagrams (8.7). Every term made from spherical and flag summands is a summand of P , so its mapping complex is concentrated in degree zero by Lemma 8.5. The resulting finite total complex has its all-spherical term in degree zero and the terms with one bad factor in degree -1 . Hence its zeroth cohomology is

$$\frac{\text{Hom}(B, F_n^{\text{ind}})}{\sum_{i=1}^n \sum_{b \text{ bad}} \text{im} \left[\text{Hom}(B_1 \boxtimes \cdots \boxtimes P_{i,b} \boxtimes \cdots \boxtimes B_n, F_n^{\text{ind}}) \longrightarrow \text{Hom}(B, F_n^{\text{ind}}) \right]}. \quad (8.13)$$

The map from the all-spherical term to this quotient is pullback along q_{Π} ; terms with two or more bad factors affect only negative degrees. The simple quotient S of Lemma 8.5 annihilates every image in the denominator, since its corresponding bad idempotent acts by zero. It detects f in the numerator. Thus the class of f survives in (8.13), proving the proposition. $\square$

Proof of Theorem 1.2. Under the assumption that no enhancement exists, Lemma 8.4 produced a finite set of selected places for which (8.10) holds. The mapping comparison of Proposition 7.8 gives the same vanishing in the Ind category, contradicting Proposition 8.6. Therefore the assumption was false.

Proposition 3.4 gives the precise resulting pair: its SL_2 -homomorphism has raising element in $\mathcal{O}_\nu$, its commuting Weil homomorphism has reductive closure and weight zero under every complex embedding, and their prescribed diagonal specialization equals the given σ_ν on the entire Weil group. That proposition also supplies continuity and a finite coefficient field. The argument used neither a restriction on the split semisimple type nor reducedness of the level divisor, so it applies to all the data of Theorem 1.2. $\square$

References

- [AB09] Sergey Arkhipov and Roman Bezrukavnikov. Perverse sheaves on affine flags and Langlands dual group. *Israel Journal of Mathematics*, 170:135–183, 2009. Section and equation locators refer to arXiv version 6.
- [Art84] James Arthur. On some problems suggested by the trace formula. In Rebecca Herb, Stephen Kudla, Ronald Lipsman, and Jonathan Rosenberg, editors, *Lie Group Representations II*, volume 1041 of *Lecture Notes in Mathematics*, pages 1–49. Springer, Berlin, 1984.
- [Art89] James Arthur. Unipotent automorphic representations: conjectures. In *Orbites unipotentes et représentations – II. Groupes p -adiques et réels*, number 171–172 in *Astérisque*, pages 13–71. Société mathématique de France, 1989.
- [BMR05] Michael Bate, Benjamin Martin, and Gerhard Röhrle. A geometric approach to complete reducibility. *Inventiones Mathematicae*, 161(1):177–218, 2005.
- [BMR20] Michael Bate, Benjamin Martin, and Gerhard Röhrle. Semisimplification for subgroups of reductive algebraic groups. *Forum of Mathematics, Sigma*, 8, 2020. e43, 10 pp.
- [BMRT13] Michael Bate, Benjamin Martin, Gerhard Röhrle, and Rudolf Tange. Closed orbits and uniform S -instability in geometric invariant theory. *Transactions of the American Mathematical Society*, 365(7):3643–3673, 2013.
- [BBD82] Alexander A. Beilinson, Joseph Bernstein, and Pierre Deligne. *Faisceaux pervers*, volume 100 of *Astérisque*. Société Mathématique de France, Paris, 1982.
- [BZCHN24] David Ben-Zvi, Harrison Chen, David Helm, and David Nadler. Coherent Springer theory and the categorical Deligne–Langlands correspondence. *Inventiones Mathematicae*, 235:255–344, 2024.
- [BIY24] Alexander Bertoloni Meli, Naoki Imai, and Alex Youcis. The Jacobson–Morozov morphism for Langlands parameters in the relative setting. *International Mathematics Research Notices*, 2024(6):5100–5165, 2024. Statement numbering as in arXiv:2203.01768v3.

- [Bez16] Roman Bezrukavnikov. On two geometric realizations of an affine Hecke algebra. *Publications Mathématiques de l’IHÉS*, 123:1–67, 2016.
- [BF08] Roman Bezrukavnikov and Michael Finkelberg. Equivariant Satake category and Kostant–Whittaker reduction. *Moscow Mathematical Journal*, 8(1):39–72, 2008.
- [dCHL18] Mark Andrea de Cataldo, Thomas J. Haines, and Li Li. Frobenius semisimplicity for convolution morphisms. *Mathematische Zeitschrift*, 289(1–2):119–169, 2018. Theorem locators refer to arXiv version 3.
- [DM18] François Digne and Jean Michel. Quasi-semisimple elements. *Proceedings of the London Mathematical Society*, 116:1301–1328, 2018.
- [Dri80] Vladimir G. Drinfeld. Langlands’ conjecture for $GL(2)$ over functional fields. In *Proceedings of the International Congress of Mathematicians (Helsinki, 1978)*, volume 2, pages 565–574. Academia Scientiarum Fennica, Helsinki, 1980.
- [EX24] Arnaud Eteve and Cong Xue. Nearby cycles commute with proper direct image on stacks of shtukas. arXiv:2409.16474v1, 2024.
- [Gai01] Dennis Gaitsgory. Construction of central elements in the affine Hecke algebra via nearby cycles. *Inventiones Mathematicae*, 144(2):253–280, 2001.
- [GLR26] Dennis Gaitsgory, Vincent Lafforgue, and Sam Raskin. Tempered vs generic automorphic functions and the canonical filtration on automorphic functions, 2026. Version 2, April 14, 2026.
- [Kal21] Tasho Kaletha. Supercuspidal L -packets. arXiv:1912.03274v2, 2021.
- [Laf02] Laurent Lafforgue. Chtoucas de Drinfeld et correspondance de Langlands. *Inventiones Mathematicae*, 147(1):1–241, 2002.
- [Laf18] Vincent Lafforgue. Chtoucas pour les groupes réductifs et paramétrisation de Langlands globale. *Journal of the American Mathematical Society*, 31(3):719–891, 2018.
- [Lun73] Domingo Luna. Slices étales. In *Sur les groupes algébriques*, number 33 in Bulletin de la Société Mathématique de France. Mémoire, pages 81–105. Société Mathématique de France, 1973.
- [MV07] Ivan Mirković and Kari Vilonen. Geometric Langlands duality and representations of algebraic groups over commutative rings. *Annals of Mathematics*, 166(1):95–143, 2007.
- [Ope26] OpenAI. Ramanujan–Arthur Decompositions of Cuspidal Functions at Full Finite Level. OpenAI Math Release preprint [OAI:Ramanujan-Arthur-Decompositions-of-Cuspidal-Functions-at-Full-Finite-Level-September-24-2026](#), 2026.
- [Ras26] Sam Raskin. Conjectures of Arthur and Ramanujan for unramified automorphic forms: announcement, 2026. Contribution to the 2026 ICM Proceedings; manuscript dated October 12, 2025, updated March 10, 2026.
- [Sal23] Andrew Salmon. Unipotent nearby cycles and the cohomology of shtukas. *Compositio Mathematica*, 159(3):590–615, 2023. <https://doi.org/10.1112/S0010437X23007017>. Preprint version consulted: arXiv:2106.04768v3.

- [The26] The Stacks project authors. The stacks project. <https://stacks.math.columbia.edu/tag/0A6A>, 2026. Tag 0A6A, Lemma 15.101.2(3). Accessed October 3, 2026.
- [Tho97] R. W. Thomason. The classification of triangulated subcategories. *Compositio Mathematica*, 105(1):1–27, 1997.
- [Xue20a] Cong Xue. Finiteness of cohomology groups of stacks of shtukas as modules over Hecke algebras, and applications. *Épjournal de Géométrie Algébrique*, 4, 2020. Article 6, 42 pp.
- [Xue20b] Cong Xue. Smoothness of cohomology sheaves of stacks of shtukas. arXiv:2012.12833v1, 2020.